Cadence Virtuoso + UMC Designkit

Tutorial for creation of an inverter cell from schematic to layout using CADENCE Virtuoso and UMC 65 nm Designkitit and hints for multi-input gates

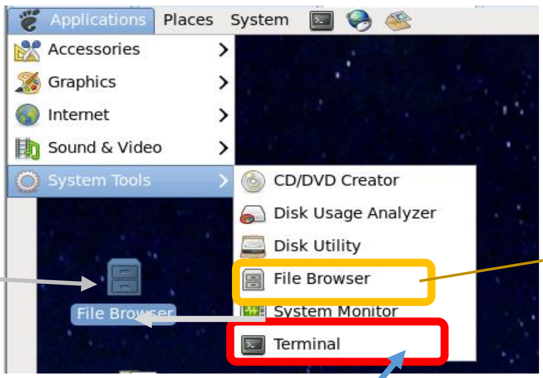
1. Using the Linux (CentOS) environment
2. Preparing a working directory for Cadence with UMC 65 nm
3. Creating a new library
4. Schematic entry
5. Creating a symbol and a testbench
6. Spice-Simulation using ADE Assembler
   - Analyses selection and setup
   - Using variables
   - Parametric simulation
7. Starting Layout XL
   - Generate layout from schematic
   - The layer window
   - Wires, vias/contacts, wells
8. Design rule check
9. Layout versus schematic check (LVS)
10. Parasitic extraction (QRC)
11. Simulation of the extracted model (Post layout simulation)

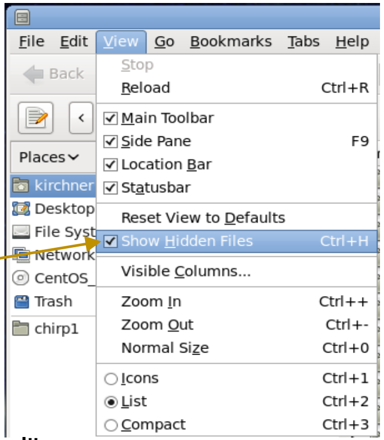
Some UNIX-commands

We use CentOS-Linux (Free version "**C**ommunity **Ent**erprise **OS**" of RedHat), because Cadence supports RedHat or Suse. We use the KornShell (ksh), because it is the standard shell at ITMZ Uni Rostock.

Our client computers use the "supported" CentOS 7, binary compatible to RedHat 7

pwd	print working directory
ls, ls –alsi, ls *dir*	list
cd -> cd /~/ske -> cd ./ske	change directory
mkdir *newdir*	make directory
cp , cp -r, cp -p	copy files/directories (recursive, preserve attributes)
mv	move or rename directories
chmod, chmod –R chmod 744 *file*	change file mode bits (security) -> -R recursive
grep	filter for pattern
use of | for pipeline -> ls | grep *pattern*	use command output as input for next command
rm, rm –r	remove/delete
rmdir	remove directory
cat file	show file at standard output
man *cmd*	show manual of cmd
history, h	show command history
r 100	run command number 100 from history





Move icon to desktop for easy use

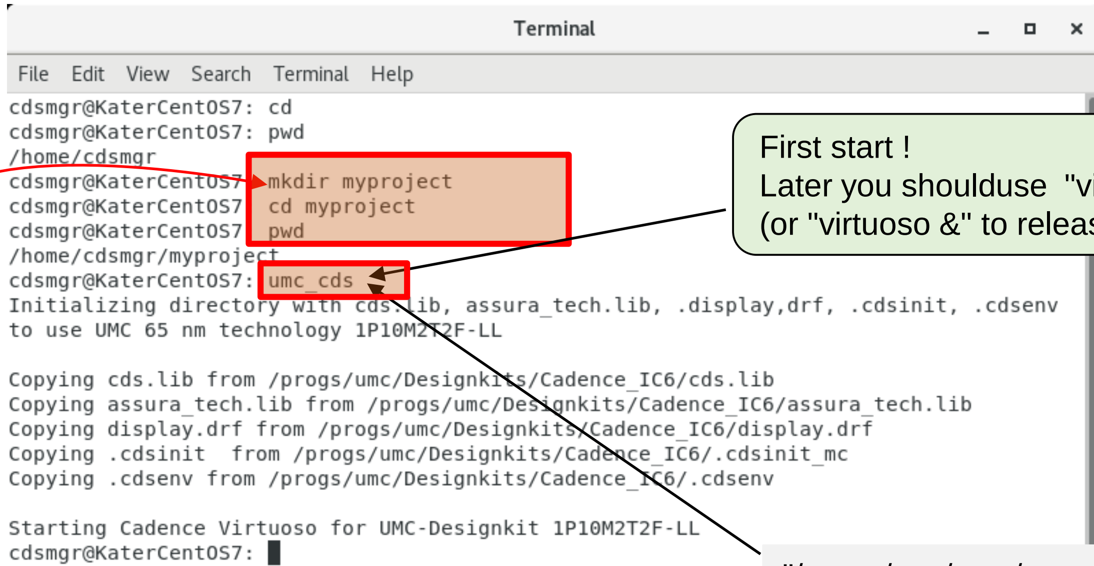
Or Right-Mouse on Background and select "Open Terminal"

To save windows images (.png) : Alt-Print (or use the screenshot tool from menue)

Unix-command to remove Cadence lock files (Close all Cadence programs before!) : **find ~/ -name "*.cdslck" -exec rm {} \;**

UMC Designkit - Start

= Starting Cadence Design Framework with Libraries and Rules for UMC 65 nm



First start !
Later you shoulduse "virtuoso"!
(or "virtuoso &" to release the terminal for further use)

"/progs/env/umc/ums_cds" is a script to initialize the working directory.
It is made at Uni_Rostock/IGS and not part of the UMC-Designkit

See the setup files shown on the next two pages! These
files will be copied to the project directory.
(.display.drf not shown)

Do not use your homedir!
Create a separate project directory!

Make your favourite directory structure
~/ske
~/ske/inverter

You may use one single directory for different designs.

Setup Files

Control of license checkout order (may be edited by CIW → Options → License → Ordering)
Adapted checkout sequence accelerates licence checkout and avoids interim error meassages

.cdsenv

```
viva.application  vivaWindowMode             string         "window"ddserv  showWhatsNew    string
"no"layout  showWhatsNew    string  "no"asimenv showWhatsNew string "no"
license           ADELicenseCheckoutOrder    string         "GXL,XL,L"
license           maestroCheckoutOrder       string         "Assembler,Explorer"
license           VLSL_UseNextLicense        string         "always"
license           VLSXL_UseNextLicense       string         "always"
license           VLSLicenseCheckoutOrder    string         "GXL,XL,L"
license           ADE_UseNextLicense         string         "always"
license           ADEXL_UseNextLicense       string         "always"
license           adeMaestroCheckoutOrder    string         "ADE,Maestro"
license           Explorer_UseNextLicense    string         "always"
license           ADEL_UseNextLicense        string         "always"
license           VSEL_UseNextLicense        string         "always"
license           VSELicenseCheckoutOrder    string         "XL,L"
```

Libraries (Cadence, UMC-Kit, User)

cds.lib

```
DEFINE cdsDefTechLib $CDS_HOME/tools/dfII/etc/cdsDefTechLib
DEFINE basic $CDS_HOME/tools/dfII/etc/cdslib/basic
DEFINE analogLib $CDS_HOME/tools/dfII/etc/cdslib/artist/analogLib

DEFINE umc65ll $umcInstallPath/umc65ll

DEFINE designLib /home/cdsmgr/myproject/designLib
```

Cadence libraries

The UMC designkit library

Additional entries after creating your own libraries

Technology libraries for physical verification using the "Assura flow"

assura_tech.lib

```
DEFINE umc65nm_DRC /eda/umc/Designkits/Cadence_IC6/RuleDecks/Assura/DRC
DEFINE umc65nm_LVS /eda/umc/Designkits/Cadence_IC6/RuleDecks/Assura/LVS
```

Setup Files

.cdsinit

```
envSetVal("spectre.envOpts" "modelFiles" 'string "/eda/umc/Designkits/Cadence_IC6/Models/Spectre/l65ll_v181.lib.scs;tt_ll_rvt12
                              /eda/umc/Designkits/Cadence_IC6/Models/Spectre/l65ll_v181.lib.scs;tt_ll_hvt12
                              /eda/umc/Designkits/Cadence_IC6/Models/Spectre/l65ll_v181.lib.scs;tt_ll_lvt12
                              …
                              /eda/umc/Designkits/Cadence_IC6/Models/Spectre/l65ll_v181.lib.scs;tt_ll_pcap25
                              /eda/umc/Designkits/Cadence_IC6/Models/Spectre/l65ll_v181.lib.scs;tt_ll_res
                                                                                    …
                              #;/eda/umc/Designkits/Cadence_IC6/Models/Spectre/l65ll_v181.lib.scs;ss_ll_rvt12
                              ….
                              #;/eda/umc/Designkits/Cadence_IC6/Models/Spectre/l65ll_v181.lib.scs;ff_ll_rvt12
                              ….
                              #;/eda/umc/Designkits/Cadence_IC6/Models/Spectre/l65ll_v181.lib.scs;fnsp_ll_rvt12
                              ….
                              #;/eda/umc/Designkits/Cadence_IC6/Models/Spectre/l65ll_v181.lib.scs;ff_ll_rvt12
                              ….
                              #;/eda/umc/Designkits/Cadence_IC6/Models/Spectre/Monte_Carlo/l65ll_v181_mc.lib.scs;mc_param
                              #;/eda/umc/Designkits/Cadence_IC6/Models/Spectre/Monte_Carlo/l65ll_v181_mc.lib.scs;mc_ll_rvt12
                              …
                              #;/eda/umc/Designkits/Cadence_IC6/Models/Spectre/Monte_Carlo/l65ll_v181_mc.lib.scs;mc_ll_pcap25
                              #;/eda/umc/Designkits/Cadence_IC6/Models/Spectre/Monte_Carlo/l65ll_v181_mc.lib.scs;mc_ll_res"
                                             …                                      ")
```

Simulation libraries for Typical

Slow NMOS Slow PMOS

Fast NMOS Fast PMOS

Fast NMOS Slow PMOS

Slow NMOS Fast PMOS

Simulation libraries for Corner Simulation

Simulation libraries for Monte Carlo Simulation

Setup for layout

```
envSetVal("layout" "gridSpacing" 'float 0.1)
envSetVal("layout" "gridMultiple" 'int 5)
envSetVal("layout" "xSnapSpacing" 'float 0.005)
envSetVal("layout" "ySnapSpacing" 'float 0.005)
envSetVal("layout"  "stopLevel"          'int 32)
envSetVal("layout"  "snapMode"           'string "anyAngle")
envSetVal("layout"  "segSnapMode" 'string "anyAngle")
envSetVal("layout"  "displayResolution"  'string        "Medium")
```

Technology libraries for physical verification using the "Assura flow"

assura_tech.lib

```
DEFINE umc65nm_DRC /eda/umc/Designkits/Cadence_IC6/RuleDecks/Assura/DRC
DEFINE umc65nm_LVS /eda/umc/Designkits/Cadence_IC6/RuleDecks/Assura/LVS
```

techRuleSets

```
ruleSet( "default"
   (DrcRules      "G-DF-LOGIC_MIXED_MODE65N-1P10M2T2F-LL-ASSURA-DRC-1.14-P3.rul" )
   (DrcSwitchesFile "switches_DRC_1P10M2T2F_option19")  )

ruleSet( "1P10M2T2F_option19"
(DrcRules      "G-DF-LOGIC_MIXED_MODE65N-1P10M2T2F-LL-ASSURA-DRC-1.14-P3.rul" )
 (DrcSwitchesFile "switches_DRC_1P10M2T2F_option19")  )

ruleSet( "1P10M_option19_antenna"
(DrcRules      "G-DF-LOGIC_MIXED_MODE65N-1P10M-Antenna-ASSURA-DRC-V10.rul" )
(DrcSwitchesFile "switches_DRC_1P10M_antenna_option19")  )
```

For design rule check (DRC)

```
ruleSet( "default"
    (LvsExtractRules     "./G-DF-LOGIC_MIXED_MODE65N-LL_LOW_K_ASSURA-LVS-1.6-P1-EXTRACT.RUL" )
    (LvsCompareRules     "./G-DF-LOGIC_MIXED_MODE65N-LL_LOW_K_ASSURA-LVS-1.6-P1-COMPARE.RUL" )
    (LvsSwitchesFile     "./switches_LVS_1P10M2T2F_option19" ))

ruleSet( "1P10M2T2F_option19"
    (LvsExtractRules     "./G-DF-LOGIC_MIXED_MODE65N-LL_LOW_K_ASSURA-LVS-1.6-P1-EXTRACT.RUL" )
   (LvsCompareRules     "./G-DF-LOGIC_MIXED_MODE65N-LL_LOW_K_ASSURA-LVS-1.6-P1-COMPARE.RUL" )
   (LvsSwitchesFile     "./switches_LVS_1P10M2T2F_option19" ))

ruleSet( "1P10M2T2F_opt19_LowK_Typ"      (RcxSetupDir        "../QRC/Typ" ))
ruleSet( "1P10M2T2F_opt19_LowK_RCmax"   (RcxSetupDir        "../QRC/RCmax" ))
ruleSet( "1P10M2T2F_opt19_LowK_RCmin"   (RcxSetupDir        "../QRC/RCmin" ))
ruleSet( "1P10M2T2F_opt19_LowK_Cmax"    (RcxSetupDir        "../QRC/Cmax" ))
ruleSet( "1P10M2T2F_opt19_LowK_Cmin"    (RcxSetupDir        "../QRC/Cmin" ))
```

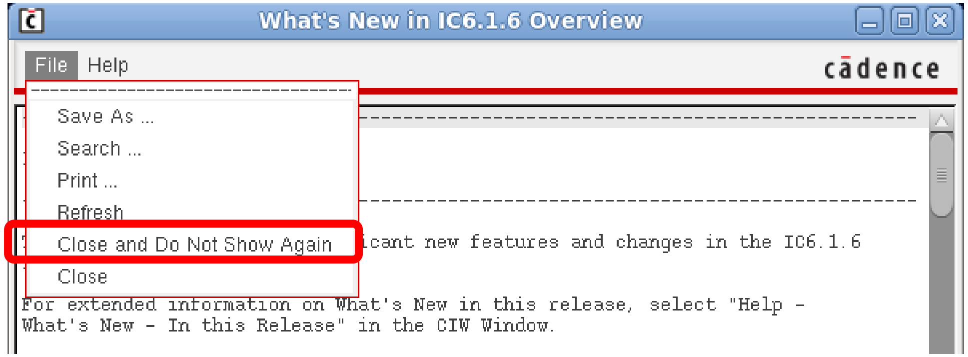
For layout versus schematic check (LVS)

For parasitic extraction (QRC)

Switch files for tool control (omit several checks like coverage for single cells etc.)

Cadence Virtuoso - Start

In Terminal: cd project_directory
virtuoso &

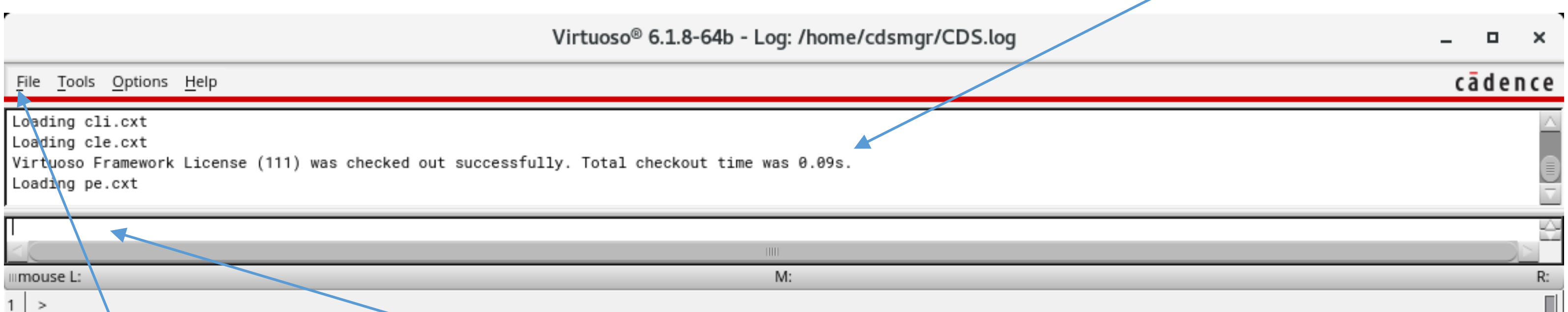


We try to supress this window, but sometimes it still opens. Cadence setup is a mystery sometimes ..

Command Interpreter Window - CIW

Stays open as long as virtuoso is running

Messages



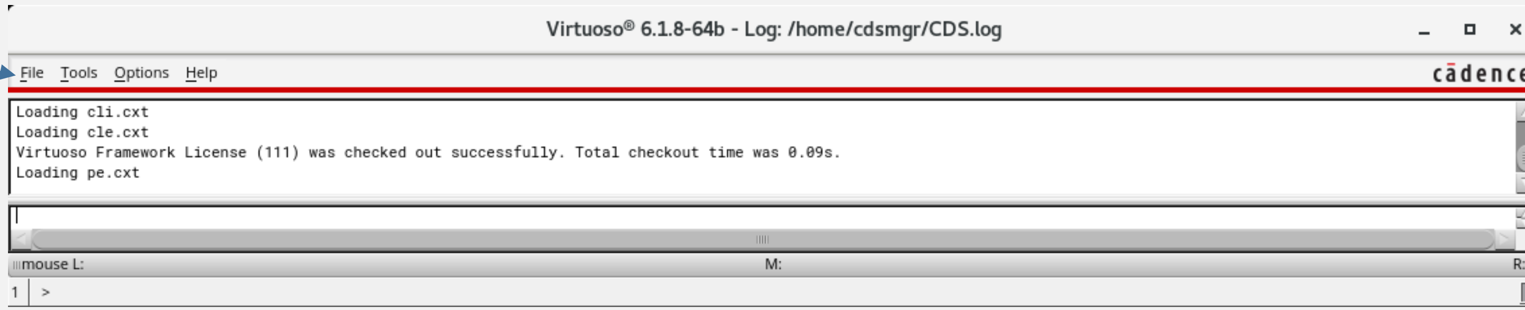
To close: File -> Exit

Command Inputs

Creating a new library (Version 2018)

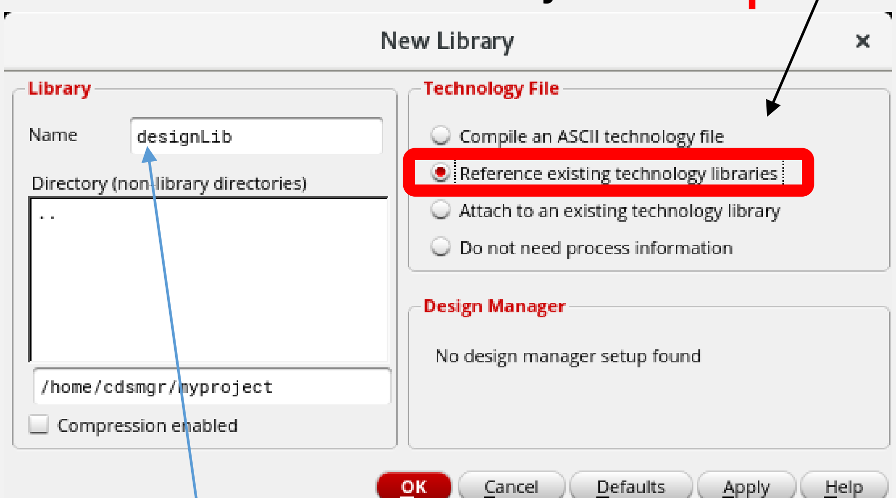
Command Interpreter Window - CIW

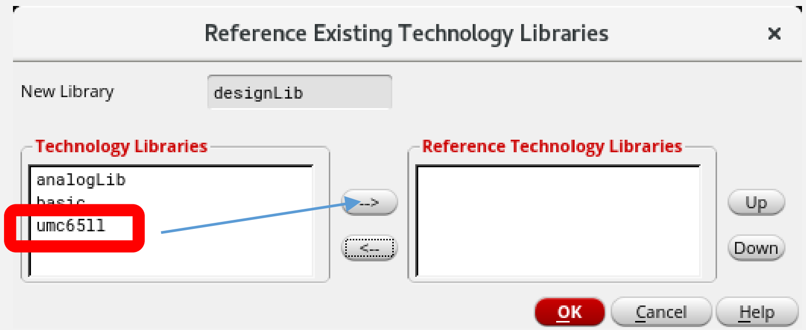
It is possible to create the library using the LIbrary Manager instead of CIW as well. The flow will differ (windows sequence).

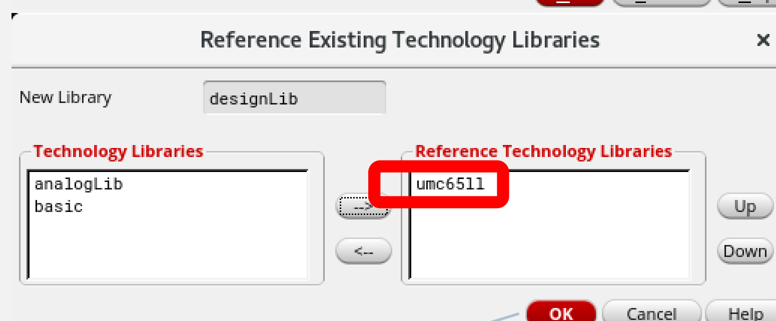


File -> New -> Library

important step!







Choose a name for your library!
(You may use one library for different designs.)

Recommended by EUROPRATICE: Select "**Reference** existing technology libraries" -> OK
For use of GDS3D (3D-Viewer) better: "**Attach** existing technology libraries"

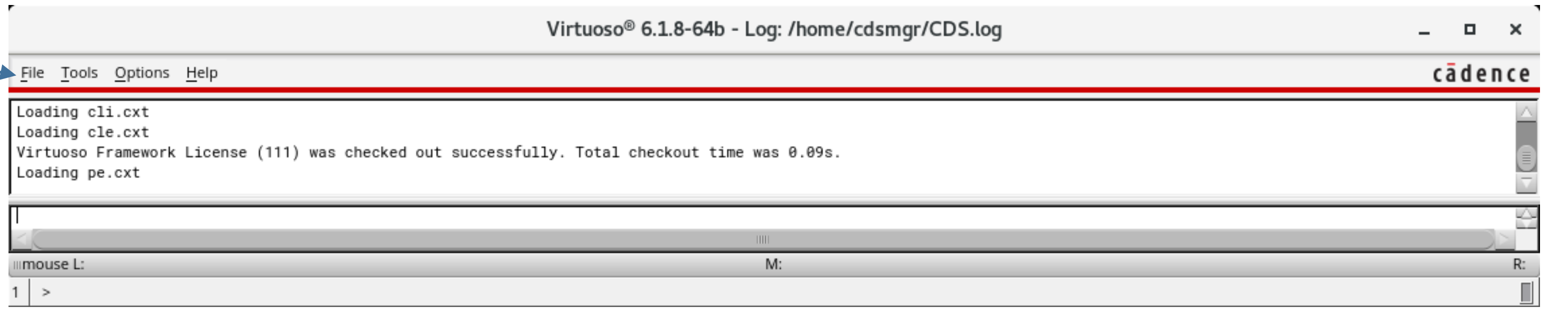
Do **not** press OK before you have selected the TECH-library!

Quit with OK!

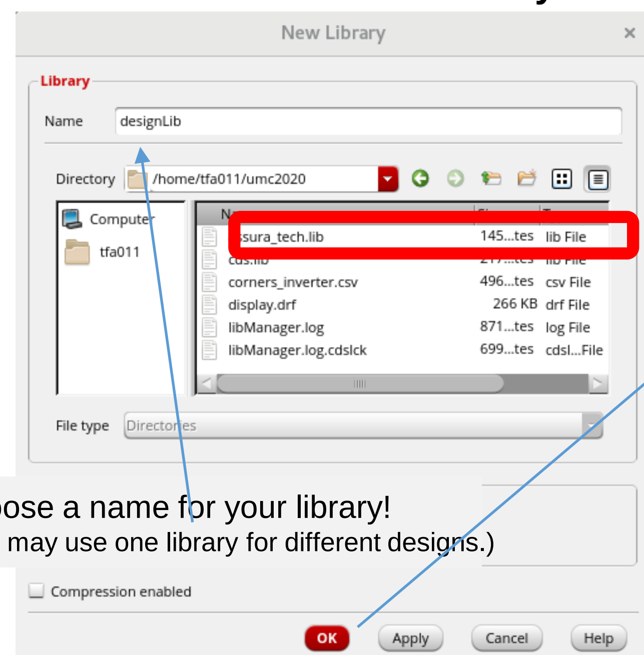
Creating a new library (Version 2019)

Command Interpreter Window - CIW

It is possible to create the library using the LIbrary Manager instead of CIW as well. The flow will differ (windows sequence).

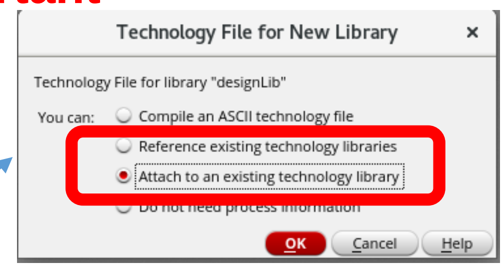


File -> New -> Library

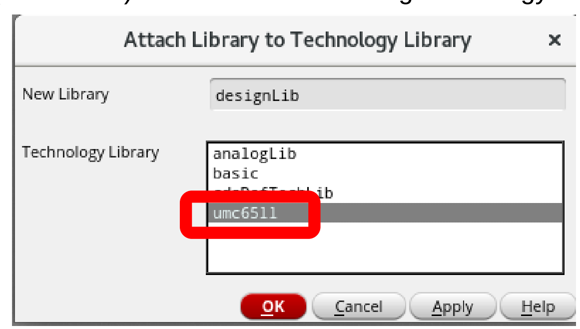


Choose a name for your library!
(You may use one library for different designs.)

important step!



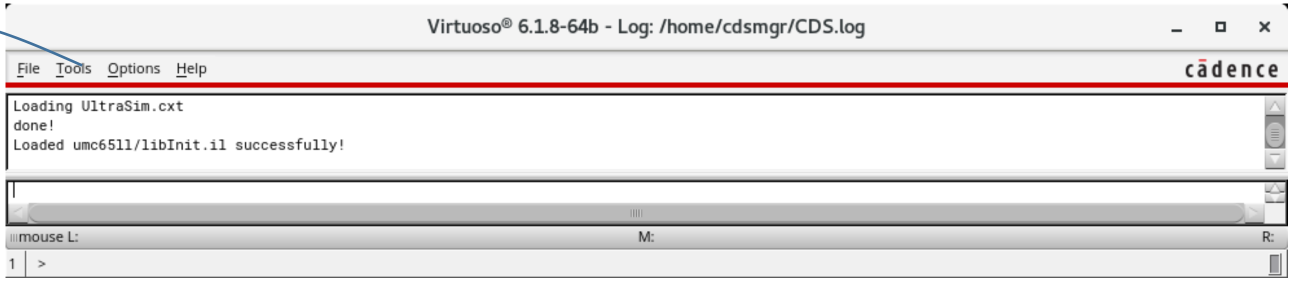
Recommended by EUROPRATICE: Select "**Reference** existing technology libraries" -> OK
For use of GDS3D (3D-Viewer) better: "**Attach** existing technology libraries"



Quit with OK!

Virtuoso - A new cellview

Command Interpreter Window CIW (You can do this in the Library Manager as well "Tools ->Library Manager -> New -> Cellview")

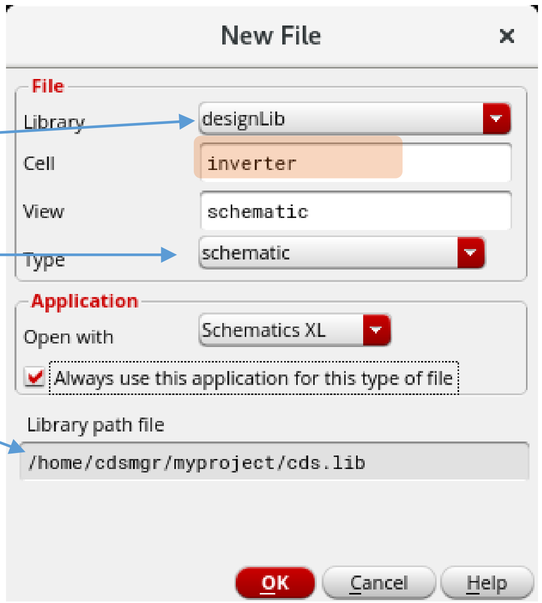


File -> New -> Cellview

Your (new) library

Start with schematic view

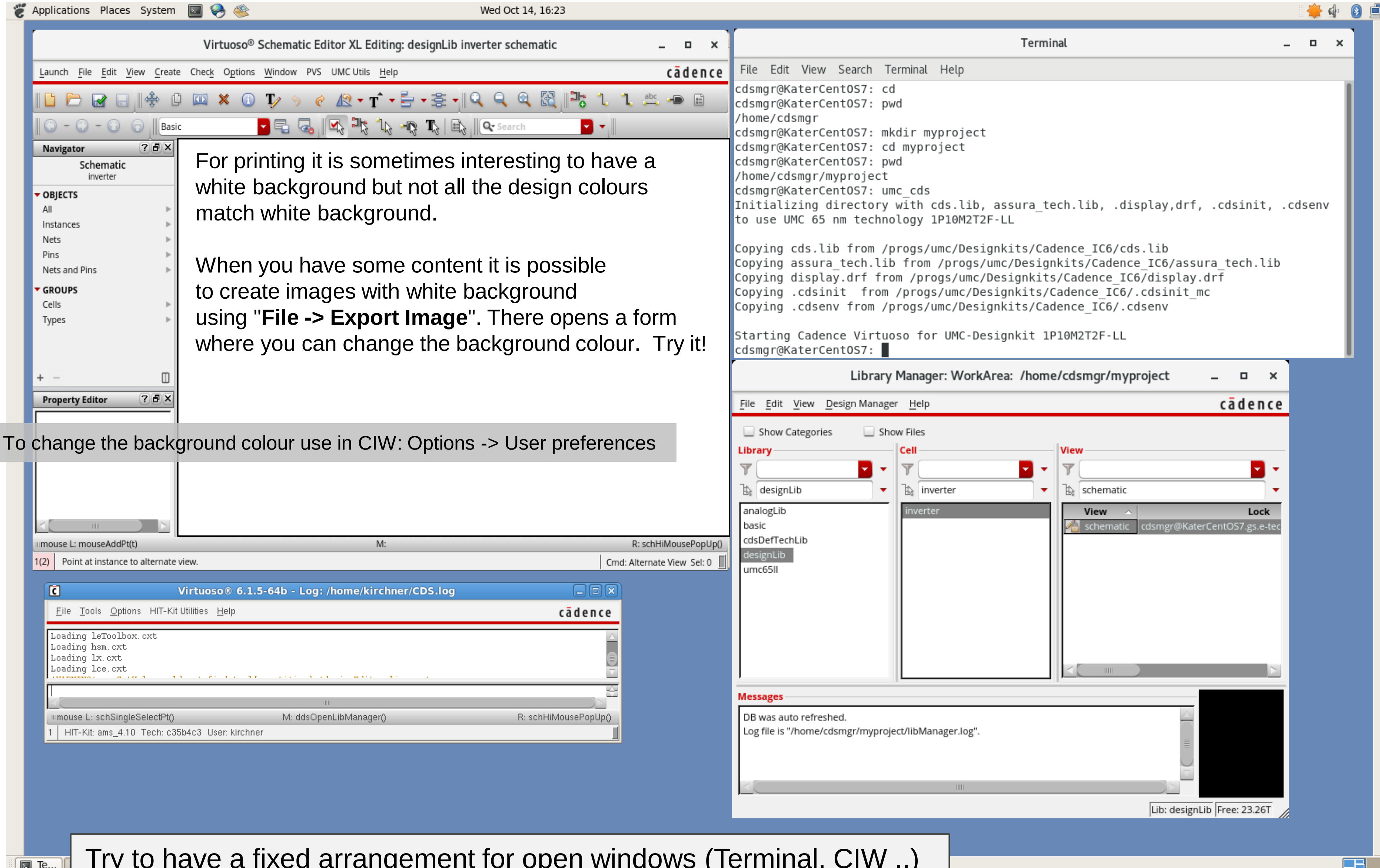
Depends on your project path



Choose a cell name, check "schematic" as cellview, Schematics **XL** as tool -> OK

Best choice for names/identifiers: use letters, numbers, underscore only

Virtuoso Schematic Editor is open



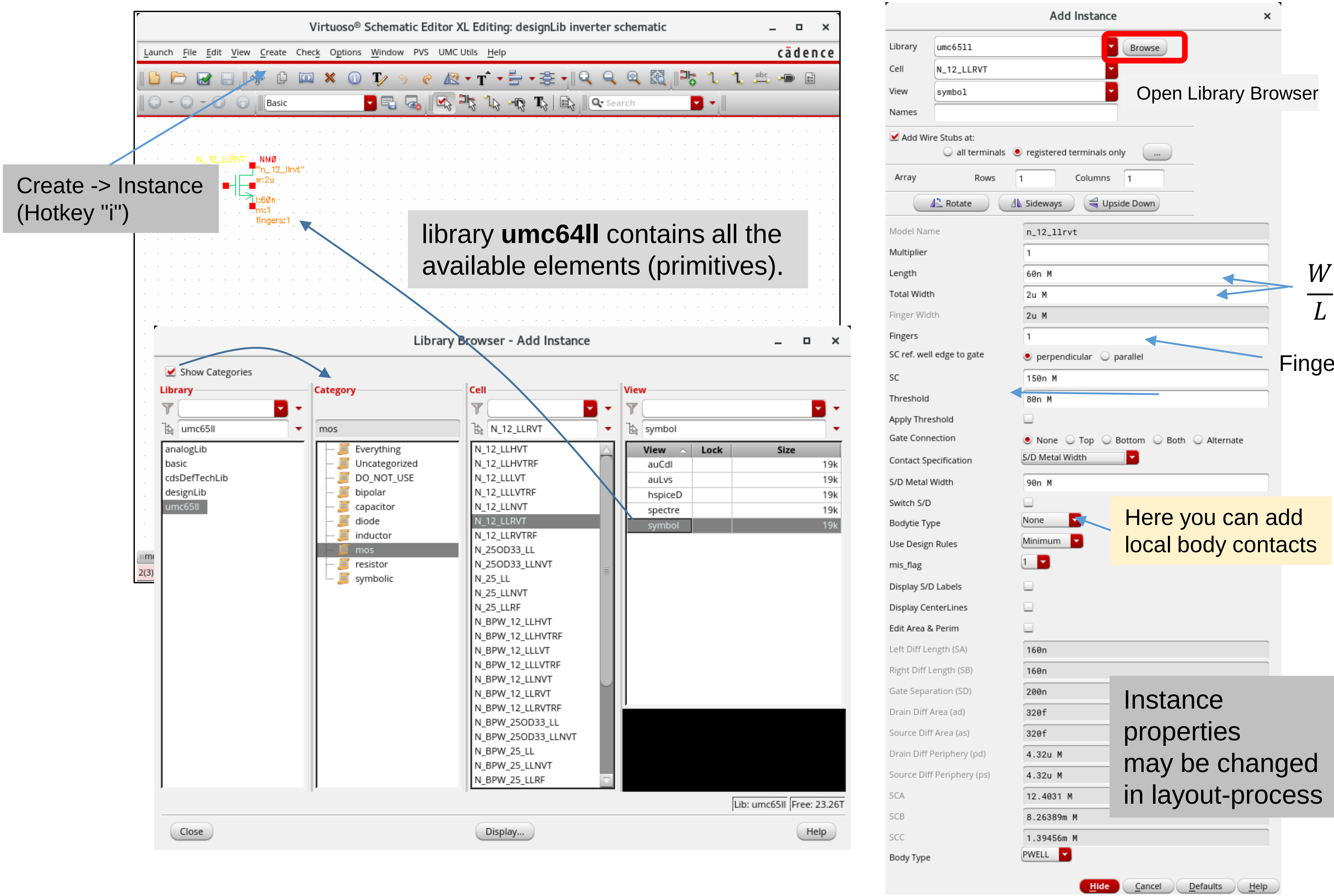
For printing it is sometimes interesting to have a white background but not all the design colours match white background.

When you have some content it is possible to create images with white background using "**File -> Export Image**". There opens a form where you can change the background colour. Try it!

To change the background colour use in CIW: Options -> User preferences

Try to have a fixed arrangement for open windows (Terminal, CIW ..)

Schematic entry - Components



Inverter schematic

transistors from library "**umc65ll**"
(also **Design-**R,C,L)

LL: Low Leakage

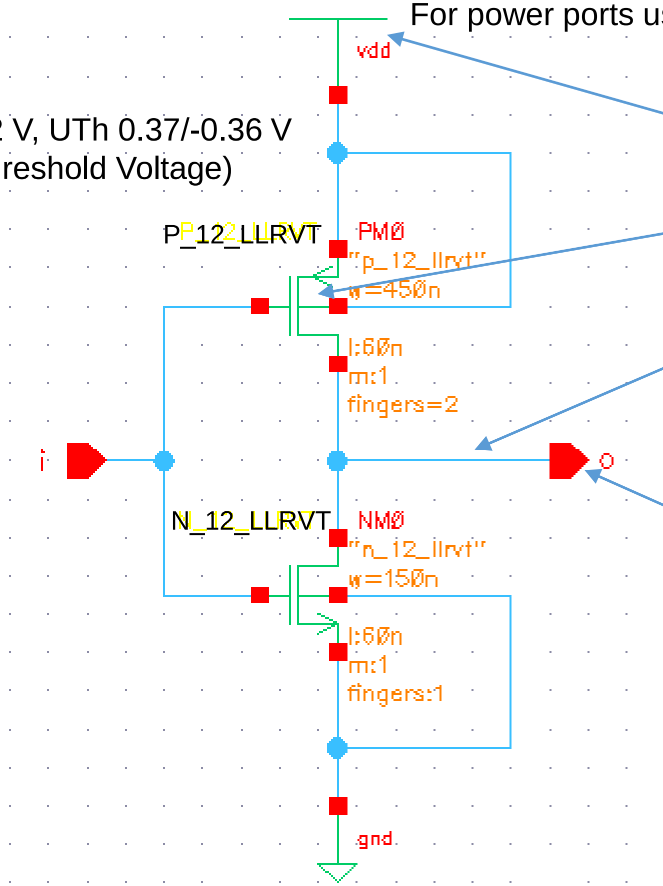
NMOS/PMOS : LLRVT Tox=26 nm , 1.2 V, UTh 0.37/-0.36 V
(RVT,LVT, HVT -> Regular/Low/High Threshold Voltage)

$$C_{ox} = 12\,\frac{fF}{\mu m^2}$$

no substrate contact for
the transistors in logic

vdd, gnd etc. from library "**analogLib/**globals"

For power ports use names including "vdd" or "vcc"



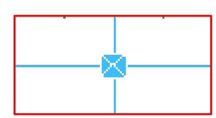
Form: name, direction ..

key	function
i	create instance
w	create wire
M	move
m	stretch (im Layout "S" !)
c	copy
q	object properties
p	add pin
l	add wire name (label)
x	check current cellview
f	fit
]	zoom in
[	zoom out

key	function
F3	show option window
u	undo
U	redo
e	descend & read
E	descend & edit
CTRL-e	return
DEL	delete

For ground ports use names including "gnd".
(Some scripts (ERC) need this naming convention.)

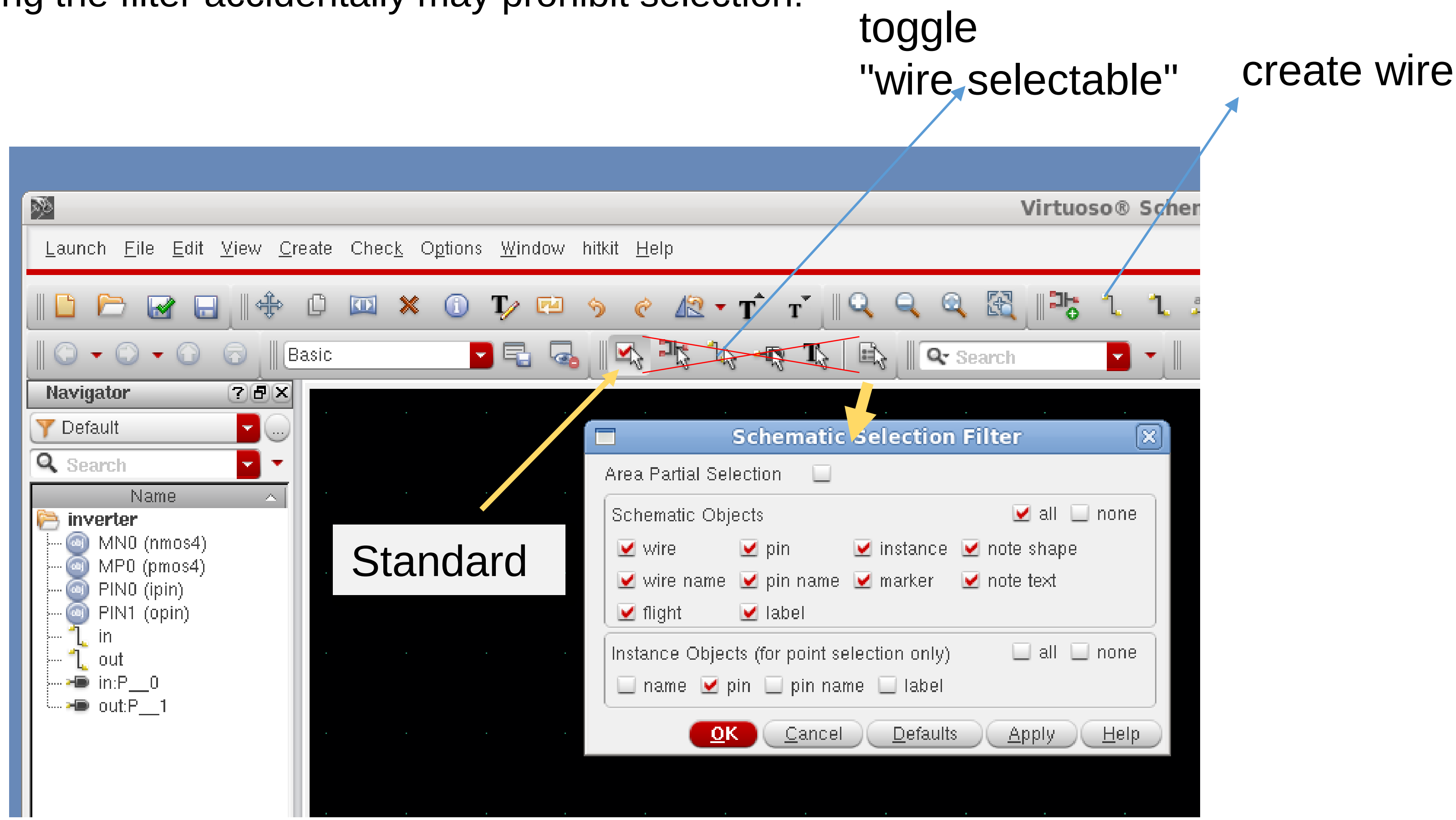
File -> Check and Save (shft-x)



Avoid crossover dots.
Warnings! May be deselected
Check-> Rules-> Physical

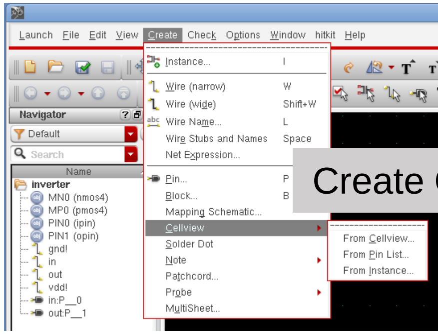
Schematic Editor: Filter

Schematic editor has a filter function for selectable objects!
Do not confuse "create" buttons with "filter" buttons!
Changing the filter accidentally may prohibit selection!

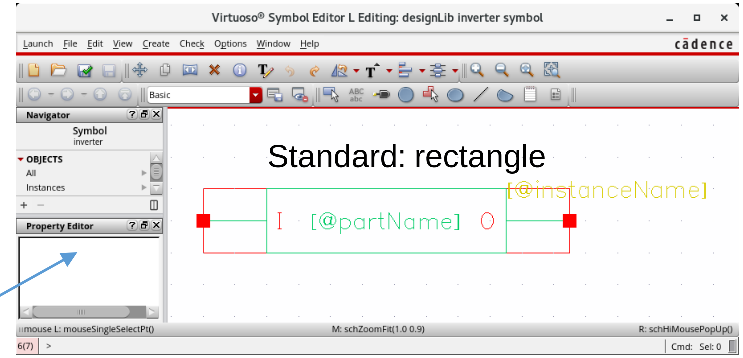


Inverter Generate Symbol from Schematic

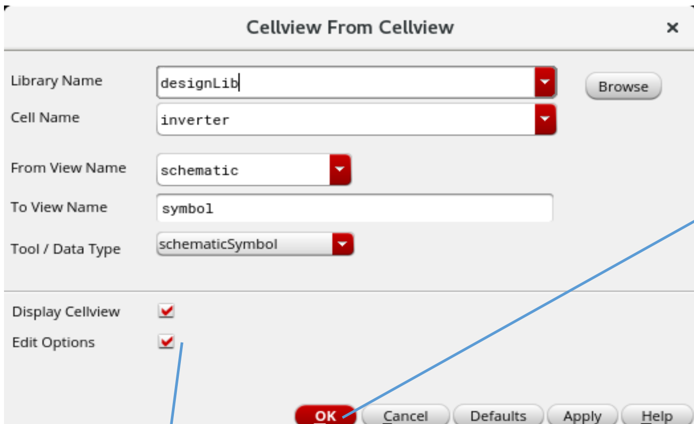
Symbol needed to use our design in higher levels of hierarchy.
First Example will be the instantiation of the design in a testbench.



Create Cellview -> From Cellview -> OK

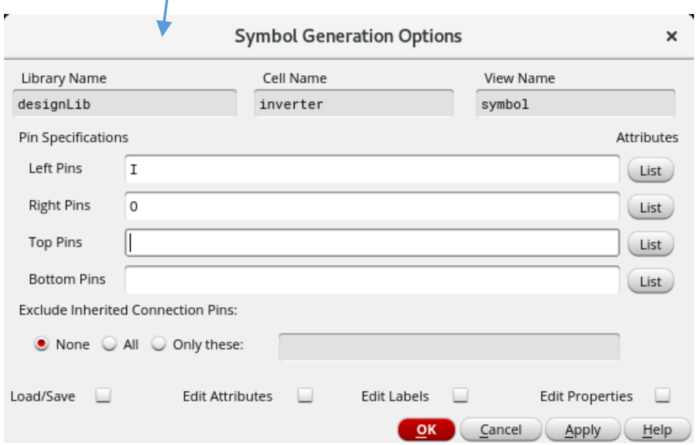


Standard: rectangle



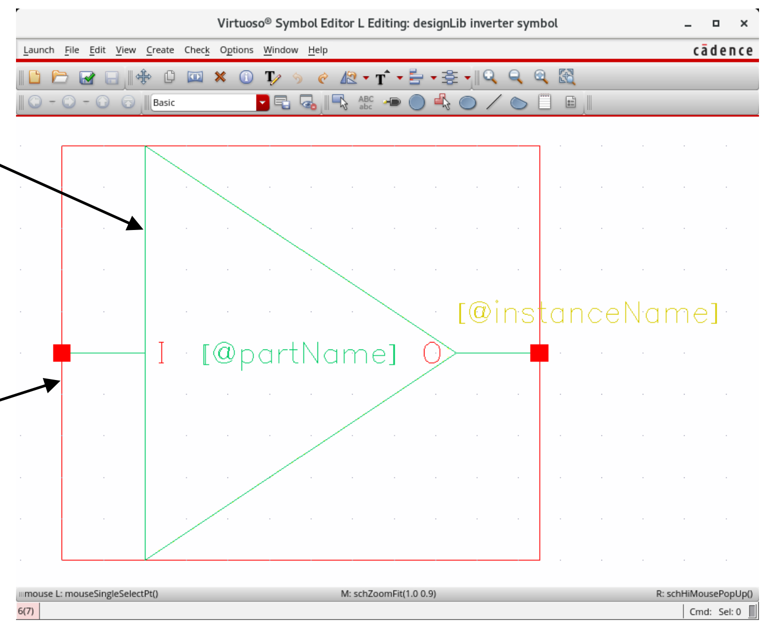
You may change shapes:
outline (green), selection box (red)
use: stretch, delete,move,create polygon

optional

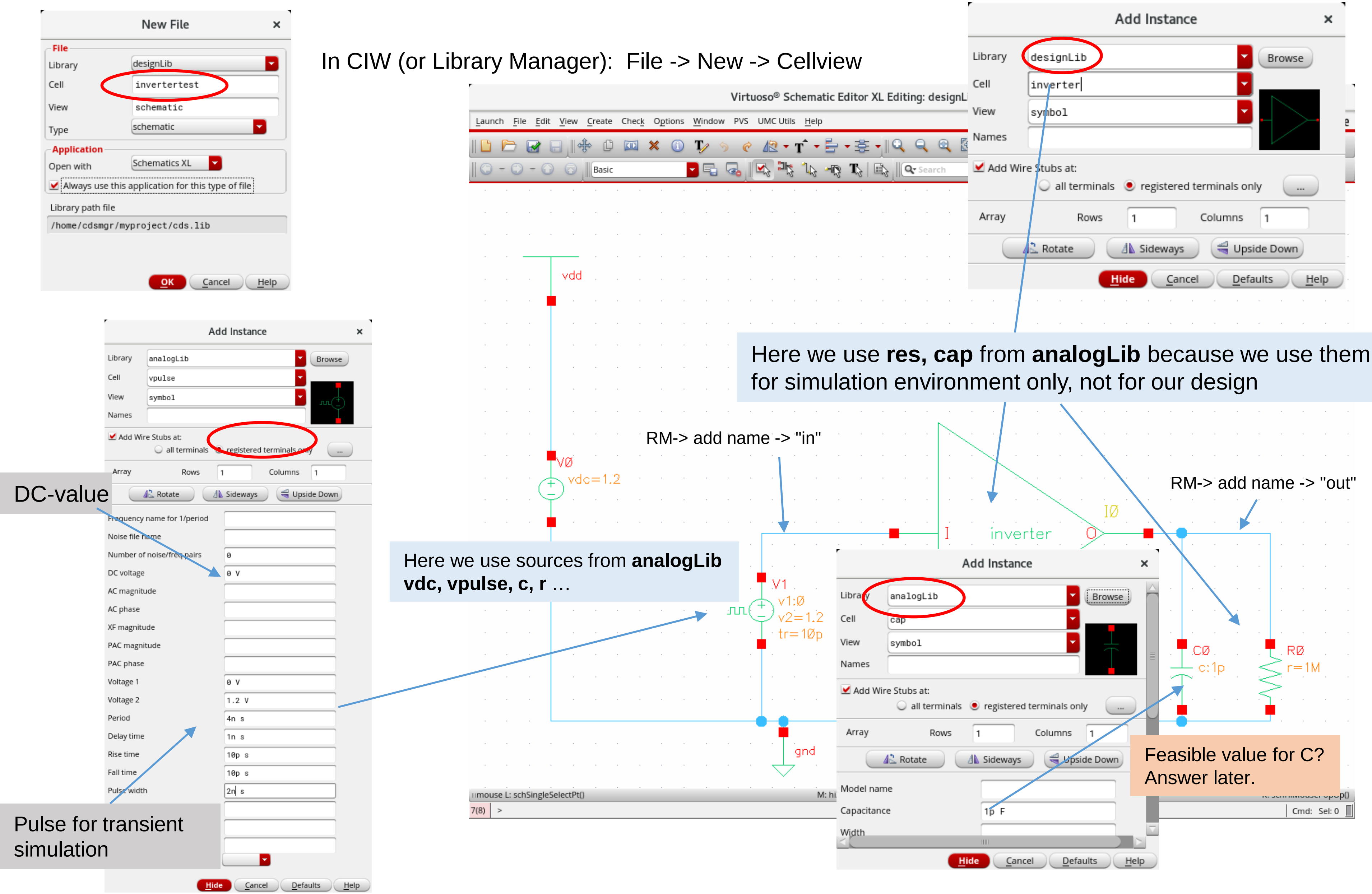


Create Shape
Polygon

Create
Selection Box
Automatic



Inverter Testbench Schematic for Simulation

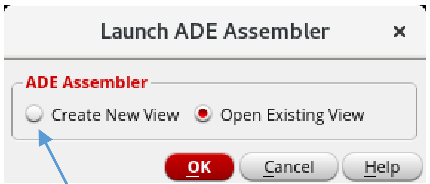


Simulation ADE Assembler Start

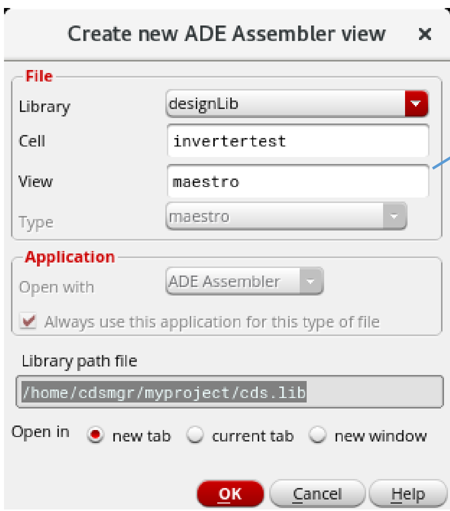
Most tutorials use older ADE L

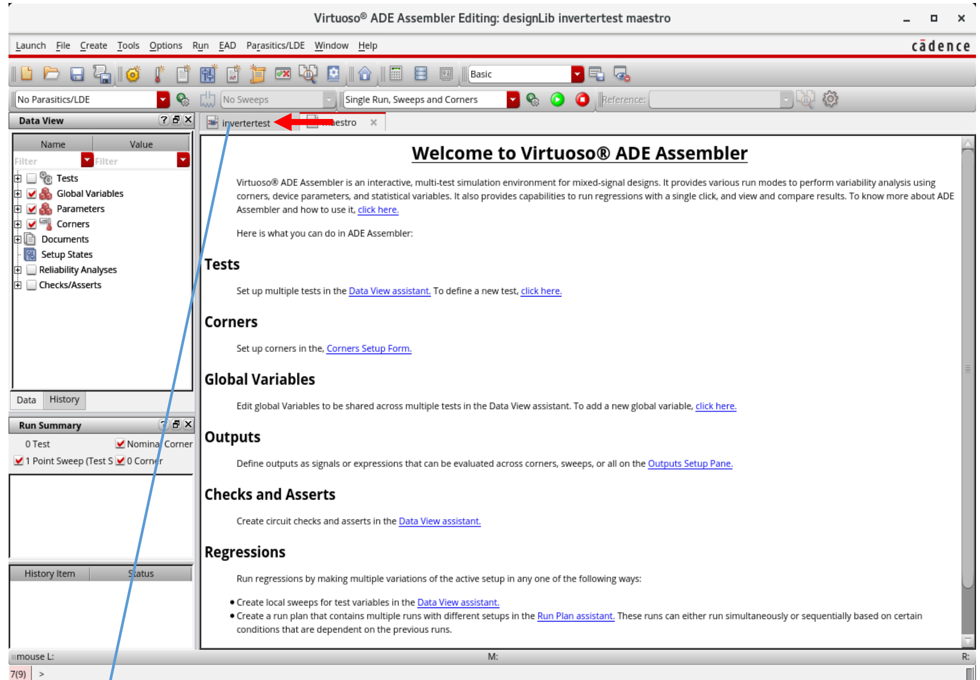
From the Schematic Editor -> **Launch -> ADE Assembler**

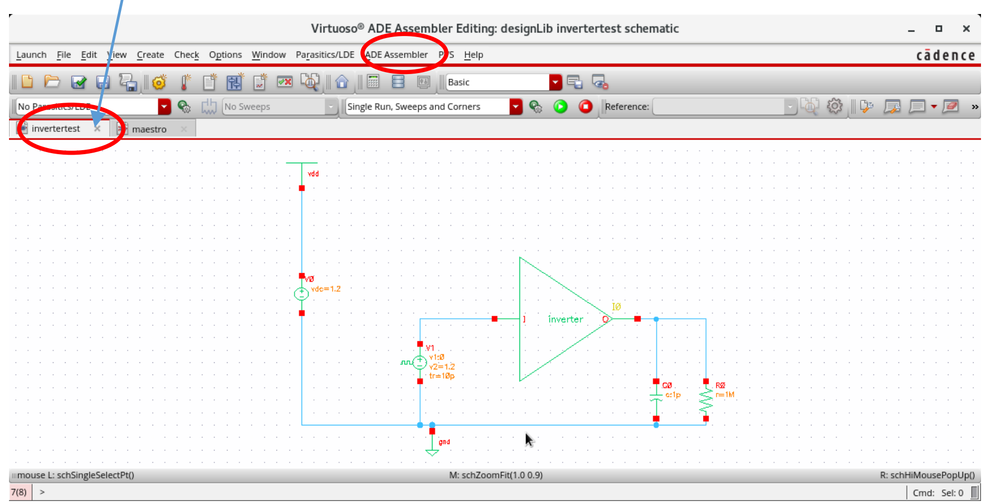
LDE Layout Dependent Effects
EAD Electrically Aware Design



ADE Assembler creates a new view "maestro"!



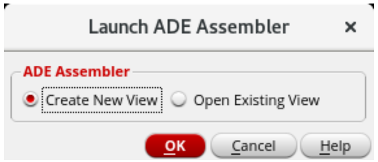




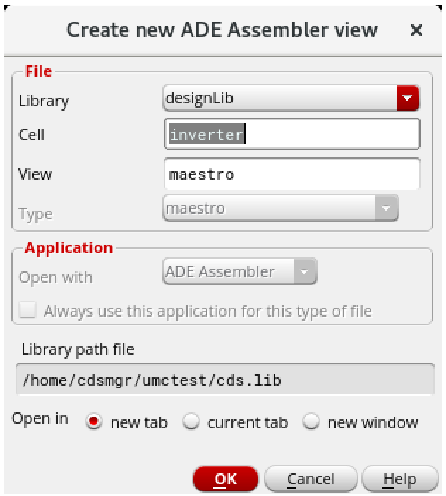
Simulation using ADE Assembler - Start from schematic view of the design

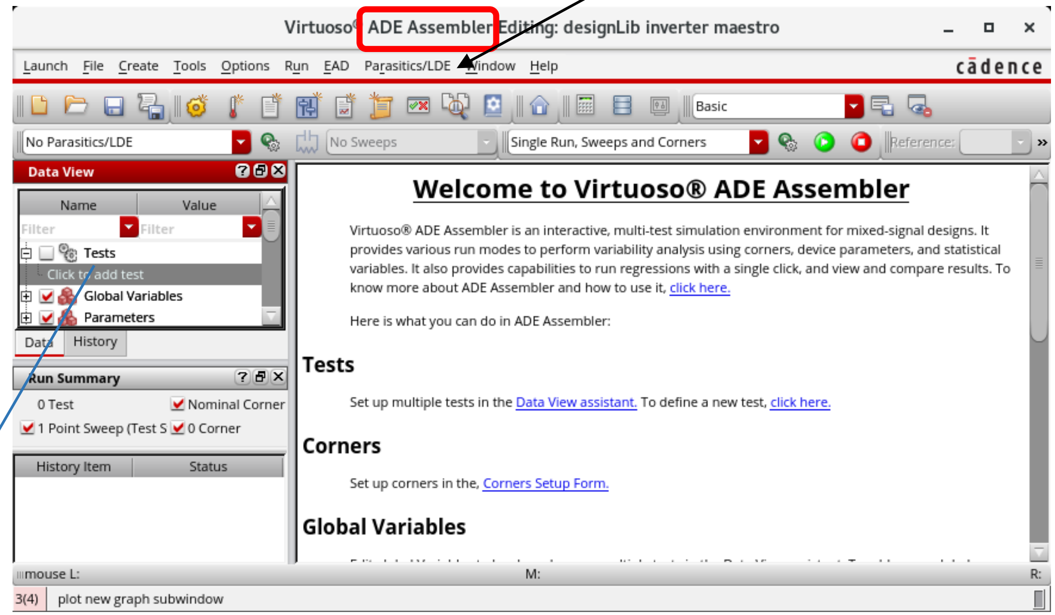
From schematic view of the design (not testbench):

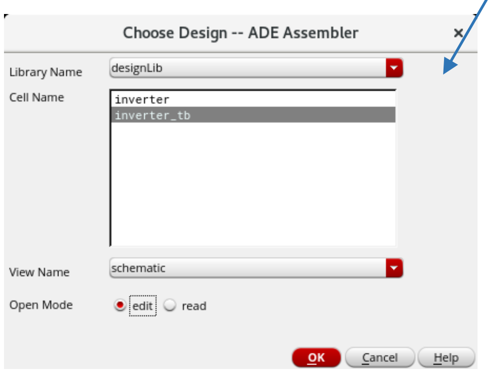
LDE Layout Dependent Effects
EAD Electrically Aware Design

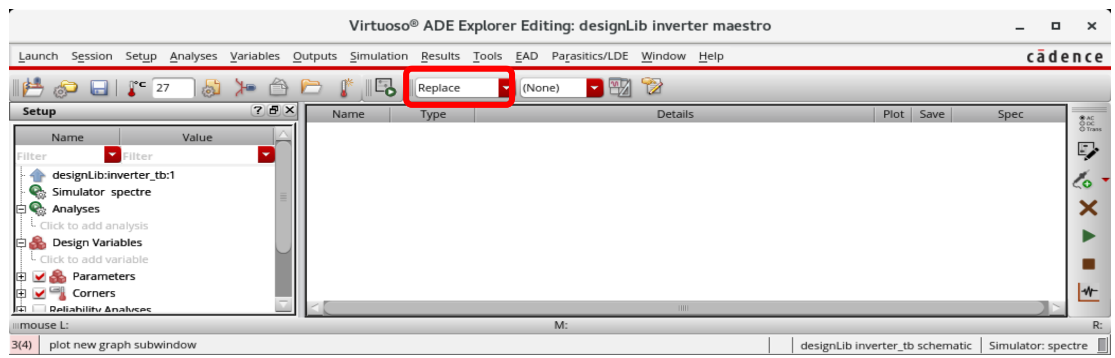


ADE Assembler creates a new view "maestro"!





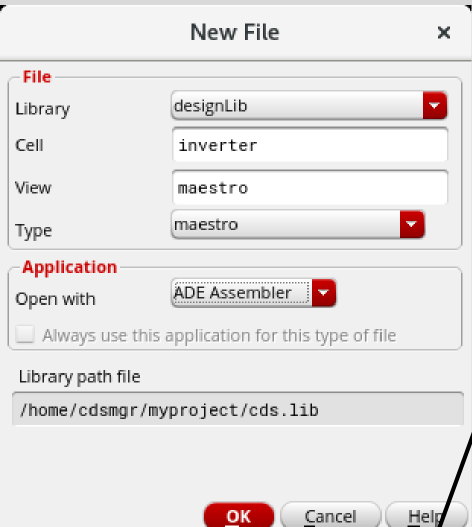


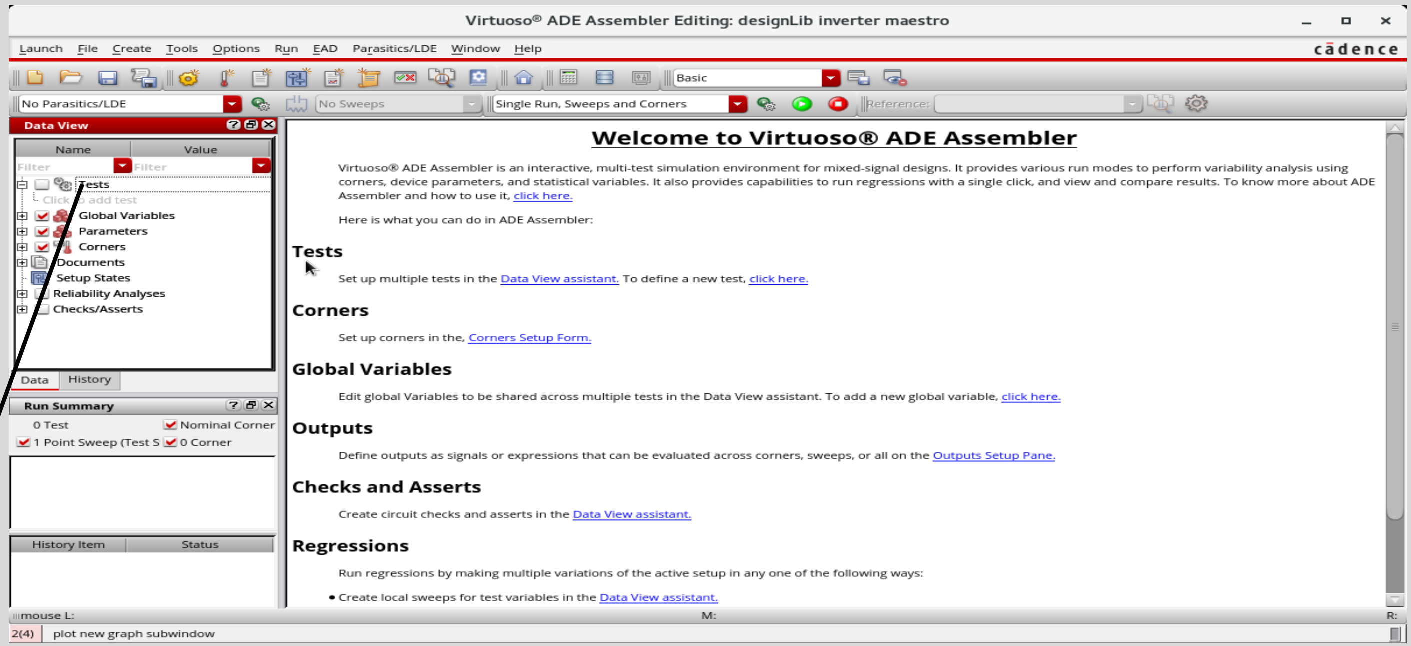


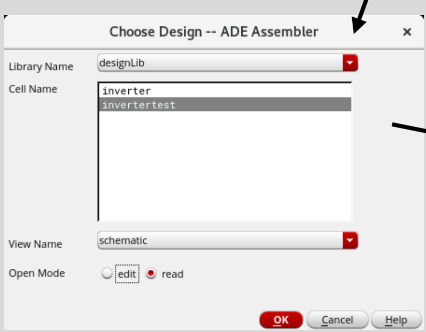
We may control different testbenches for the same design in the maestro view with ADE Assembler..

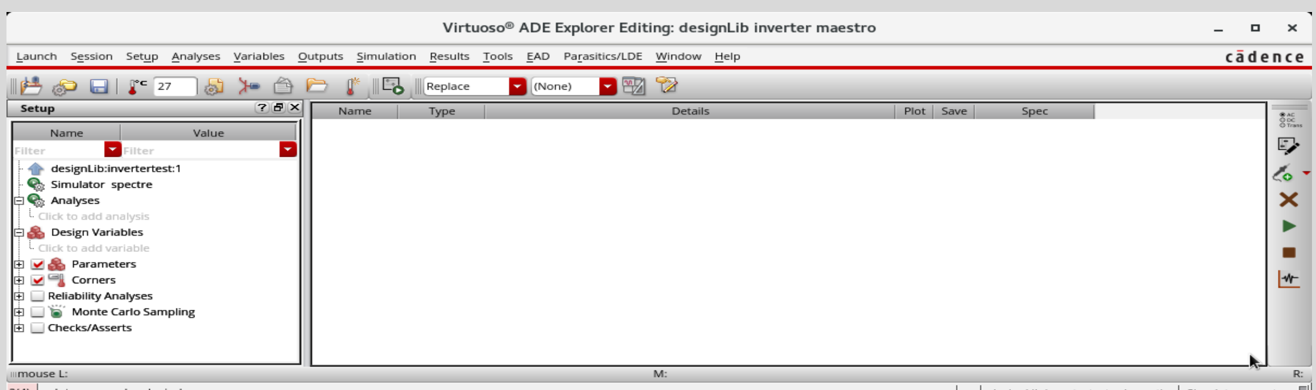
Simulation ADE Assembler Start via library manager

Library Manager → File → New → Cellview



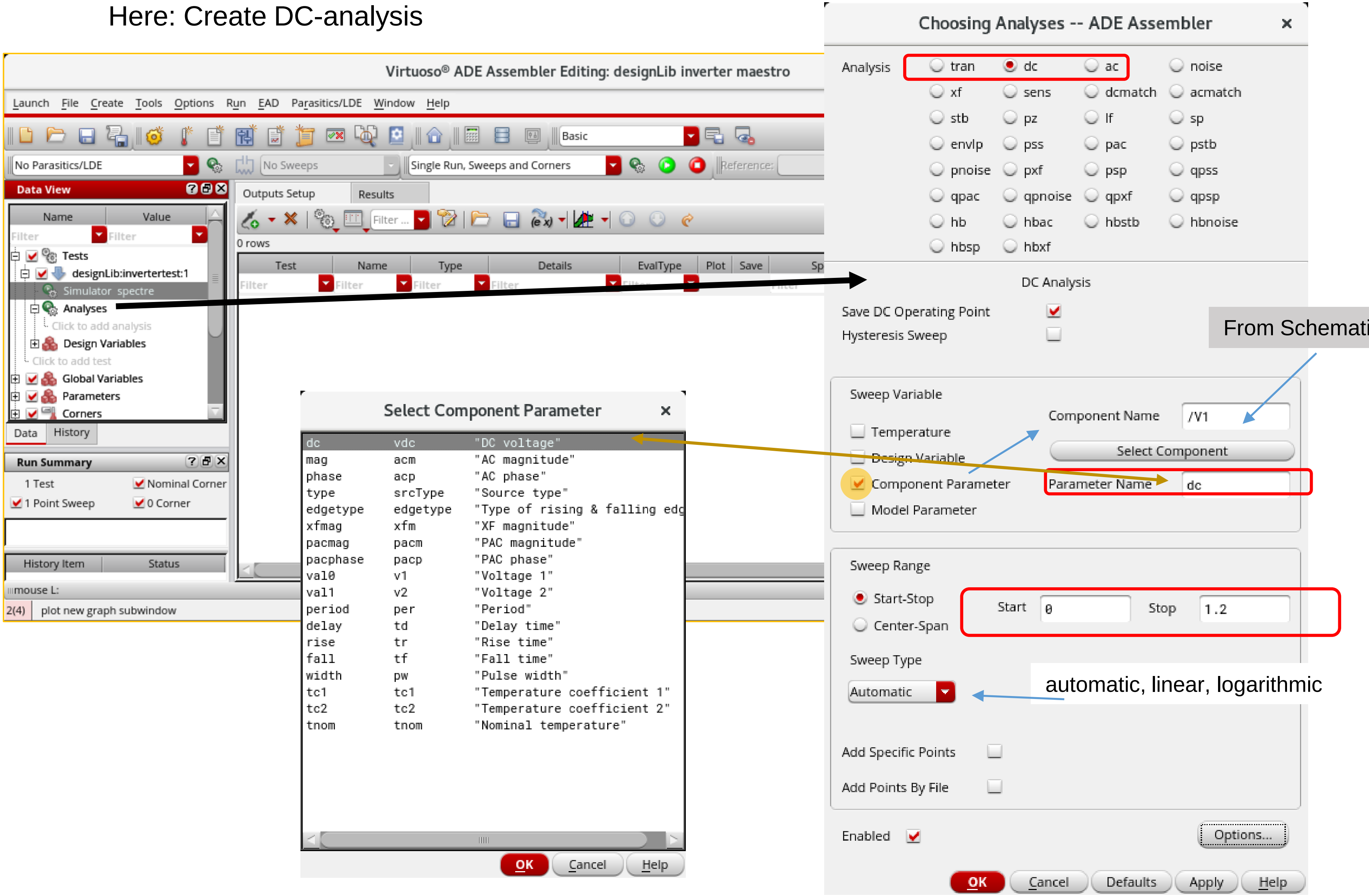




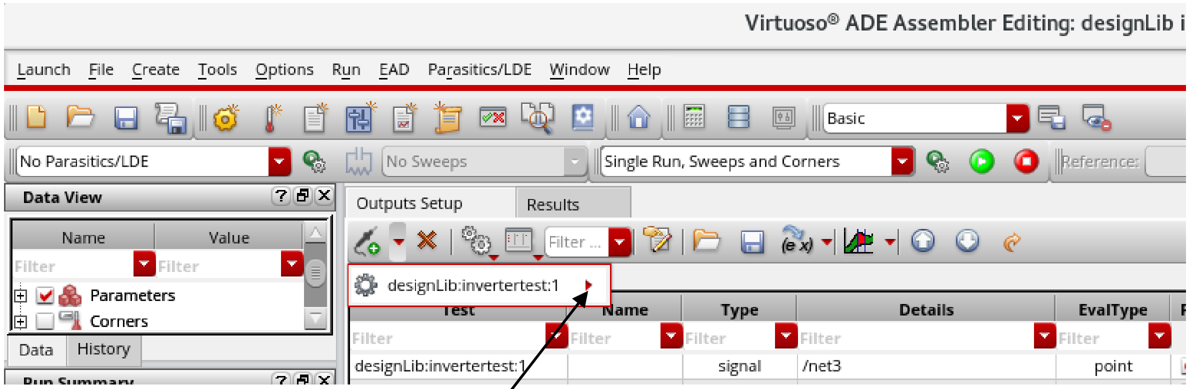


Preparing a Test (Analyses, Design Variables, Outputs)

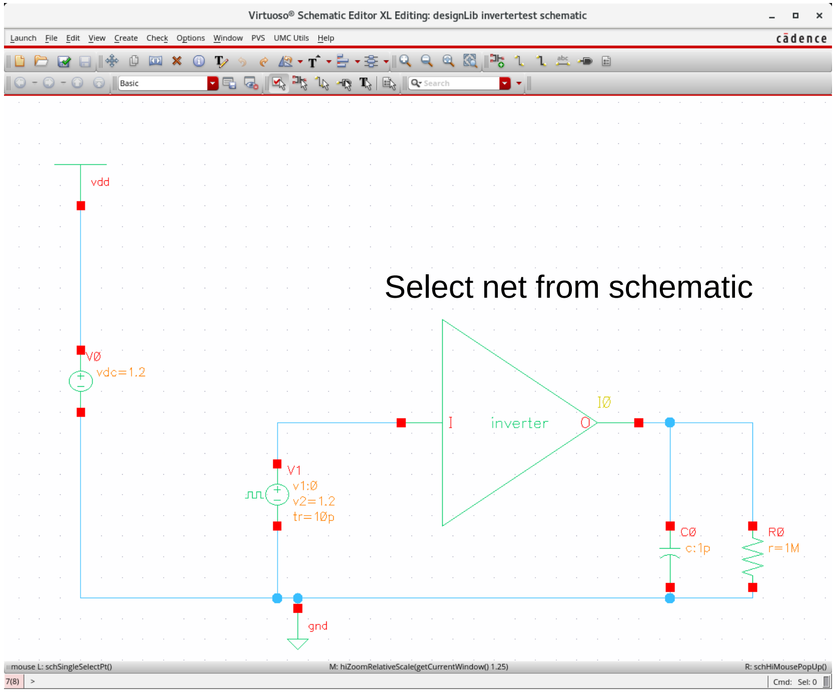
Here: Create DC-analysis



Setting the Outputs of Simulation



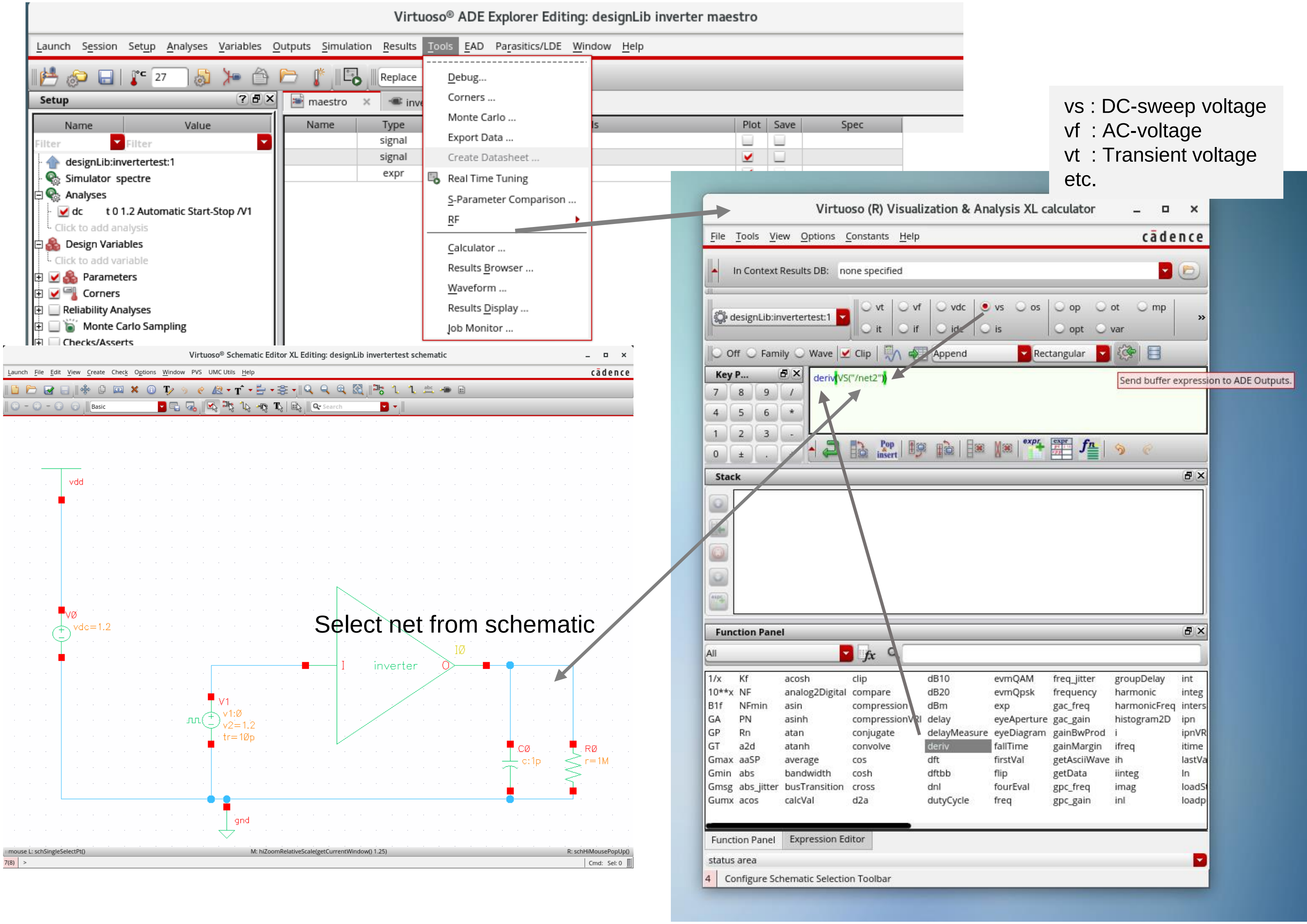
Open a new pulldown menue here and select "Signal".
The schematic will open and you may select the signal(s) there.



Select net from schematic

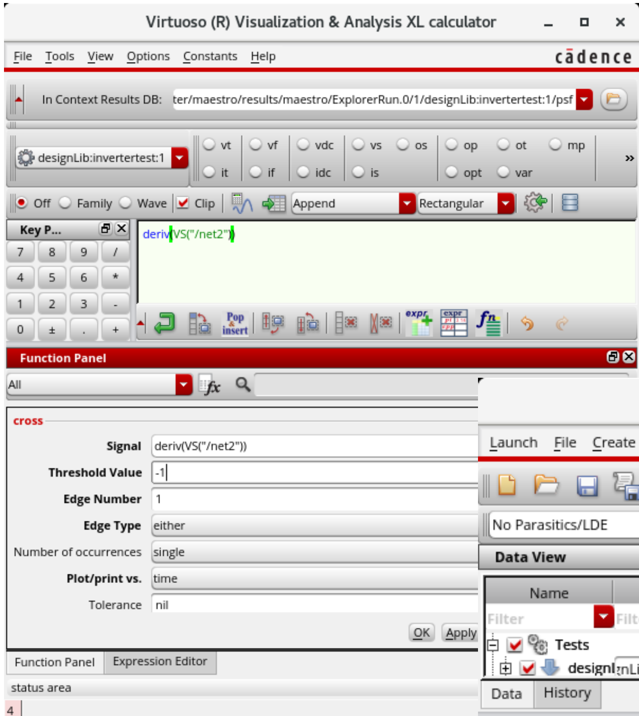
Setting up an output expression using the calculator

vs : DC-sweep voltage
vf : AC-voltage
vt : Transient voltage
etc.

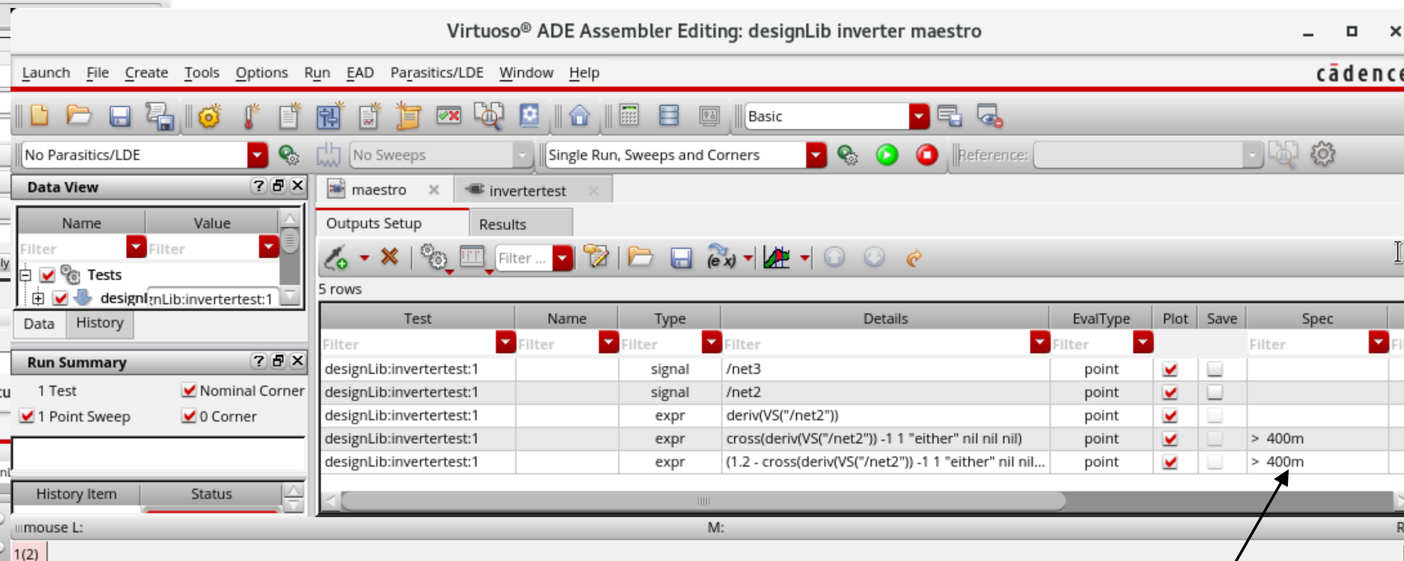


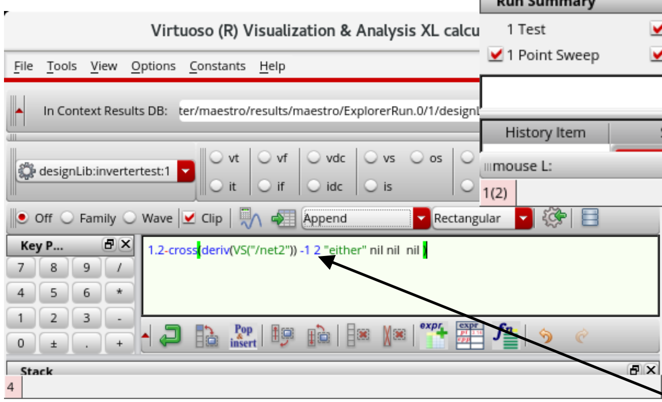
Determine Noise Margins for a Logic Gate

Here we have an example how we can evaluate special expressions. The same method may be used in other designs for bandwith, phase margin etc.



1st event of "derivate is -1" → Low level noise margin is difference to GND



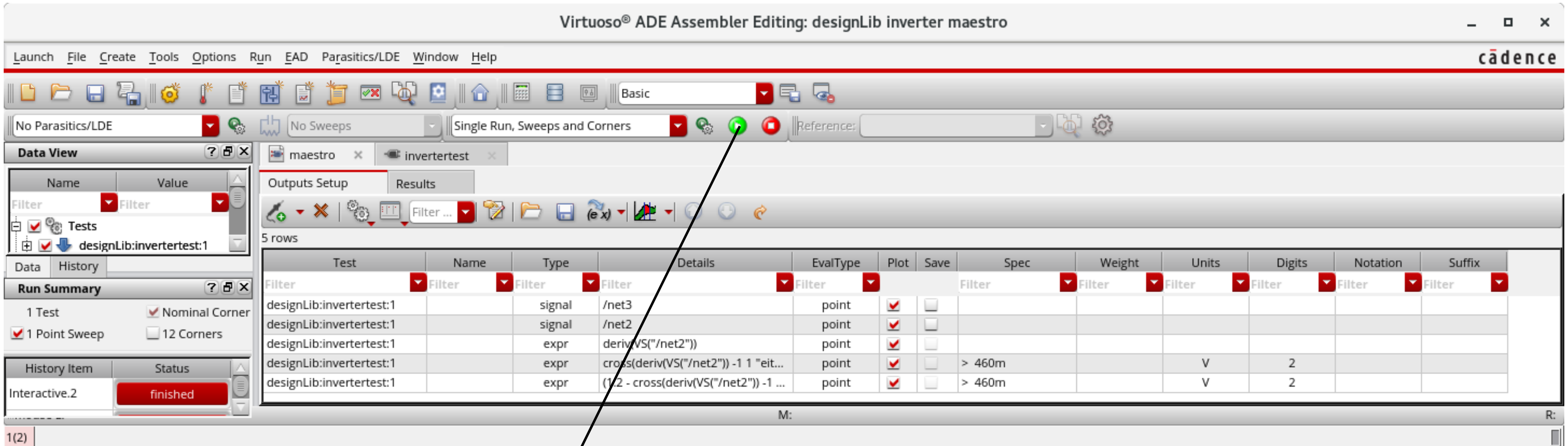


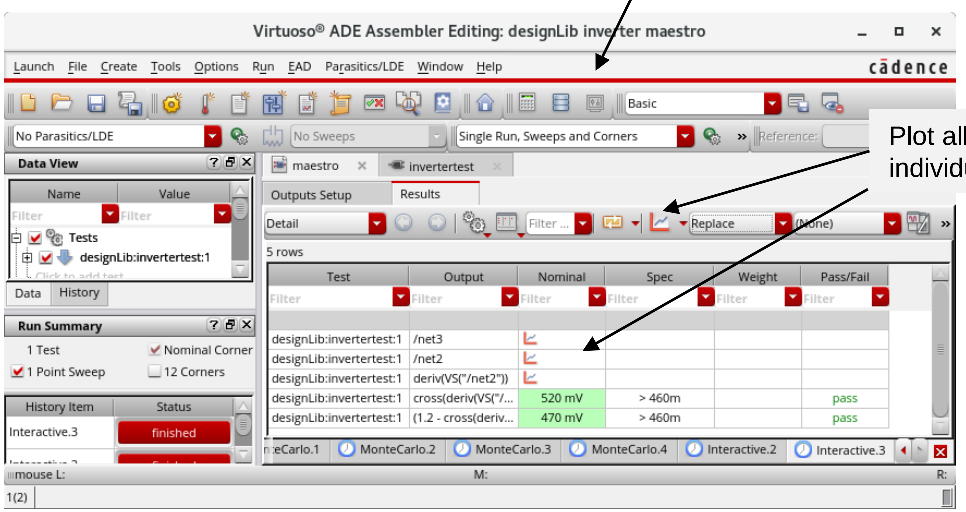
Set specification

2nd event of "derivate is -1" → High level noise margin is difference to VDD

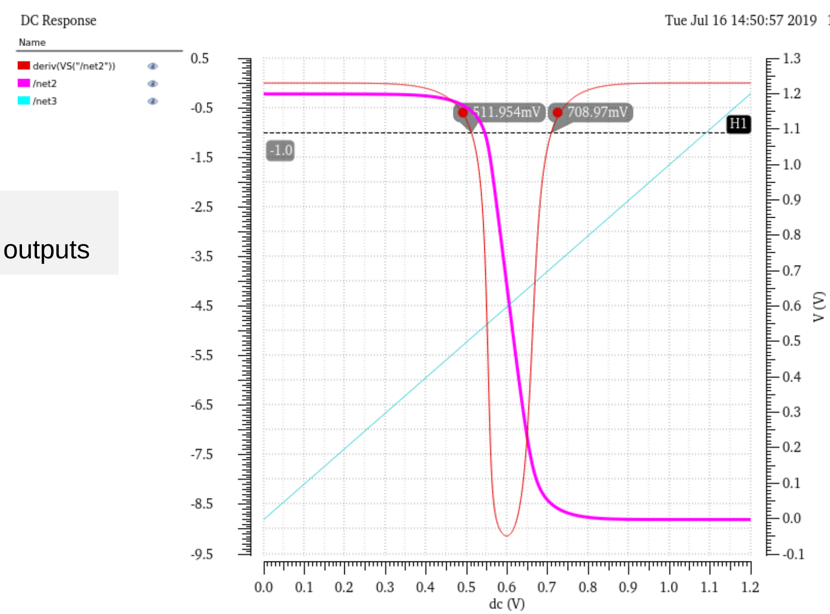
K.-P. Kirchner, Uni Rostock, Institut GS

Run Simulation with Nominal Values





Plot all or individual outputs



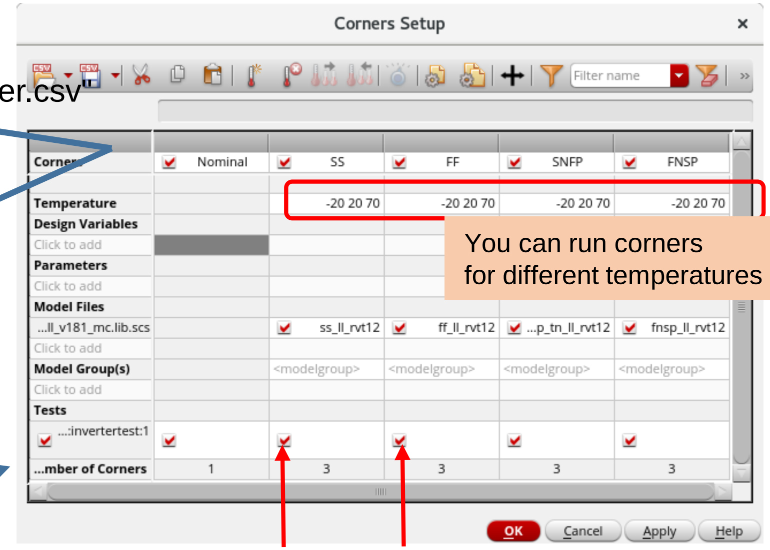
Result with horizontal bar at -1 level of derivate and interceptions on

Run Corners Simulation

Import "/progs/umc/Designkits/Cadence_IC6/corners_inverter.csv
or select the library and section
manually.

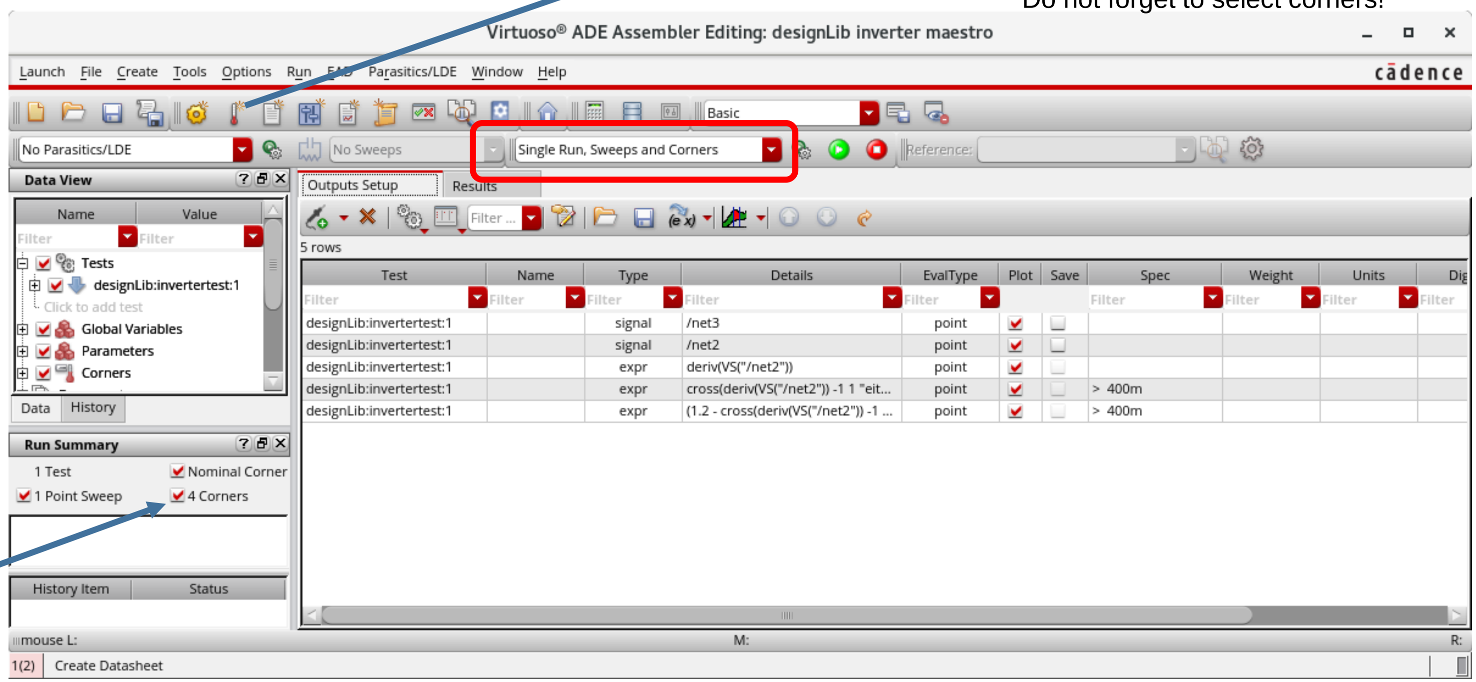
Corner model
See header of .lib.scs - file

/eda/umc/Designkits/Cadence_IC6/Models/Spectre/Monte_Carlo/l65ll_v181_mc.lib.scs



You can run corners for different temperatures

Do not forget to select corners!

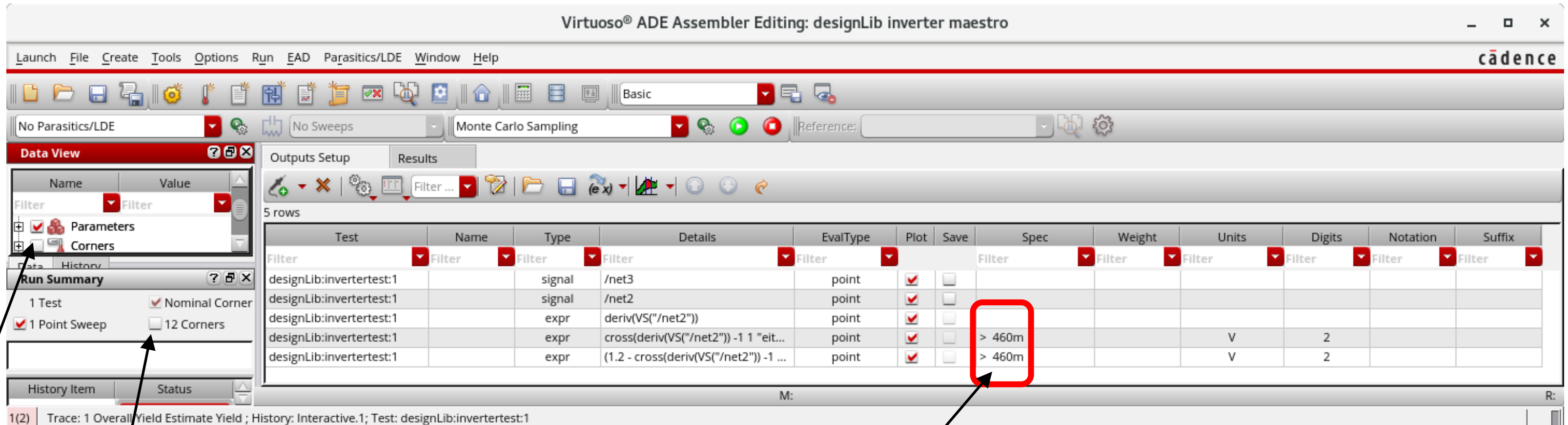


Select the 4 corners

K.-P. Kirchner, Uni Rostock, Institut GS

Yield prediction using Monte Carlo Sampling

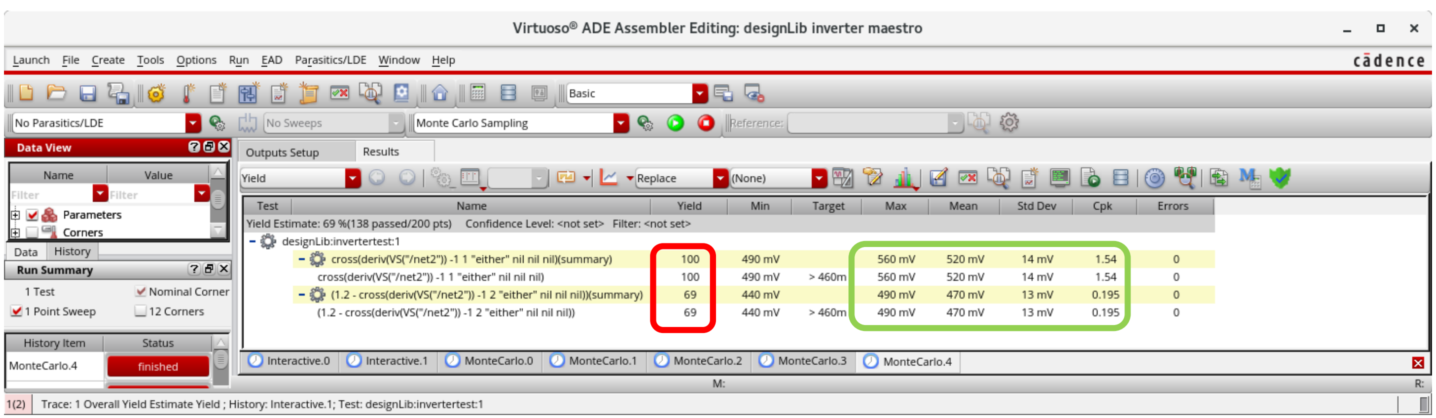
In our simulation environment we use the Monte Carlo libraries for the UMC 65 nm process. These libraries are used for single runs as well. So we do not change the simulation libraries for different simulations. See **.cdsinit** file.



Disable corners here

Our **spec**ification here:

Expecting a noise margin of 460 mV for both high and low level

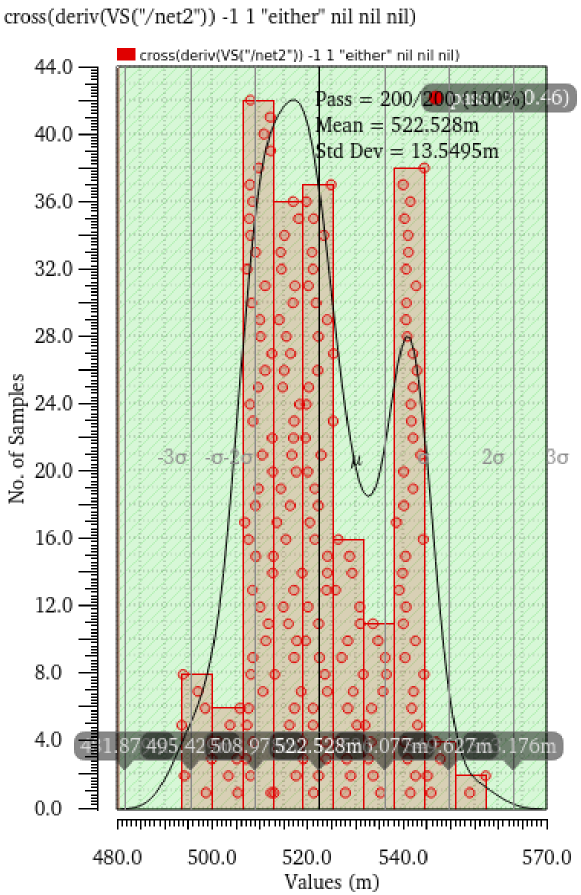


Noise margin OK. for low level but not for high level.
Only 69% of samples fit.

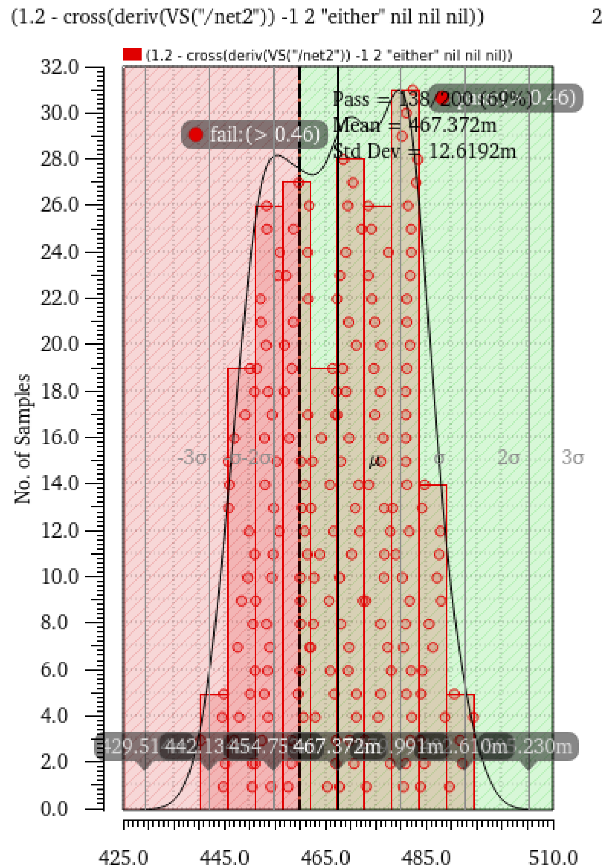
Cpk sould be > 1

Histogram of Inverter Noise Margin

Input low (first tangent)



Input high (second tangent)

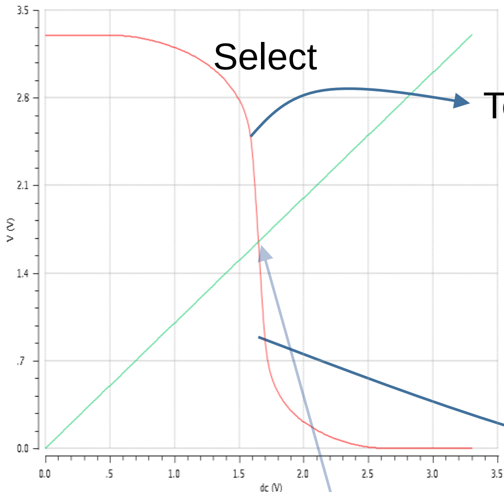


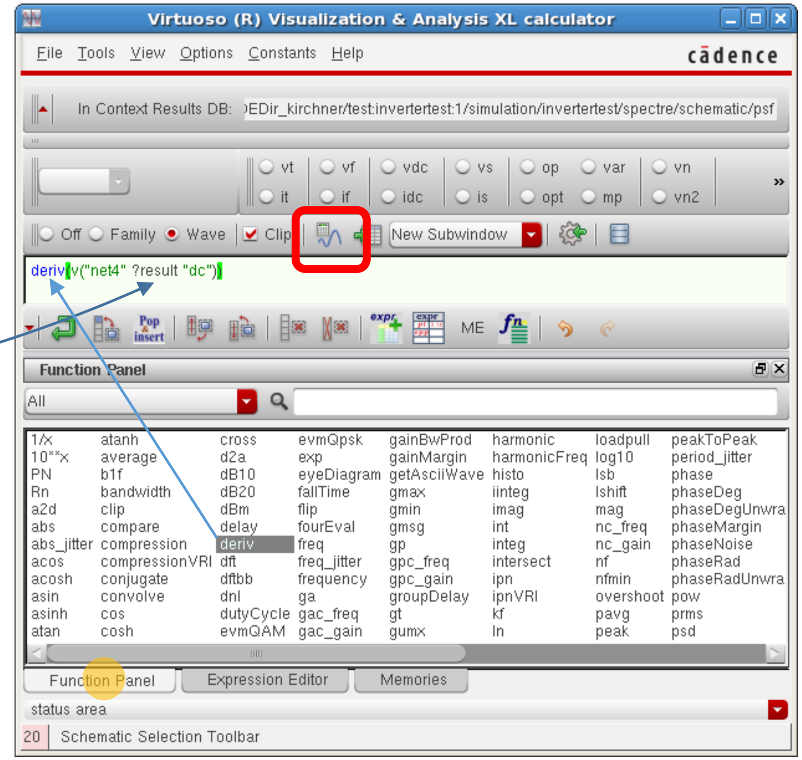
Red area: Failed, because noise margin < 460 mV

Simulation Evaluate Results

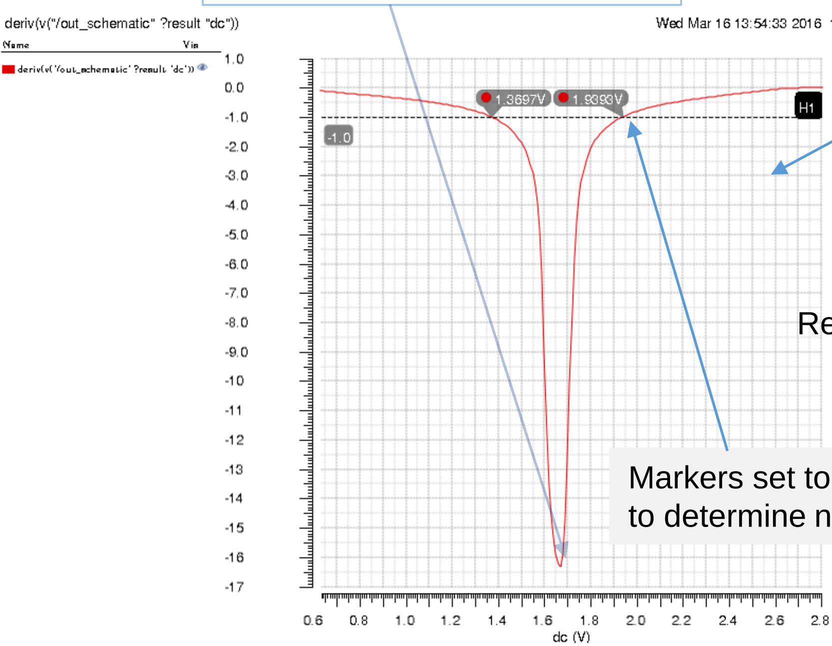
Select

Tools –> Calculator





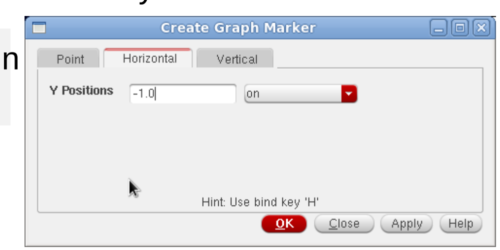
Inflexion point (max. abs. gain value)



Recommendation: Move derivation plot to new subwindow (RM-> Move to)

In Visualization&Anlaysis Window -> Marker -> Create Marker

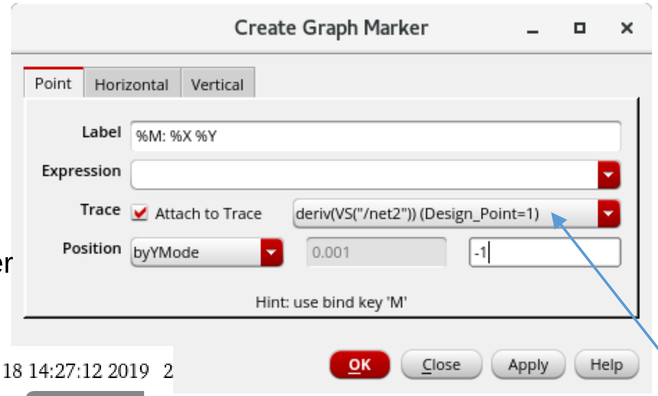
Markers set to find the -1 gain to determine noise margins.



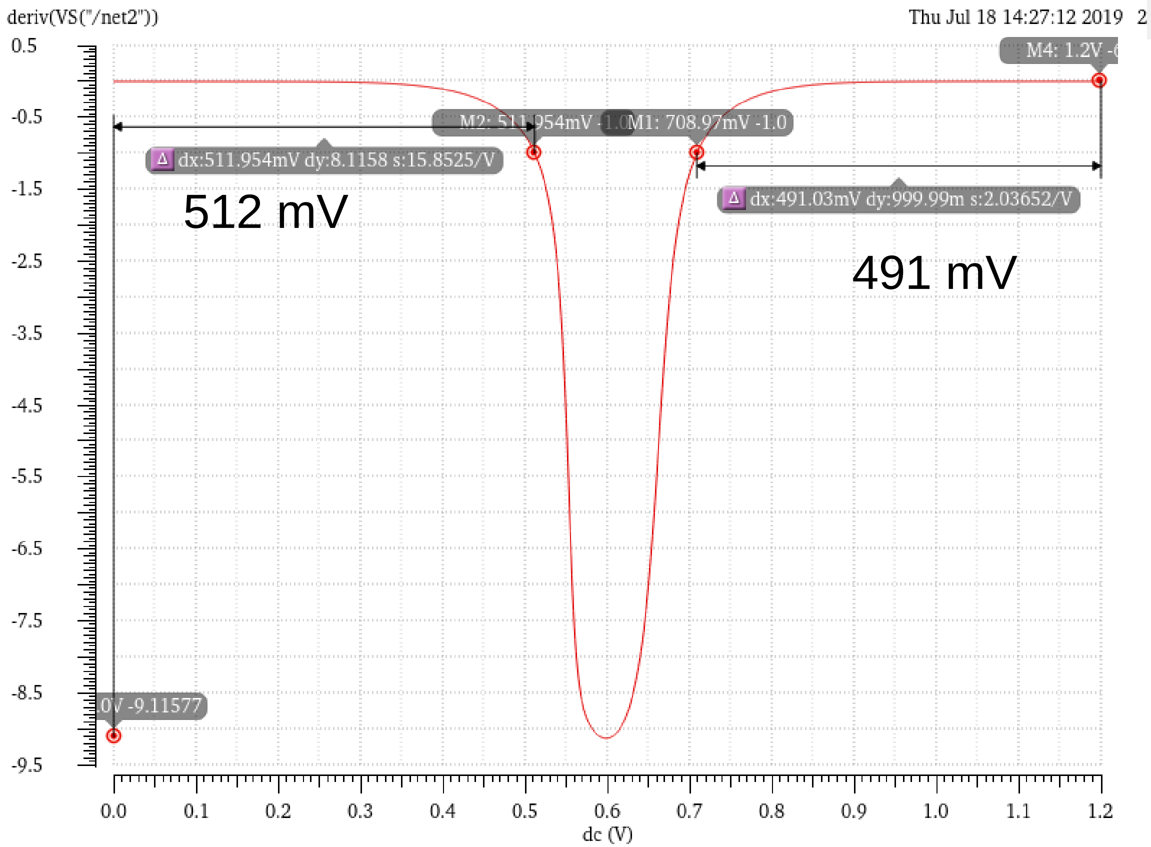
Simulation Evaluate Results using Delta Markers

Show the logic margins

1. Marker -> Create Marker -> Point byYMode for $\frac{dv_{out}}{dv_{in}} = -1$
2. Marker -> Create Marker -> Point
   (The two markers overlay at first "y = -1" point]
3. Move and edit second point marker to second "y = -1"
4. Make point markers by XMode for x = 0 V and x= 3.3 V
5. Select marker pairs (use Ctrl) and choose Marker -> Create Delta Marker
6. The dx-values are the logic margins



Select derivative trace

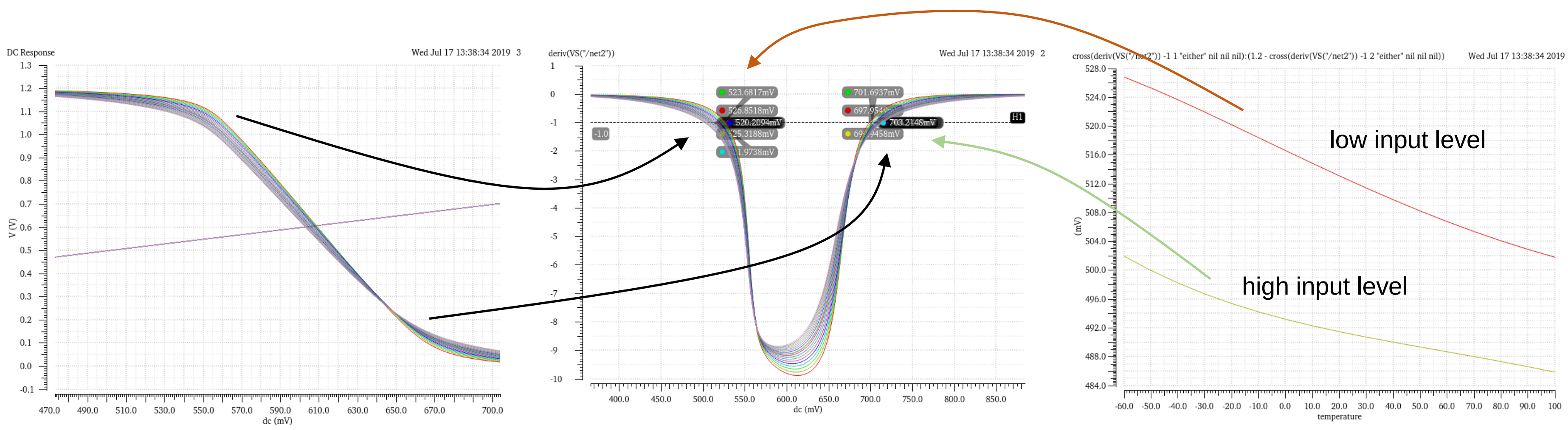


Simulation Parametric Analysis – "Temperature"

The ADE Assembler/Explorer pair provides Variables (global or per design) and Parameters to perform simulation sweeps. If a variable "temperature" is declared it will be used for the temperature in the models. The varible can be set as a single value, a range or a list of values. (If there is a variable "temperature" we must not select temperature values in the corner setup!)

Variables could be used for component parameters like width of transistors etc. In this case the variable can be used in the schematic view instead of fixed values for the component parameter. The disadvantage is, that we would have to replace the component parameter from variable to fixed value before we continue with the implementation.

Using component parameters directly we can have a fixed value in the schematic whereas we can vary the value in simulation.



$V_{out}(V_{in})$

$$\frac{\partial V_{out}}{\partial V_{in}}$$

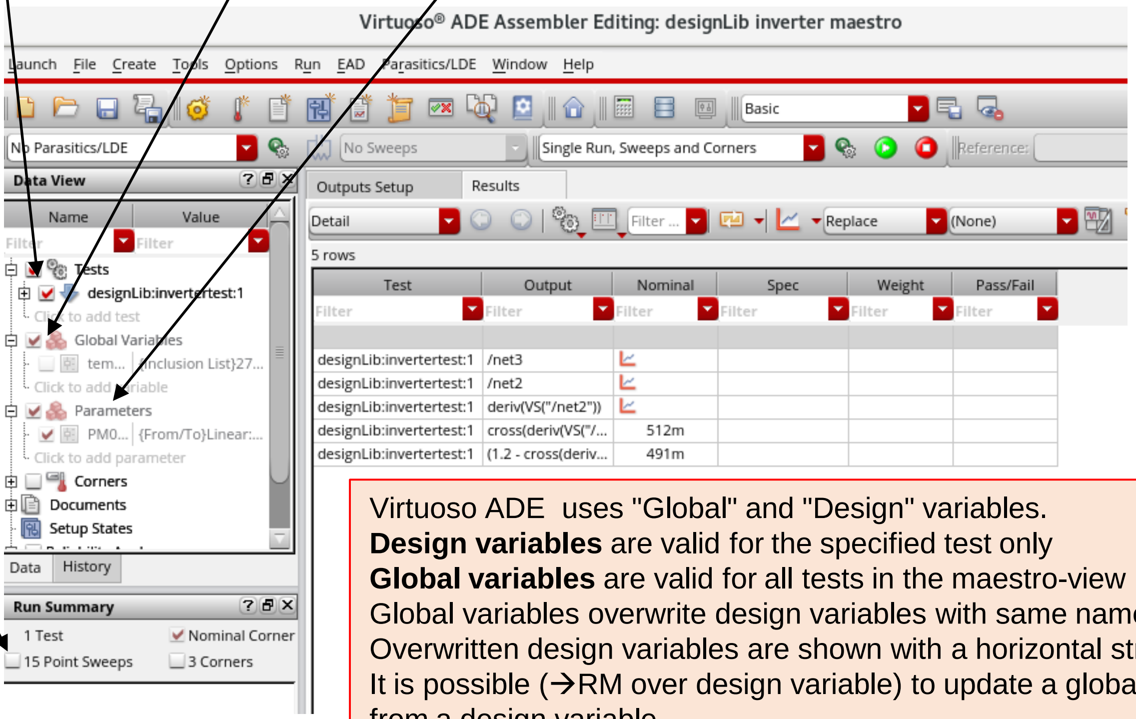
Logic margins
over temperature

Number of Tests and Point Sweeps

If we have defined different tests we can select and deselect them for simulation runs.

Different Variables, Parameters, Corners may lead to a huge number of simulation runs. We can reduce them by:

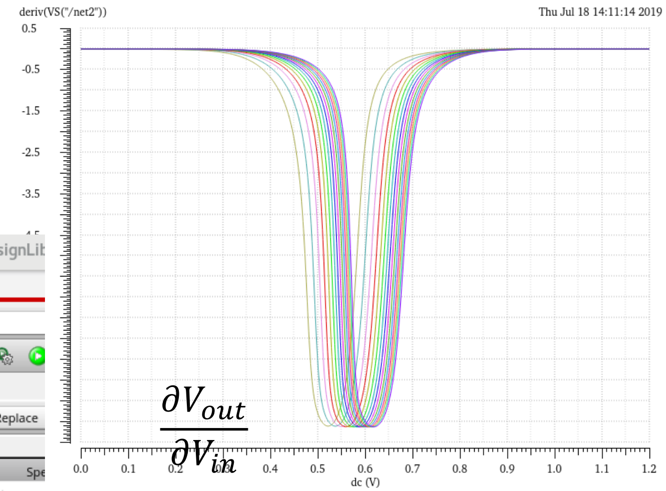
- Selecting variables (global or per designtest) or parameters
- Selecting corners or point sweeps



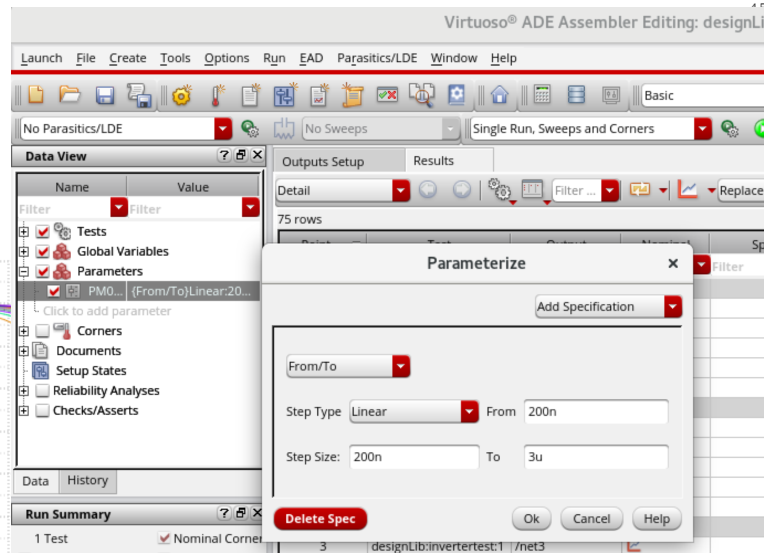
Virtuoso ADE uses "Global" and "Design" variables.
Design variables are valid for the specified test only
Global variables are valid for all tests in the maestro-view
Global variables overwrite design variables with same name
Overwritten design variables are shown with a horizontal strike
It is possible (→RM over design variable) to update a global variable from a design variable

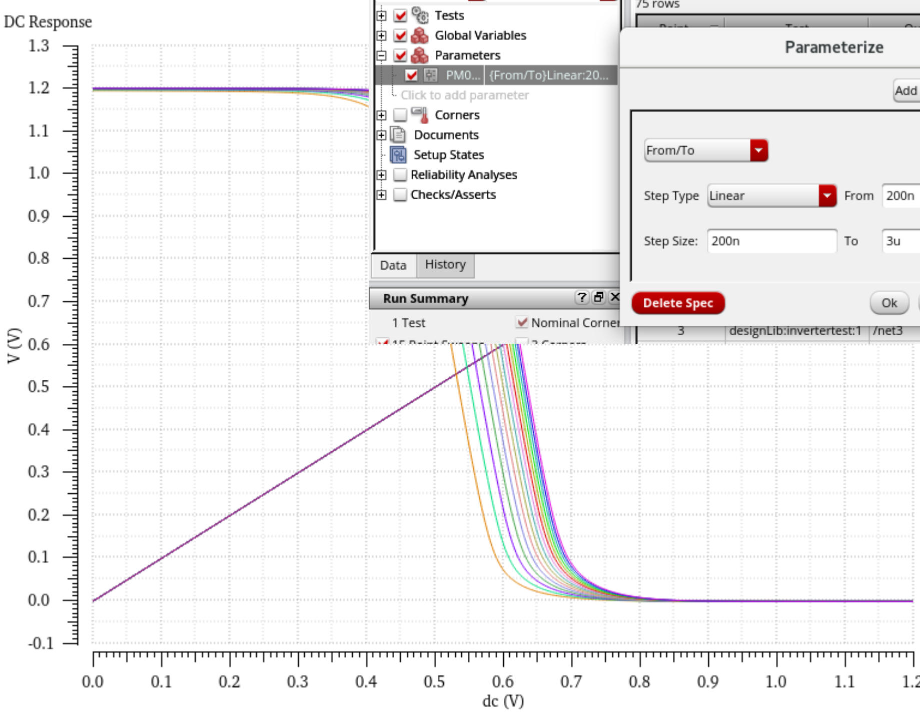
Parameters are always "global"

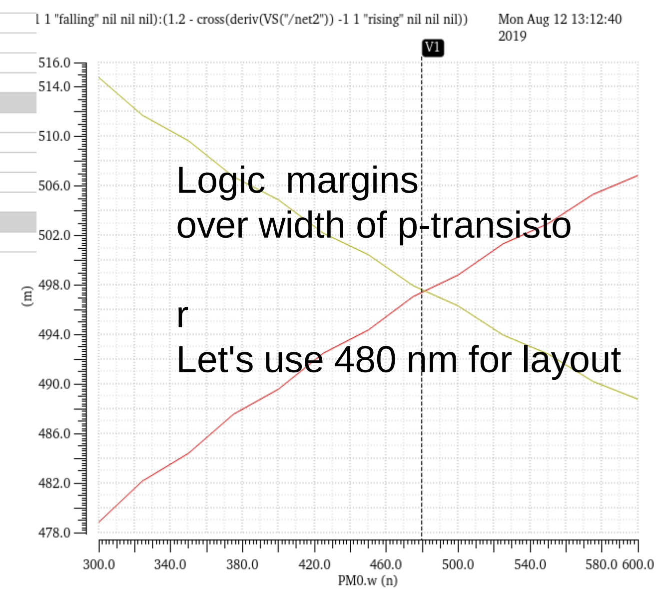
Simulation Parametric Analysis – "Width of p-channel transistor""



$$\frac{\partial V_{out}}{\partial V_{in}}$$





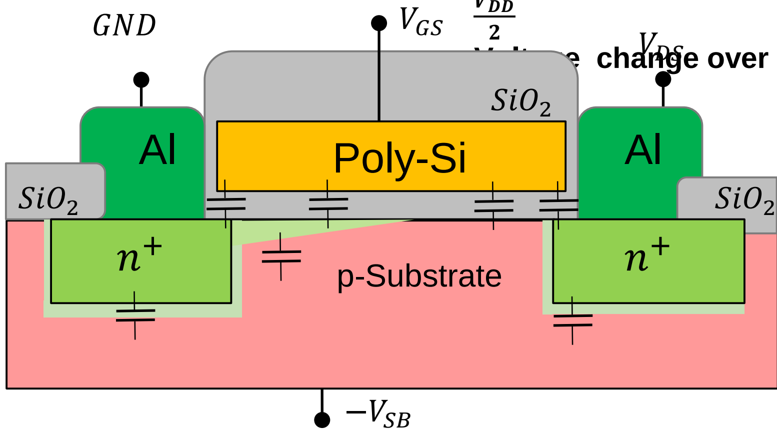


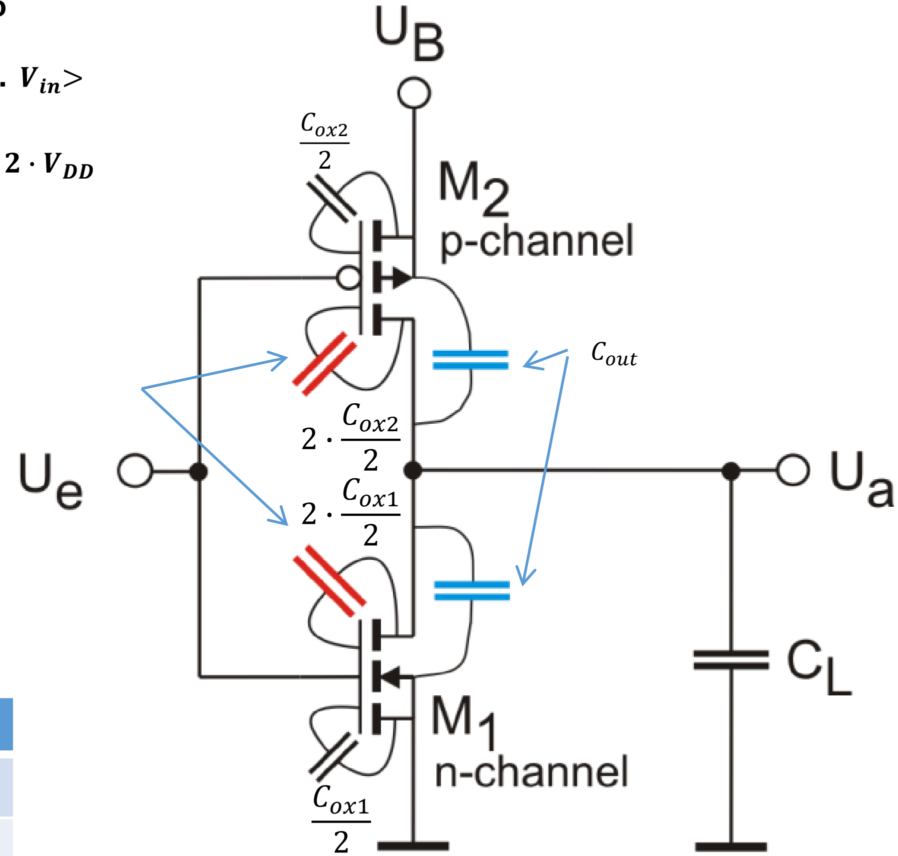
Logic margins
over width of p-transisto
r
Let's use 480 nm for layout

CMOS Inverter Capacitances and Delay

$$C_{in} = \frac{3}{2}(C_{ox1} + C_{ox2})$$

Miller-Capacitor When to consider?
→ only, if $V_{in} > V_{SP}$, resp. $V_{in} > \frac{V_{DD}}{2}$
Voltage change over C: $2 \cdot V_{DD}$





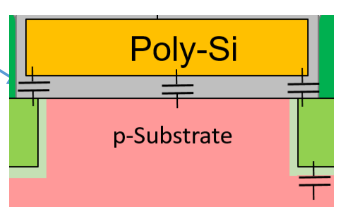
Real circumstances more complex:
Gate capacitances to drain, source, channel, bulk ->
additional $C_{Overlap}$, C_{in} nonlinear:

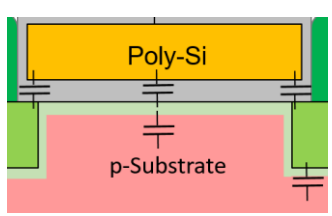
Region	C_{GB}	C_{GS}	C_{GD}	C_{in}
off	C_{ox}	0	0	$C_{ox} + 2 \cdot C_{ov}$
NPO	0	$\frac{1}{2} \cdot C_{ox}$	$\frac{1}{2} \cdot C_{ox}$	$C_{ox} + 2 \cdot C_{ov}$
PO	0	-> $\frac{2}{3} \cdot C_{ox}$	-> 0	$\frac{2}{3} \cdot C_{ox} + 2 \cdot C_{ov}$

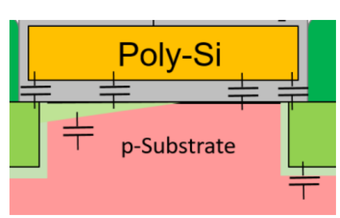
$$C_{out} \approx (C_{ox1} + C_{ox2})$$

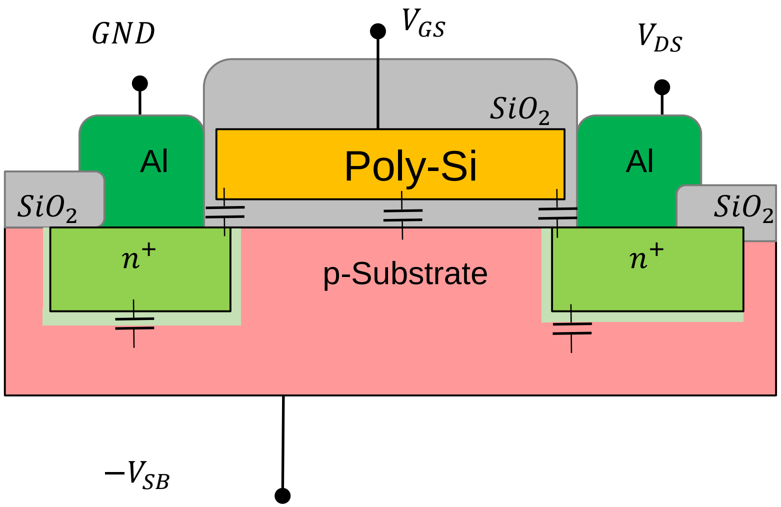
Originally the junction capacitance
between drain and bulk
In the range of C_{ox} if source and bulk
are interconnected
Nonlinear $\sim \frac{1}{\sqrt{V}}$

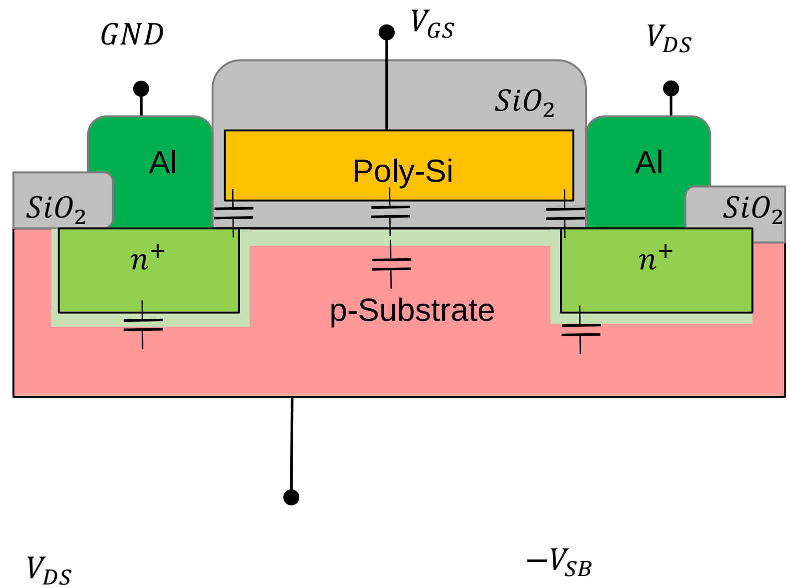
C_{ov}

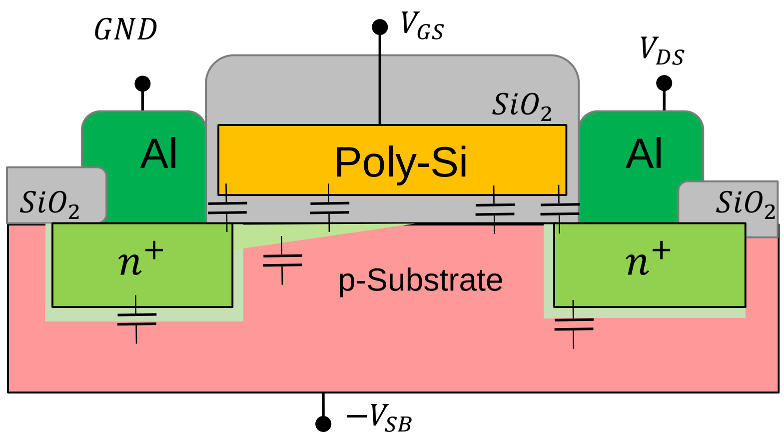




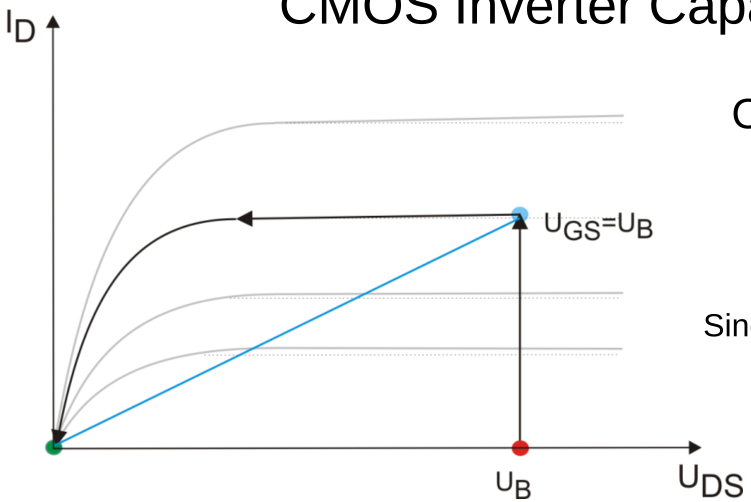



GND
V_{GS}
V_{DS}
SiO_2
Al
Poly-Si
Al
SiO_2
SiO_2
n^+
p-Substrate
n^+
$-V_{SB}$


GND
V_{GS}
V_{DS}
SiO_2
Al
Poly-Si
Al
SiO_2
SiO_2
n^+
n^+
p-Substrate
$-V_{SB}$


GND
V_{GS}
V_{DS}
SiO_2
Al
Poly-Si
Al
SiO_2
SiO_2
n^+
p-Substrate
n^+
$-V_{SB}$

CMOS Inverter Capacitances, Output Resistance and Delay



Output resistance of FET for charge reversal

Single transistor at V_{DD} with input voltage at V_{DD}

$$I_D = \frac{\beta_n}{2}(V_{DD}-V_{thn})^2 = \frac{KPN}{2}\frac{W}{L}(V_{DD}-V_{thn})^2 = \frac{170\ \mu\text{A}}{2\ \ \text{V}^2}\cdot\frac{1\ \mu\text{m}}{0.35\ \mu\text{m}}\cdot(3.3\ \text{V}-0.5\ \text{V})^2 = 0.68\ \text{mA}$$

$$R_n = R_p = \frac{V_{DD}}{I_D} = \frac{3.3\ \text{V}}{0.68\ \text{mA}} = 4.85\ \text{k}\Omega$$

Capacity and time constant-> Delay time (50%-cross distance)

$$C_{in} = \frac{3}{2}W\cdot L\cdot C'_{ox}$$

$$C_{out} = W\cdot L\cdot C'_{ox}$$

$$C_{in} = 4.5\ \frac{\text{fF}}{\mu\text{m}^2}\cdot\frac{3}{2}(1+3.2)\ \mu\text{m}^2 = 28.35\ \text{fF}$$

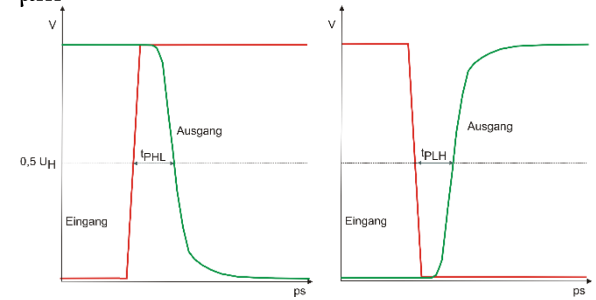
$$C_{out} = 4.5\ \frac{\text{fF}}{\mu\text{m}^2}\cdot(1+3.2)\ \mu\text{m}^2 = 18.9\ \text{fF}$$

next gate
(+ wire capacitances)

$$t_{pHL} = t_{pLH} = 0.7\cdot R_a\cdot\left(C_{inn}+C_{inp} + C_{outn}+C_{outp}\right)$$

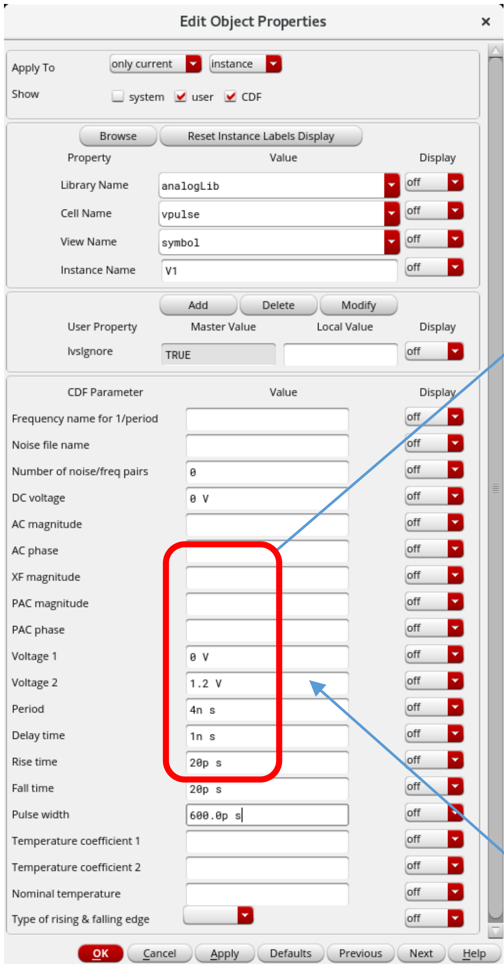
$$t_{pHL} = t_{pLH} = 0.7\cdot R_a\cdot C'_{ox}\cdot\frac{5}{2}\left(W_nL_n+W_pL_p\right)$$

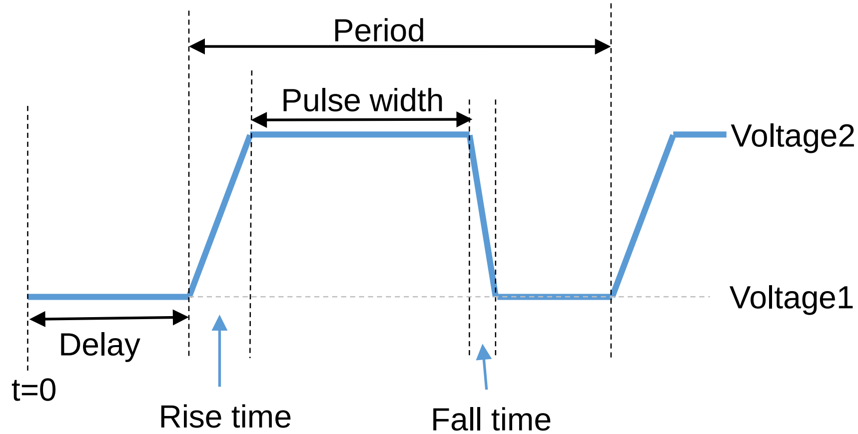
$$t_{pHL} = t_{pLH} = 0.7\cdot 4.85\ \text{k}\Omega\cdot 4.5\ \frac{\text{fF}}{\mu\text{m}^2}\cdot\frac{5}{2}(1+3.2\ )\mu\text{m}^2 \boxed{= 160\ \text{ps}}$$



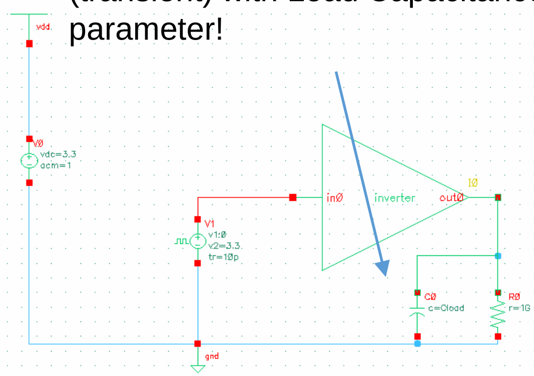
Simulation Transient Analysis

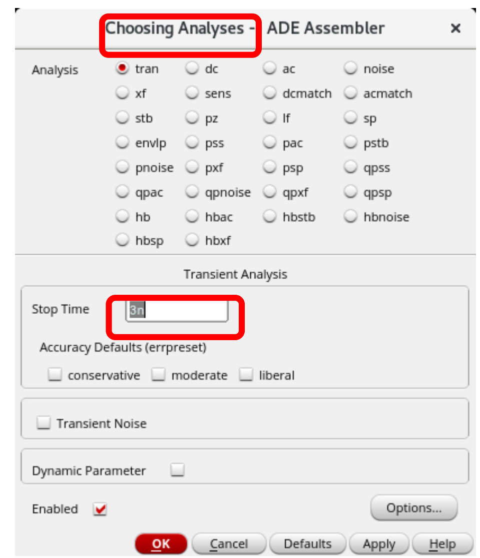
Replace the vdc-source by a vpulse-source. (It would be possible to use this from scratch even for dc-simulation!)





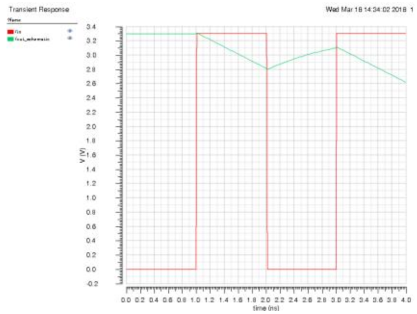
Try another parametric simulation (transient) with Load Capacitance as a parameter!





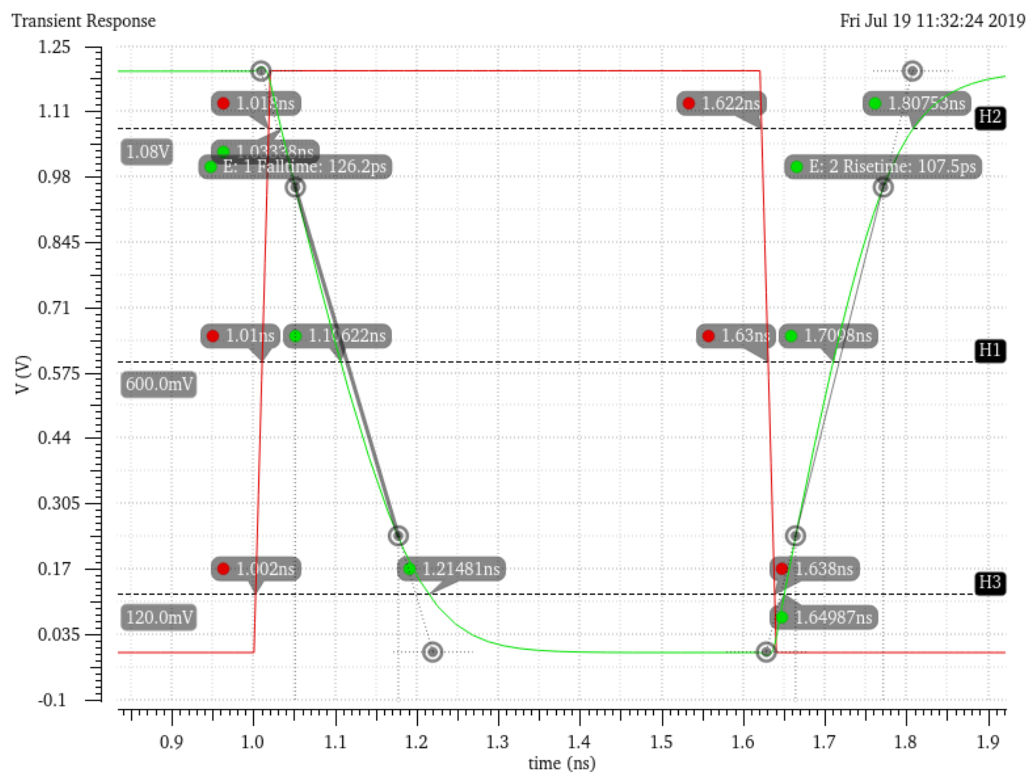
How to find proper time values?

Simulation Transient Analysis



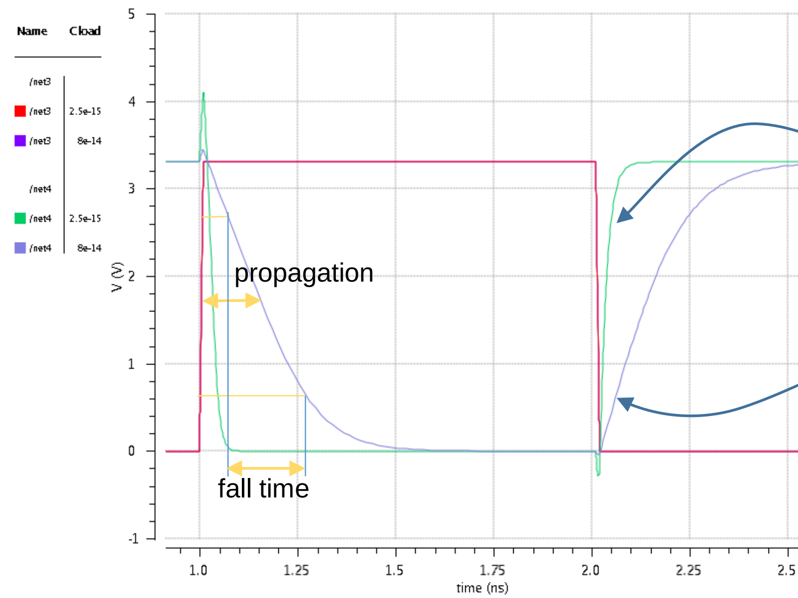
Oops!, looks strange …
1pF load is too much for our small gate

To display cross points: RM on marker line > Intercepts -> On



Simulation Transient Analysis

Comparasion of simulation and fab data sheet. Here for AMS 0.35µ. Do not have exact date from UMC 65 nm.



INV0

www.ams.com

Databook Build Date: Wednesday Jun 18 17:26 2014

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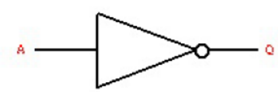
Conditions for characterization library **c35_CORELIB_TYP**, corner **c35_CORELIB_TYP_typical**: Vdd= **3.30**V, Tj= **25.0** deg. C .
Output transition is defined from **20%** to **80%** (rising) and from **80%** to **20%** (falling) output voltage.
Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Cell Area	36.400 um^2
Equation	Q = "!A"
Type	Combinational
Input	A
Output	Q



State Table	
A	Q
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		2.50	80.00	2.50	80.00
A to Q	fall	0.05	0.53	-0.20	1.07
	rise	0.08	0.94	0.86	2.24

Output Transition [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		2.50	80.00	2.50	80.00
A to Q	fall	0.05	0.71	0.56	1.48
	rise	0.10	1.39	0.54	1.87

Capacitance [fF]	
A	3.8210

Leakage [pW]
0.21

Uups!, explain the negative value!

Dynamic Power Consumption [nW/MHz]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		2.50	80.00	2.50	80.00
A to Q	fall	5.98	6.28	160.14	131.01
	rise	34.83	35.09	256.12	205.46

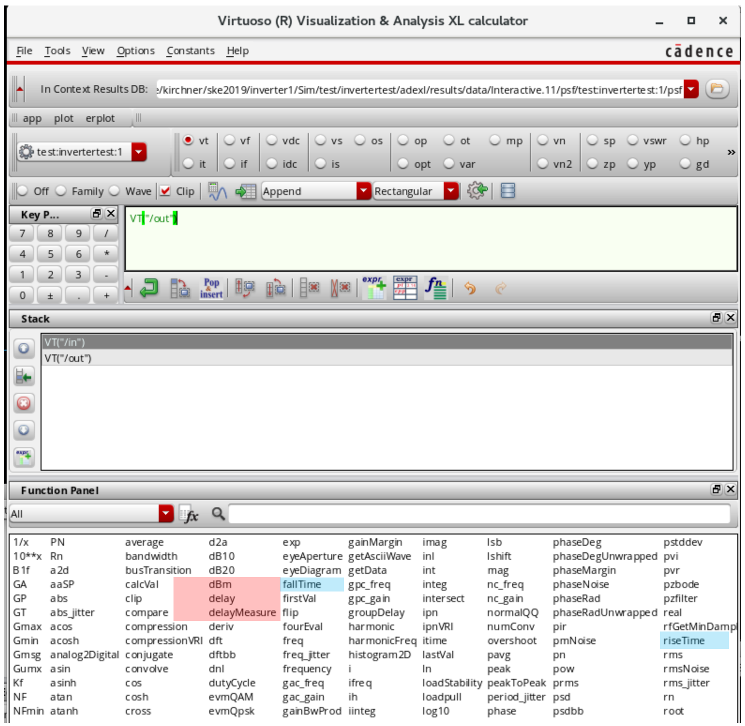
Simulation Transient Analysis

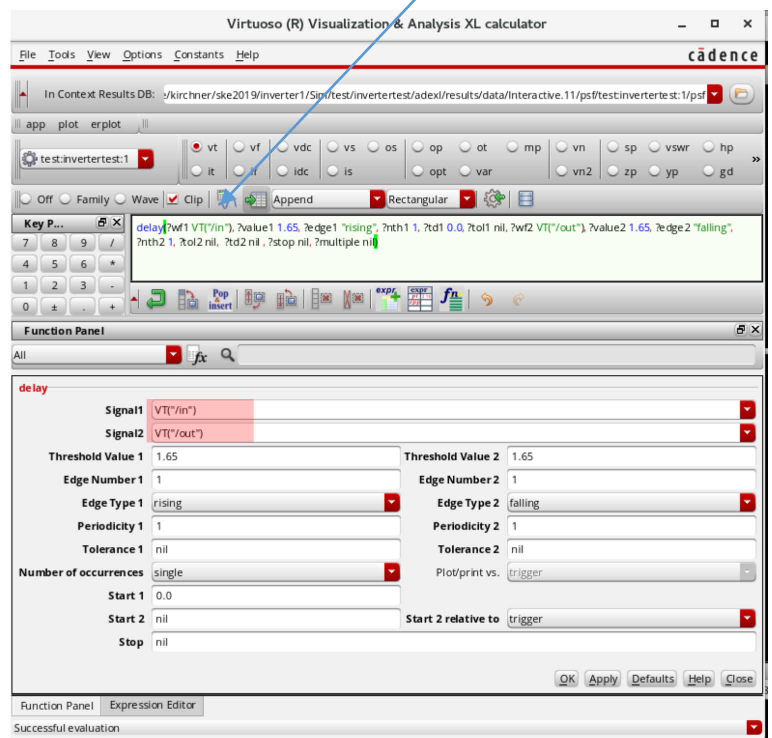
Using Calculator and functions to show delay, fall time, rise time

Calculator works with stack in RPN

Input and output from voltages transient simulation
VT("/in"), VT("/in") used in the function "delay"

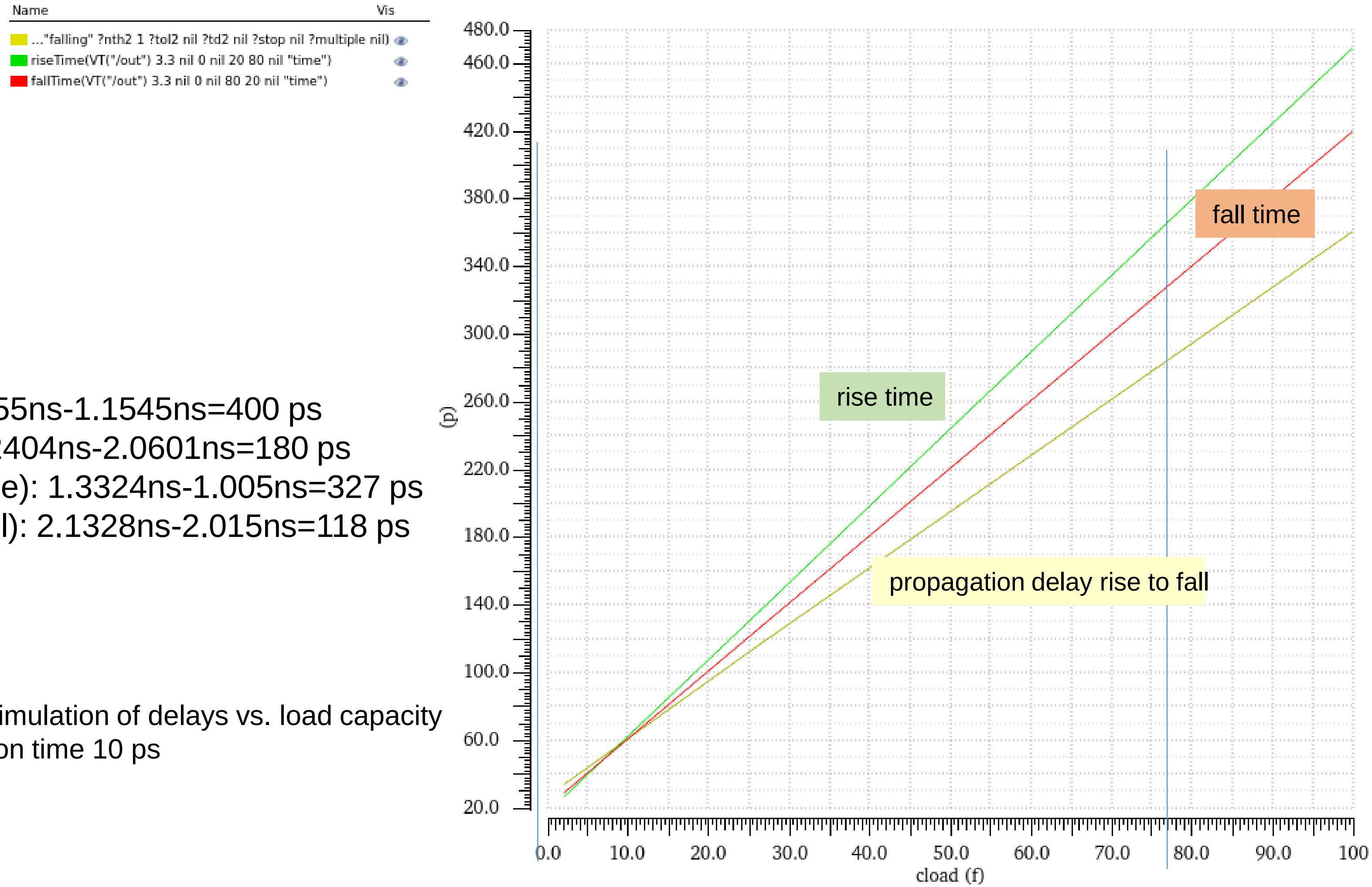
To wave





Simulation Transient Analysis

Similar to the input-output-delay you may use the risetime and falltime functions



falltime: 1.555ns-1.1545ns=400 ps
risetime: 2.2404ns-2.0601ns=180 ps
delay(inp rise): 1.3324ns-1.005ns=327 ps
delay(inp fall): 2.1328ns-2.015ns=118 ps

Parametric simulation of delays vs. load capacity
Input transition time 10 ps

Typical capacitance of a gate input in 65 nm technology: 4 … 10 nF

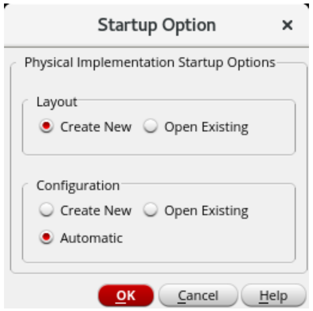
Layout Start and Generate from Schematic I

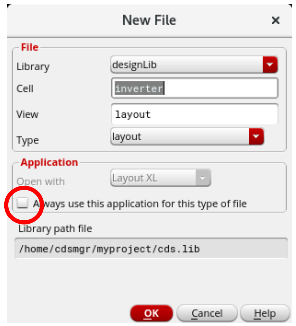
Replace all the Variables in the design with fixed values.
Variables have their scope in simulation only.

It is the better choice to use Parameters in the Simulation!

Variables in the testbench are not affected, because they will not be involved in the layout.

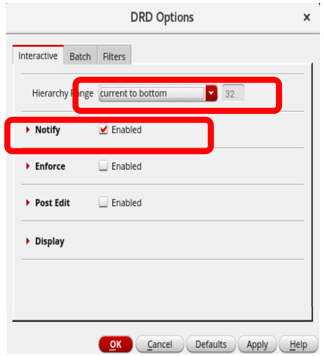
In Schematic Editor: **Launch -> Layout XL**





In Layout XL: Enable Design Rule Driven Edit:

Options -> DRD-Edit



In Layout XL: **Connectivity -> Generate ->All From Source**

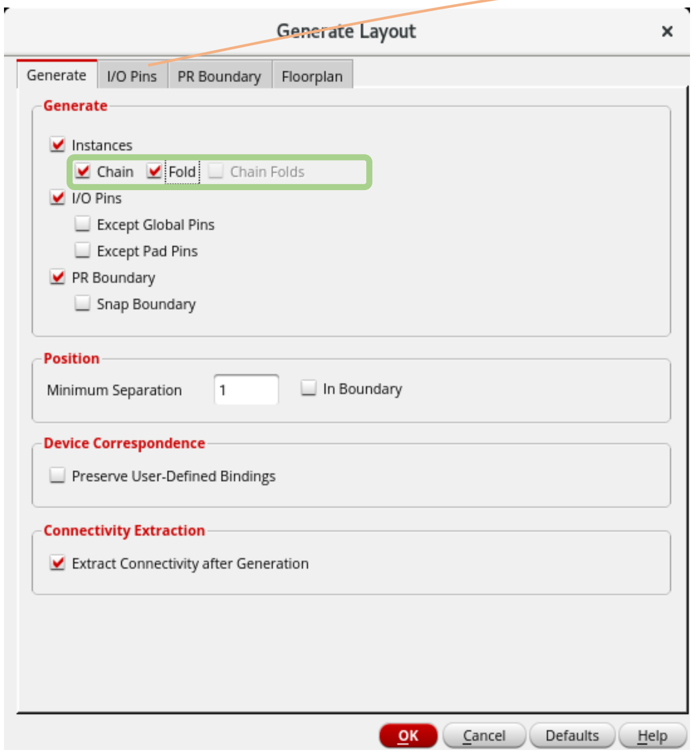
Generates a layout with all instances placed outside a rectangle (estimated cell boundary)

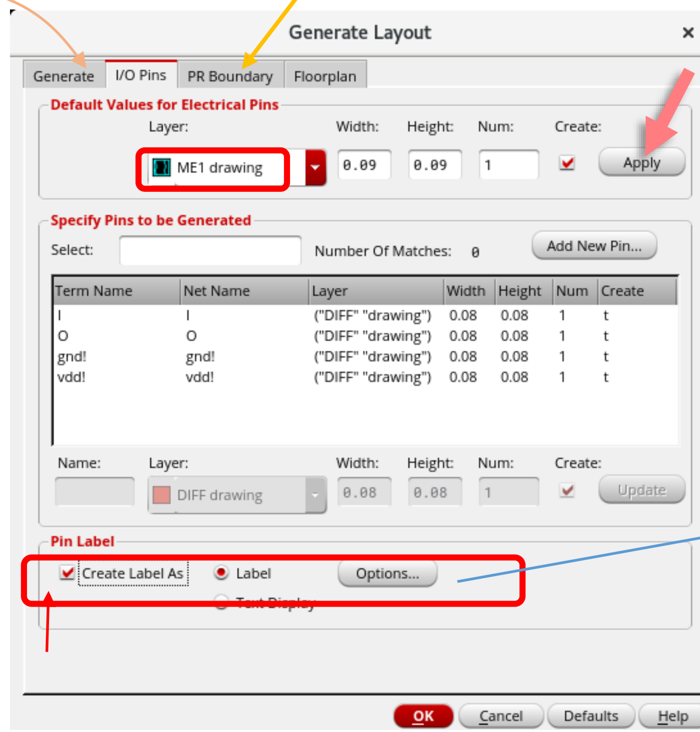
Layout Start and Generate from Schematic II

In Layout XL: C**onnectivity -> Generate ->All From Source**

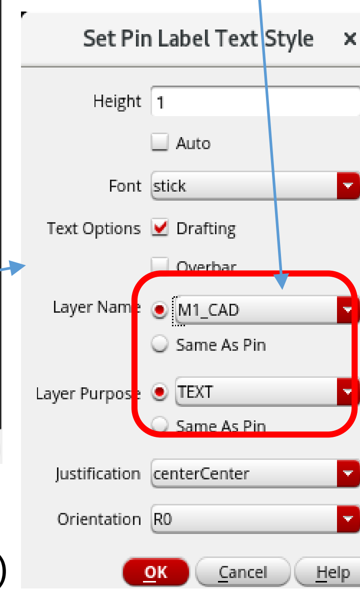
Generates a layout with all instances placed outside a rectangle (estimated cell boundary)

Here we may select the cell boundary:
origin, utilization, aspect ratio **or** polygon corners





For correct pin recognition in LVS.
Technology kit specific!



Generates a layout with all instances placed outside a rectangle (estimated cell boundary)

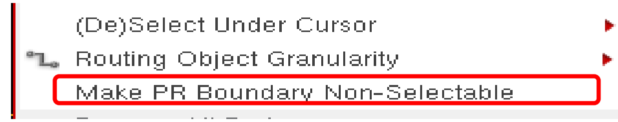
Layout Floorplaning AMS 0.35µ

To see all layers: Options -> Display -> Display Levels -> Set "stop" to 32 (Shift-f) Save to ~/.cdsenv

Suggested cell area

20µm x6 µm cell

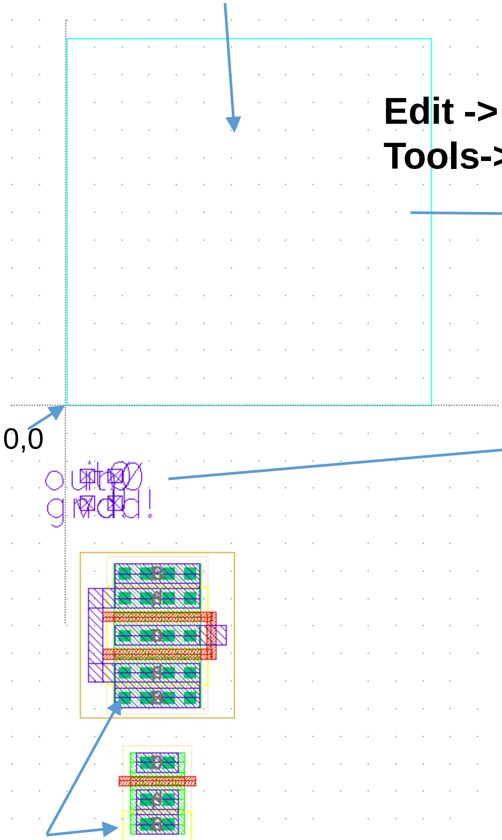
Fix it using RM
In Layout-XL,
<u>not</u> in Layout-L !

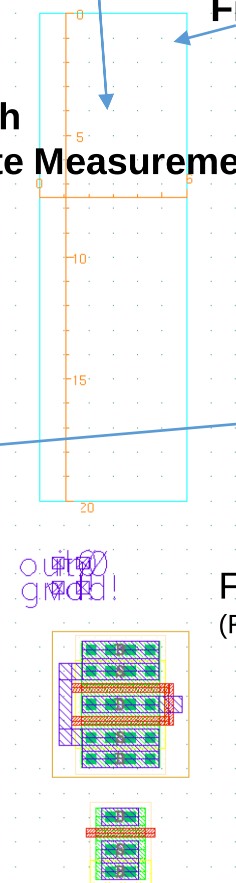


Edit -> Stretch
Tools-> Create Measurement*

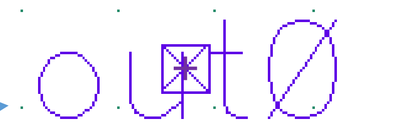
Tools -> Clear All Measurements*m
Create -> Shape -> Rectangle

***Ruler** in older versions





Point 0,0



When moving pins include all parts.
Best practice: Snap to the little "+"!

F3 opens move options (e.g. orthogonal, all angels etc.)
(F3 opens options for all commands)

If you have lost prBoundary you can generate a new one by:
Create -> P&R Objects -> P&R Boundary
(prBoundary is an object, not a simple shape!)

Don't have Bulk ?
Delete both transistors in layout.
Open schematic and change transistor property there (add substrate contact).
In Layout XL use Connectivity -> Update Components and Nets.

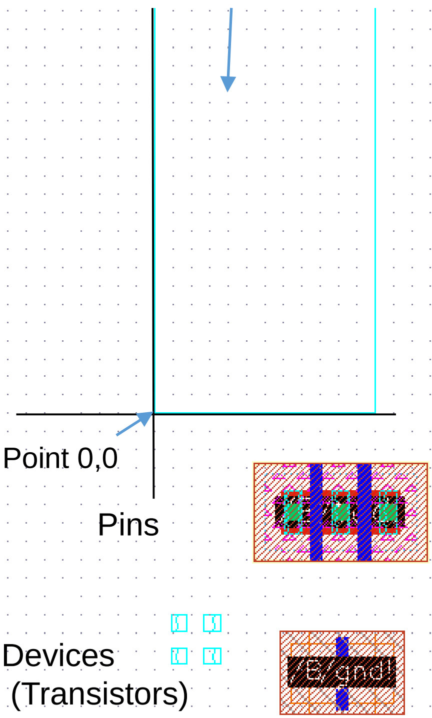
Way to change something in the schematic and make it actual in Layout

Layout Floorplaning UMC 65n

To see all layers: Options -> Display -> Display Levels -> Set "stop" to 32 (Shift-f)
You may also select the objects to display (axis, nets, names etc.)

PR Boundary = **P**lace and **R**oute Boundary

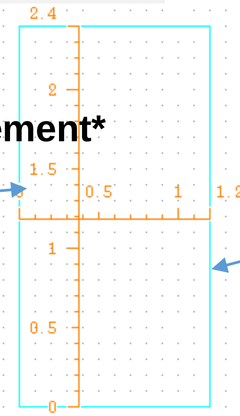
Suggested cell area "prBoundary"



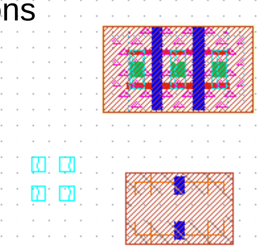
Edit -> Stretch
Tools-> Create Measurement*

2.4 x 1.2 µm cell

Typical cell libraries in 65 nm:
Hight: 1.4 … 2.4 µm
Witdth pitch: 0.2 … 0.3 µm
Power rails: 0.2 … 0.3 µm

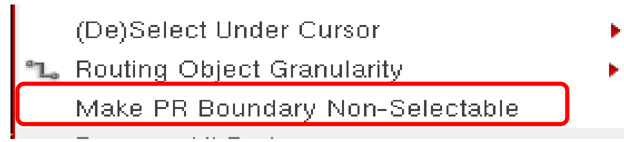


*__Ruler__ in older versions



Lock PR Boundary using RM ->

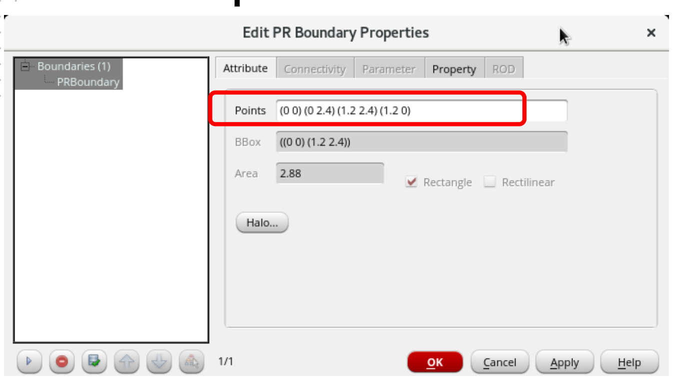
In Layout-XL, NOT available in Layout-L !



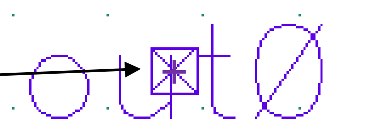
If you have lost prBoundary you can generate a new one by:
Create -> P&R Objects -> P&R Boundary
(prBoundary is an object, not a simple shape!)

Edit Boundary Shape (when not locked):

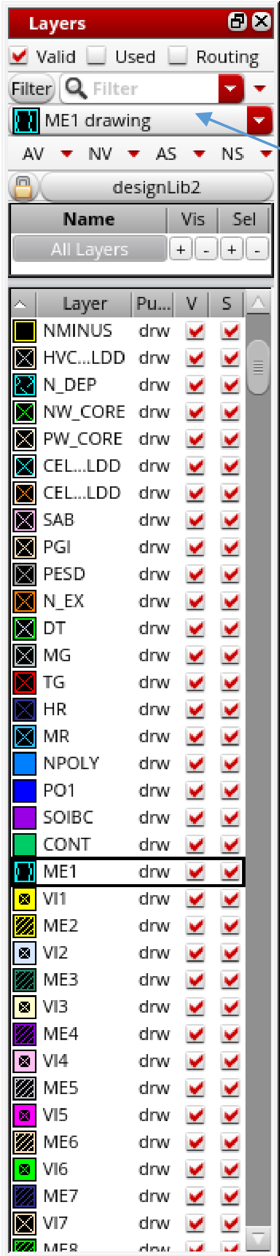
RM -> Properties



When moving pins include all parts!
Most important is the little "+" in the center.



Layout Edit



actual layer

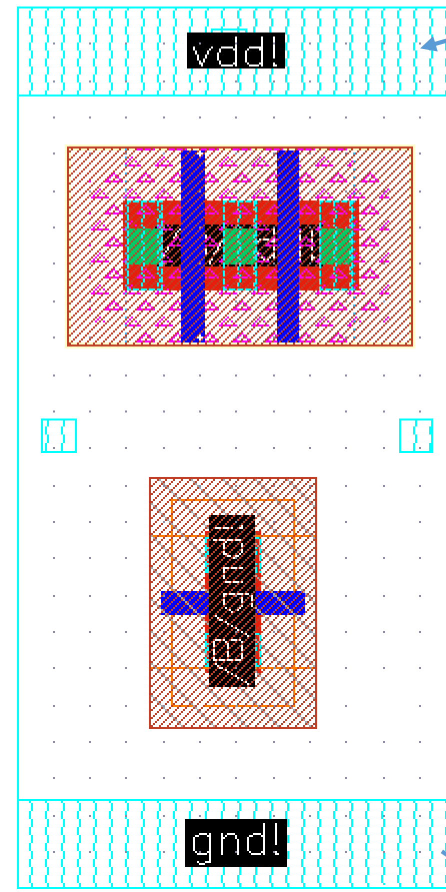
visible/selectable

Create -> Shape -> Rectangle

Input with **mouse**:
First corner 0:0
Opposite Corner 1.2:0.25
or input in **CIW**:
0:0 enter -> 1.2:0.25
enter

1.2:2.4

0:2.15



1.2.0.25

0:0

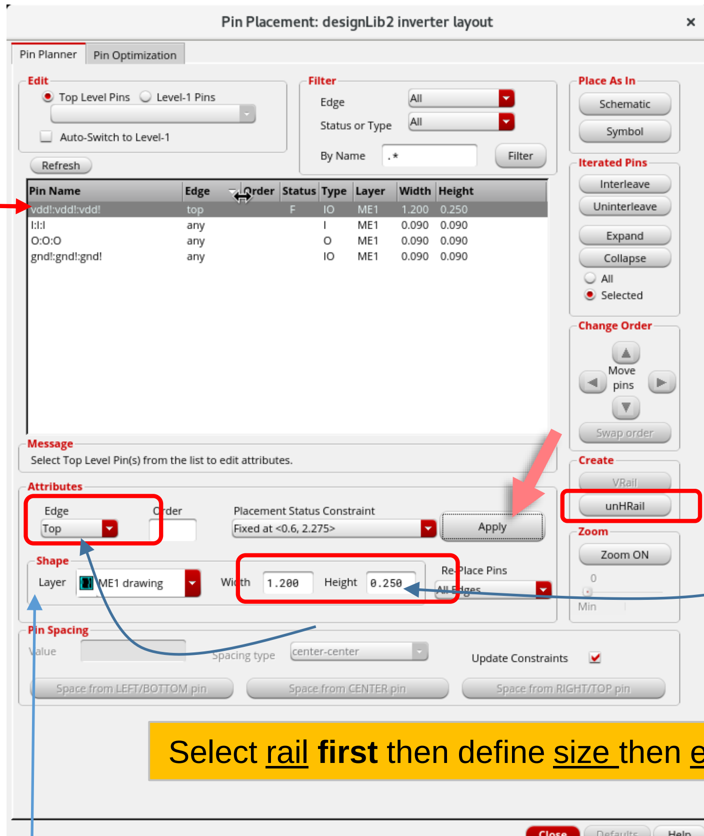
Edit -> Copy

F3 opens move options (e.g. orthogonal, all angels etc.)
(F3 opens options for all commands)

Layout (Automatic) Pin Placement (optional)

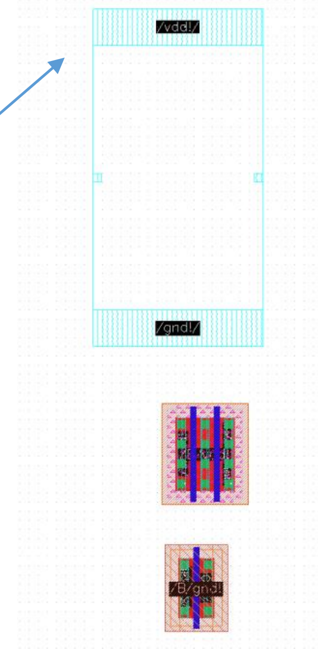
In Layout **XL**: **Place -> Pin Placement**

Form offers a choice to place pins at top (vdd!) bottom (gnd!) left (inputs) and right (outputs)
For top (vdd!) and bottom (gnd!) the pins can be extended to rails with the width up to the cell width and a defined hight.



Select rail **first** then define size then egde

VDD-pin 1.2x0.25 µm
at top of the cell



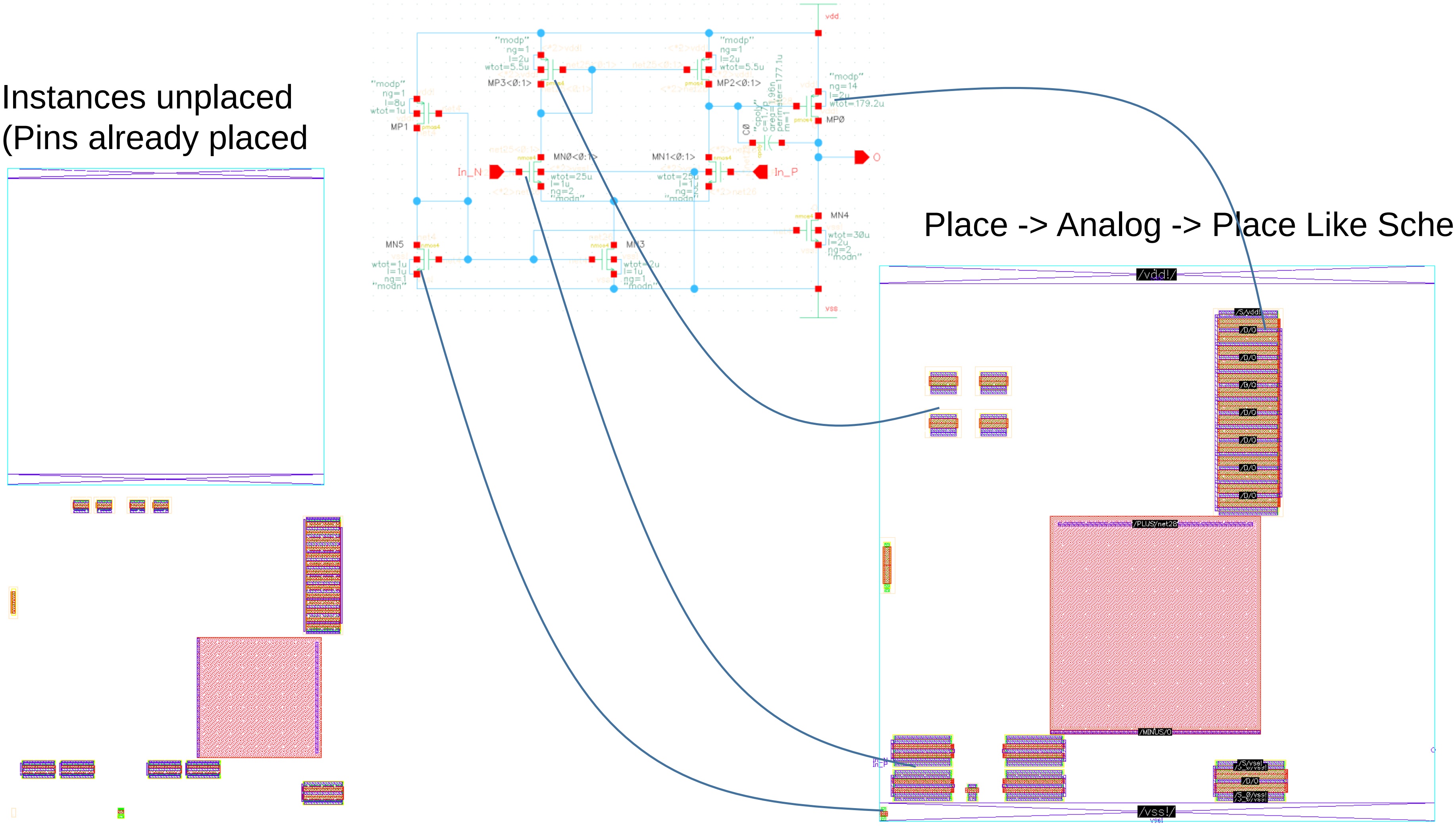
Signal pins may be placed left or right first, then change the placement to "any" and move to the appropriate location. As long as they have the attribute "left" or "right" they are locked to the edge.
Other way : Properties -> Placement Status -> Unplaced

Layout (Automatic) Placement of Instances (optional)

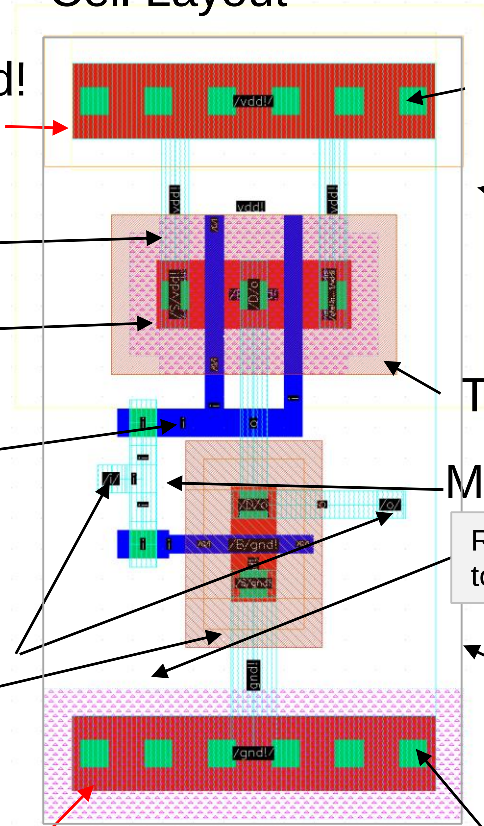
For a simple inverter with two tansistors it is easy to identify the transistors and to place them in an appropriate pattern in the layo
Fore more complex designs some kind of automatic placement may be used. Here shown for an operational amplifier (OPA) and
the placement option "like schematic". Helps to identify the instances/parts.

Instances unplaced
(Pins already placed

Place -> Analog -> Place Like Schema



Cell Layout



ME1-rail for vdd!
must enclose vdd!-pin

M1-NWELL - VIAs

MET1

pmos-Transistor
in n-well (folded)

n-Well(NWELL) extended for vdd
connection and merging with adjacent cells
Remind usage in (flipped) raws

Transistors NWELL

POLY1

M1_POLY - VIA

F3 to change
wire width

RM and Via up/down
to change layer

PIN in ME1

Hight: 2.4 µm, Width: 1.2 µm
PR-Boundary

nmos-Transistor

ME1-rail for gnd!
must enclose gnd!-pin

M1_PSUB - VIAs

gnd!, vdd! -> Exclamation marks:
Global signal names valid through design hierarchy

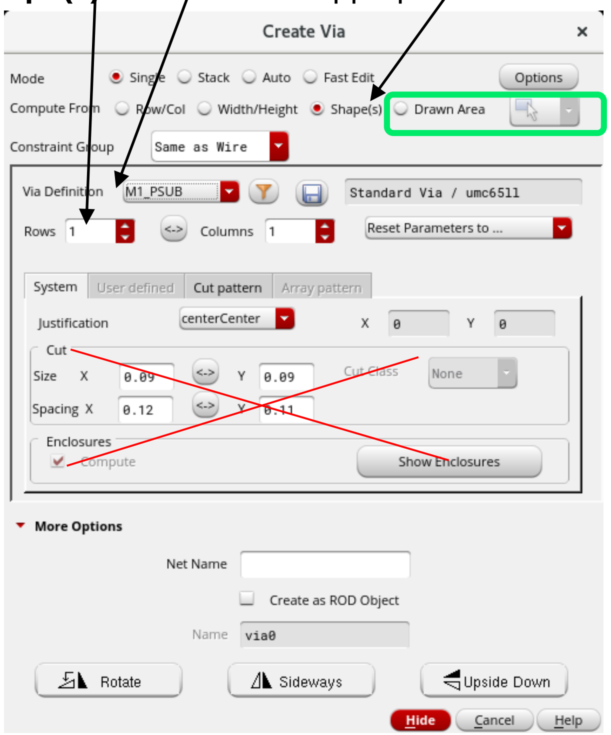
Layout - VIAs

Creating VIAs:

M1_PSUB : p-Substrat-Metal1
M1_NWEL : n-Well-Metal1
M1_POLY : Poly1-Metal1
M2_M1 : Metal1-Metal2
M3_M2 : Metal2-Metal3
….

Make them for the power stripes **before** routing of wires.

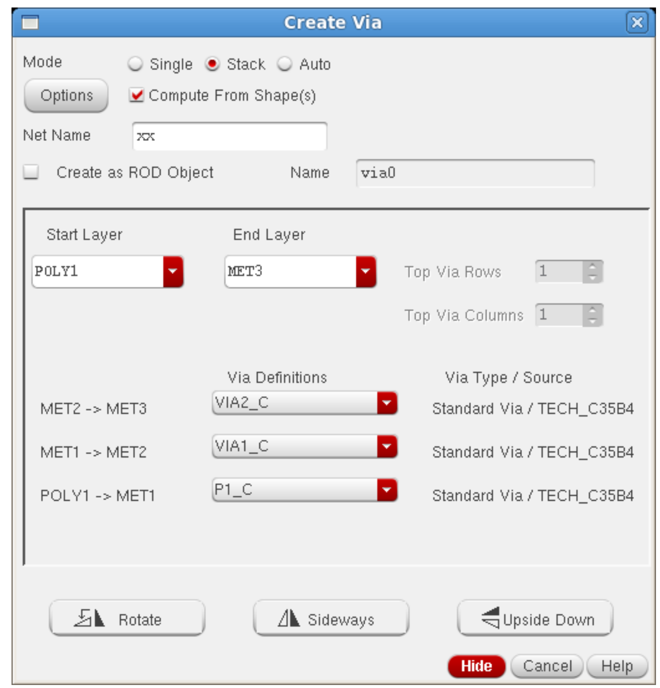
Layout of single contact predefined. Do not change! Number and alignment variable. Using "**Compute from Shape(s)**" fills area with appropriate number of contacts.



Create Shapes fills **ALL** shapes of the net!

To select a special area use "Drawn Area" and select the area using the mouse.

Stacked VIAs connecting more than two layers available:

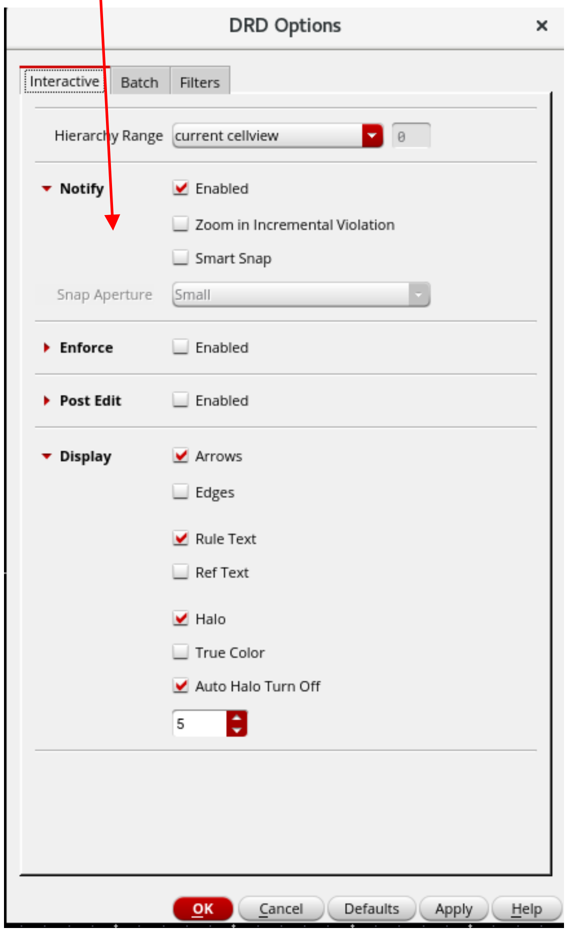


Cell Layout – Design Rule Driven Design -DRD (optional)

To avoid design rule violations when moving instances or wiring we can switch on the DRD.

Options -> DRD-Edit

You must select one mode here!



"Enforce" stops edit action (route a wire, move an element) if a design rule is violated.

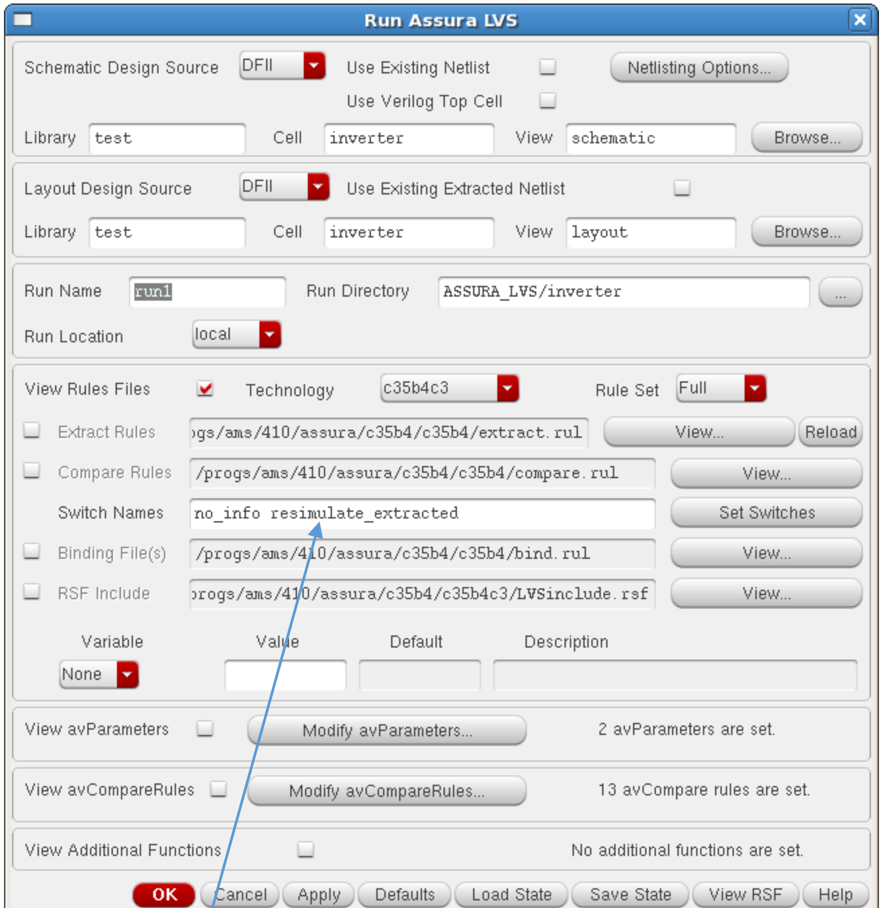
"Notify" writes a warning if a design rule is violated.

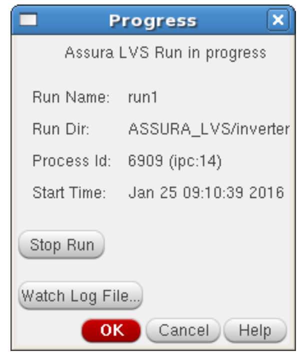
"Post-Edit" sets markers only if a design rule is violated.

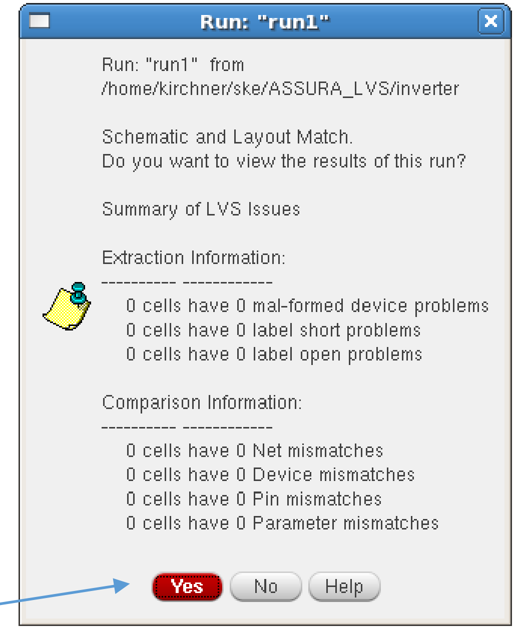
Cadence Layout - LVS

Compare Layout Versus Schematic

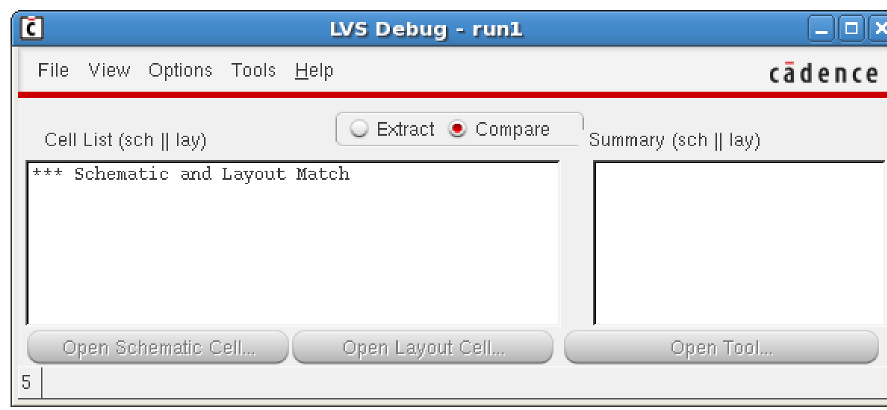
Assura -> Run LVS







wait!



Necessary for QRC parasitic extraction

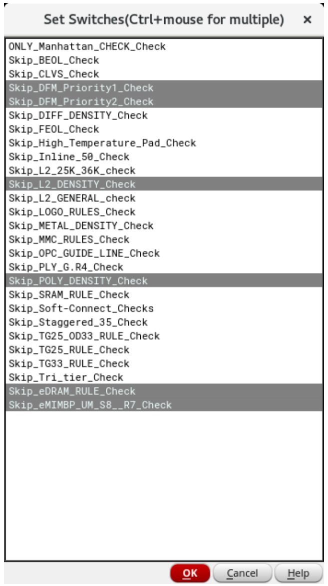
That's what we want to see!

ams says: "**Do NOT use** the **resimulate_extracted** switch because in that case substrate shorts (soft connections between different nets via substrate) might not be found correctly!

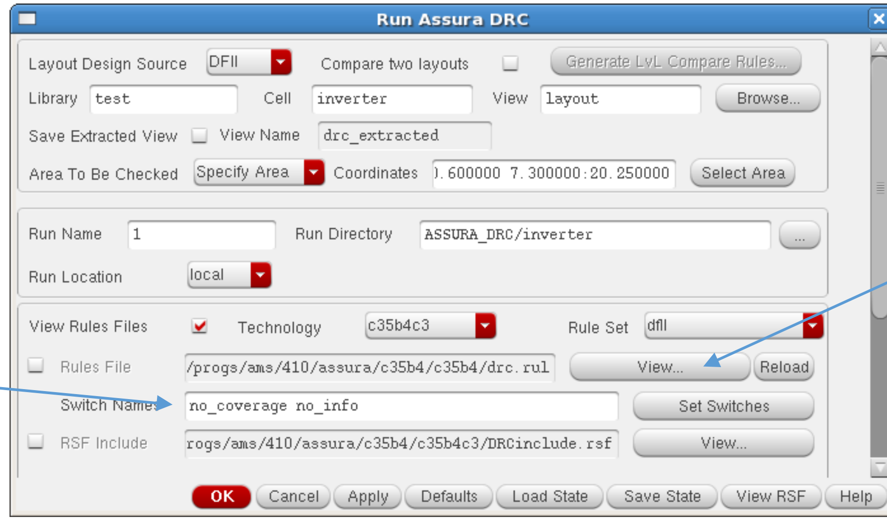
Dann fehlen im av_extracted die design layer

Cadence Layout - DRC

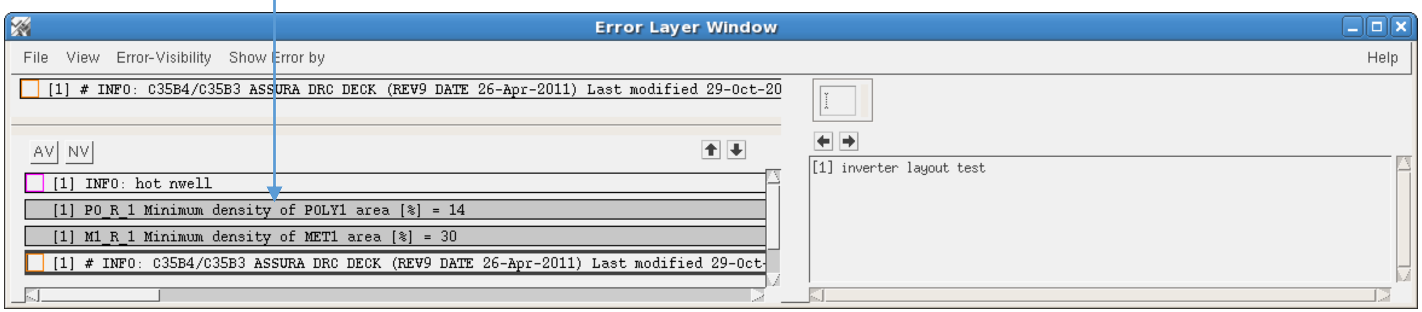
Assura -> Run DRC

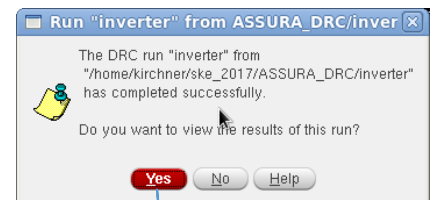


Set these switches to avoid some messages (see below!). What does "Coverage" mean?



Here you may open the rule set file for information. Contains all the rules like distances, overlap, min. width et cetera.





That's what we would like to see!

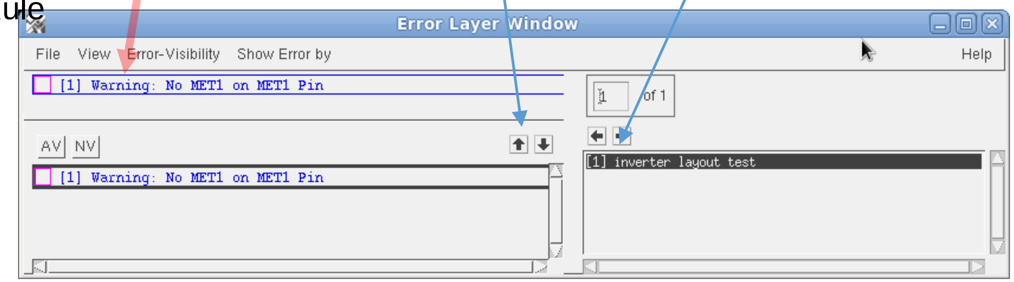
Cadence Layout – DRC with Errors remaining

Example for an Error Message:
Here: Pin is not fully covered by metal

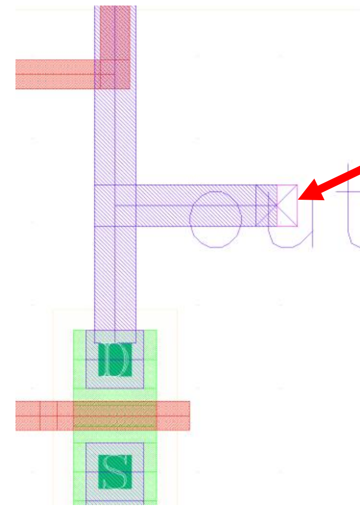
Violated Rule

Browse violated rules

Browse Errors (if multiple)

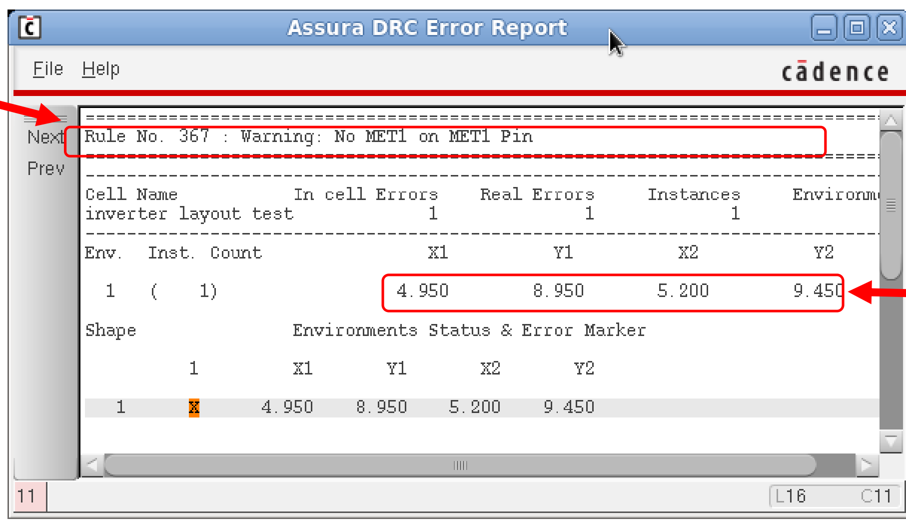


Position in Layout



View -> Error Report

Violated Rule

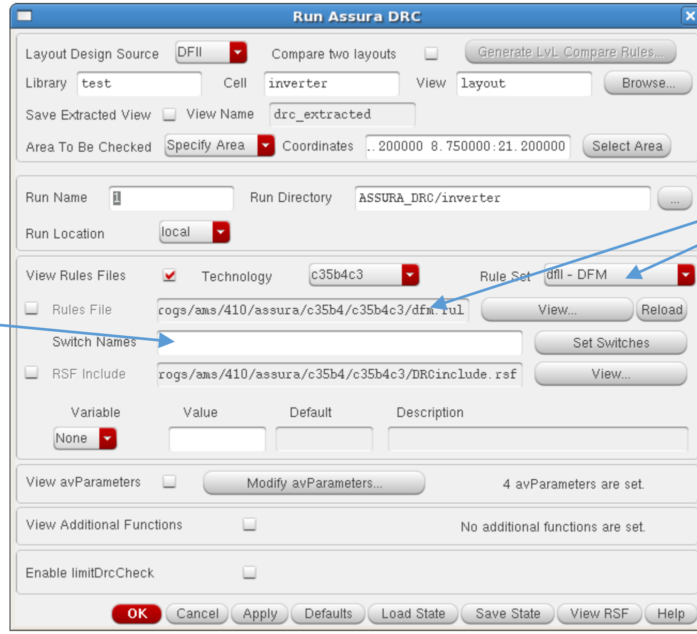


Position in Layout

Solutions:
Extend wire to the right or
add a rectangle in MET1 covering the Pin

Cadence Layout - DFM

Assura -> Run DRC

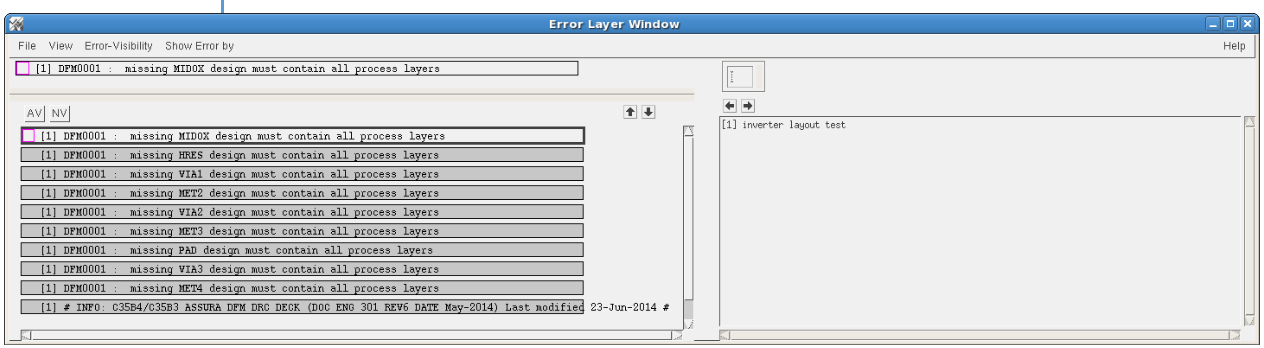


OPtion: extended ruleset DFM: **D**esign **F**or **M**anufacturing ->

Coverage of all but upper metal layer et cetera.

no switches!

Errors because we did not use some layers.
Interesting for a top layout.



Cadence Layout - ERC

Electrical rule check is included in DRC.
There is no separate rule check in the ams-HITkit.

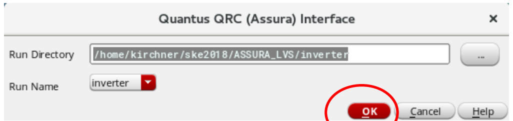
Starting ERC with Assura -> Run ERC opens a form similar to DRC with no technology selected and either rule files.

Up to now no verification for this.
Separate ERC-usage found nowhere.

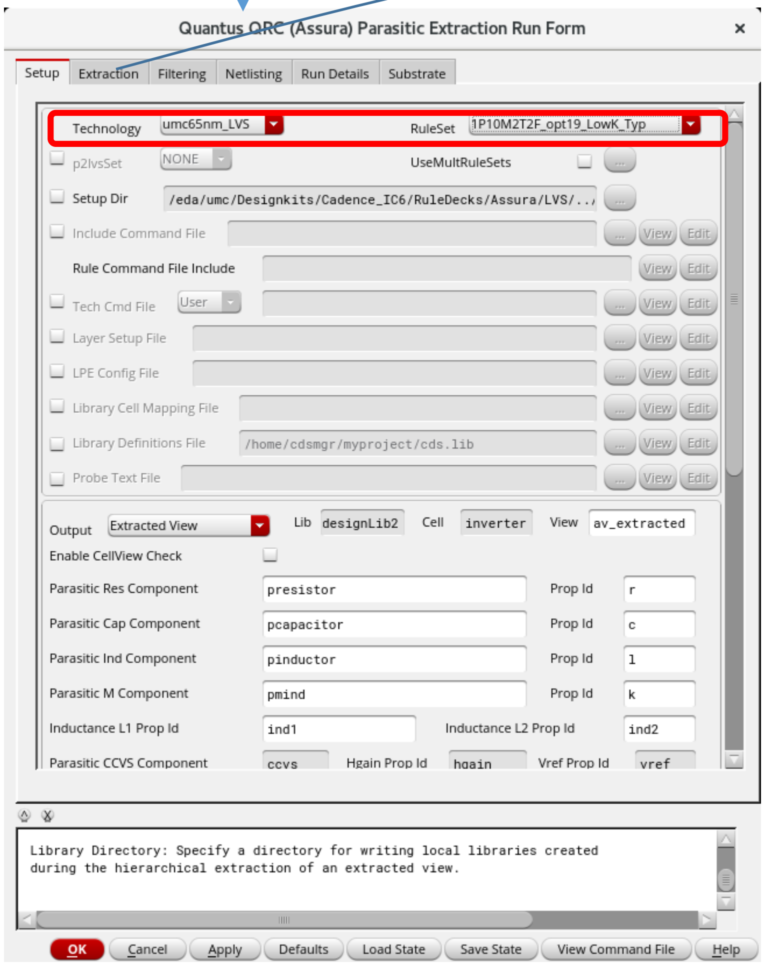
Cadence Layout – QRC – Parametric Extraction

QRC -> Run Assura Quantus -QRC

no changes



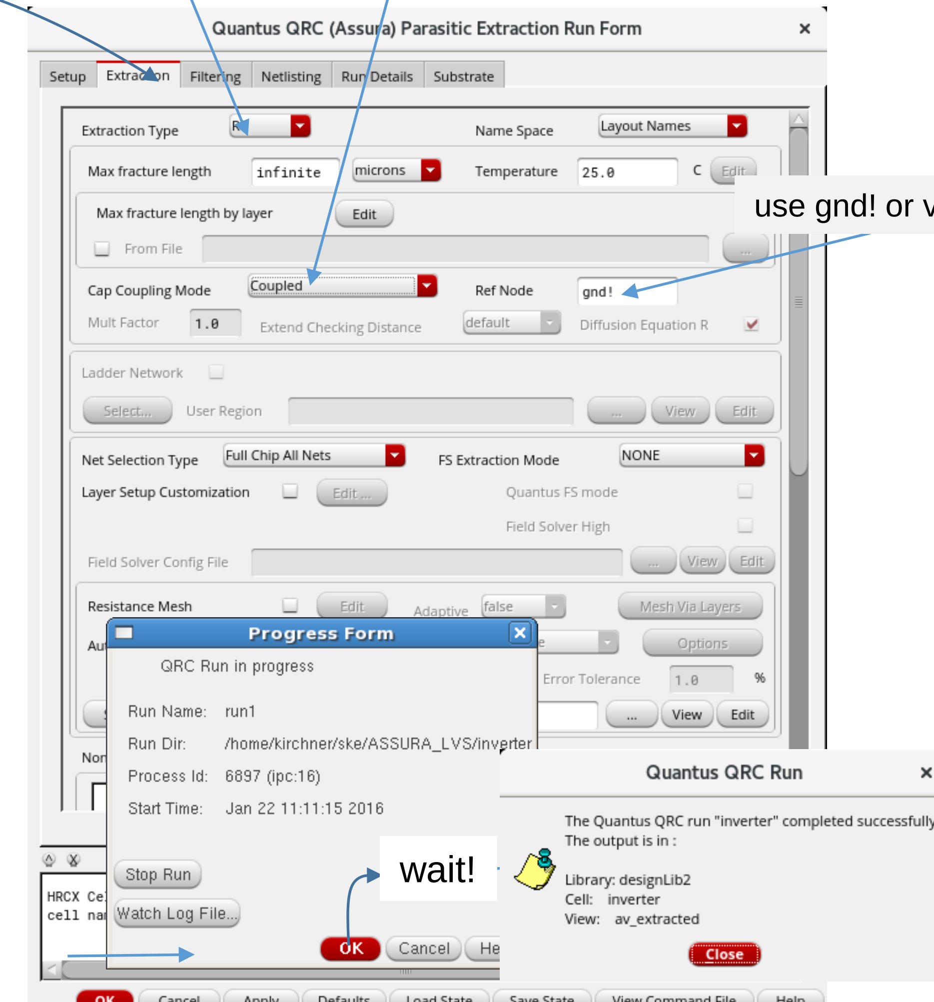
seit 6.1.7 dazwischengeschaltet



RC

Coupled C: net to net (signals)
Decoupled C: net to power and ground only

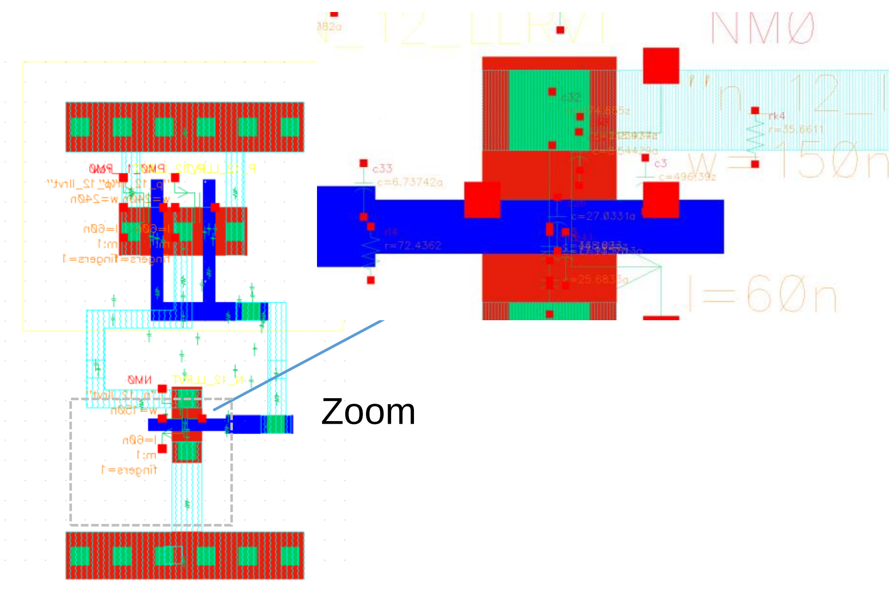
use gnd! or vcc!

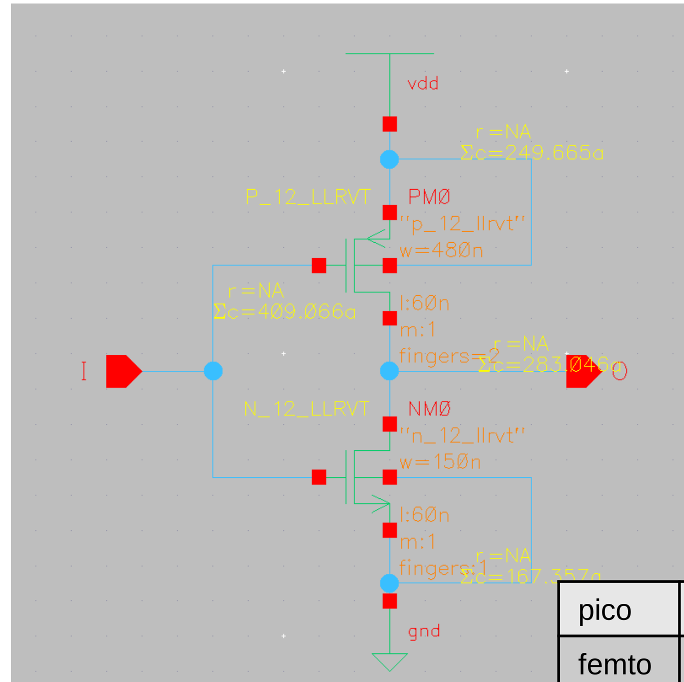


Cadence – QRC – Extracted view

Open View "av_extracted" in Library Manager

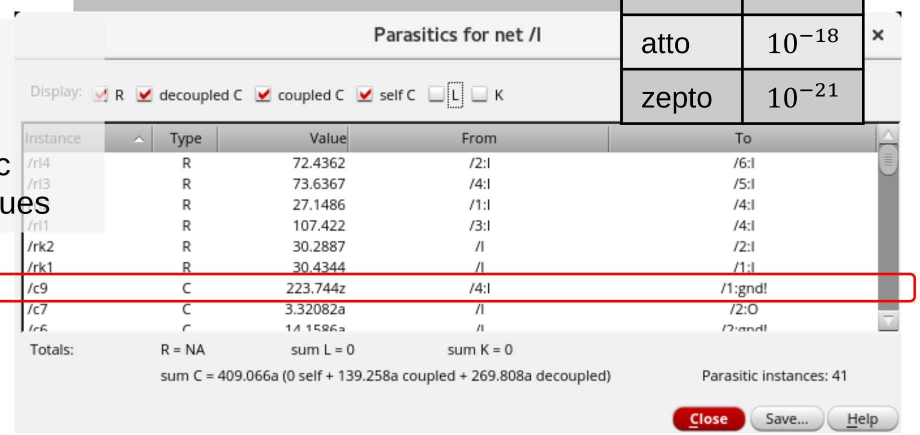
Maybe you must select all layers (up to 32) in Options -> Display once more





pico	10^{-12}
femto	10^{-15}
atto	10^{-18}
zepto	10^{-21}

In **Schematic** and/or **av_extracted** view:
Launch -> Parasitics : New Tab "Parasitics"
In **Setup** give reference nodes gnd! and vdd
Parasitics -> Report Parasitics -> net : Select in schematic
In Schematic **Parasitics -> Show Parasitics** : Lumped values


Parasitics for net /I

Display: R decoupled C coupled C self C L K

Instance	Type	Value	From	To
/rl4	R	72.4362	/2:I	/6:I
/rl3	R	73.6367	/4:I	/5:I
	R	27.1486	/1:I	/4:I
/rl1	R	107.422	/3:I	/4:I
/rk2	R	30.2887	/I	/2:I
/rk1	R	30.4344	/I	/1:I
/c9	C	223.744z	/4:I	/1:gnd!
/c7	C	3.32082a	/I	/2:O
/c6	C	14.1586a	/I	/2:gnd!

Totals: R = NA sum L = 0 sum K = 0
sum C = 409.066a (0 self + 139.258a coupled + 269.808a decoupled) Parasitic instances: 41

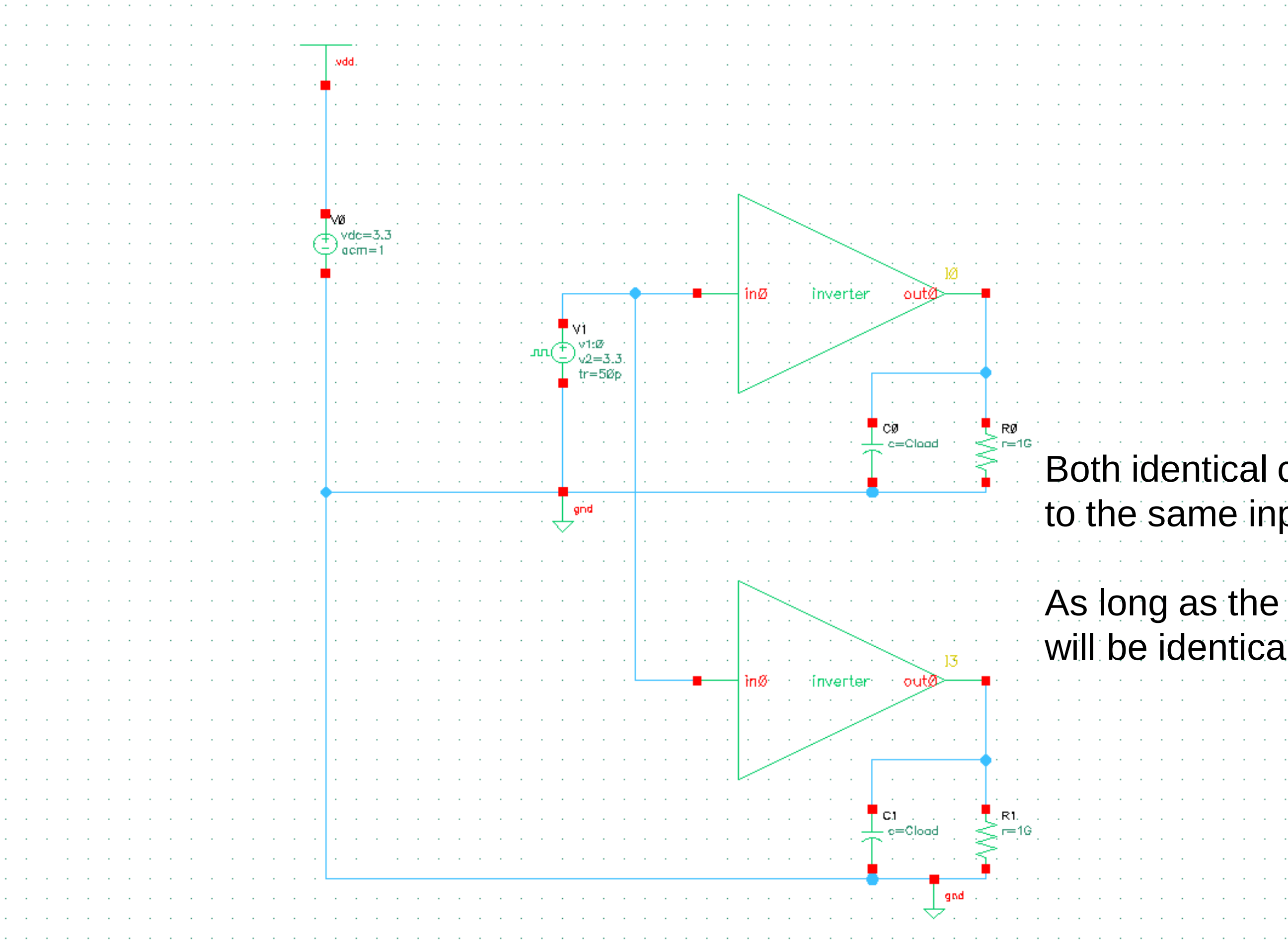
Close Save... Help

Try also: net to net
terminal to terminal (r)

Viewing parasitics may be interesting.
More important is their use in postlayout simulation.
They will be included in simulation netlists.

A schematic to simulate both the schematic entry and the extracted view

New cell schematic "invertertest_postlayout"



Both identical cells are connected
to the same inputs and have identical loads.

As long as the models are identical the results
will be identical as well.

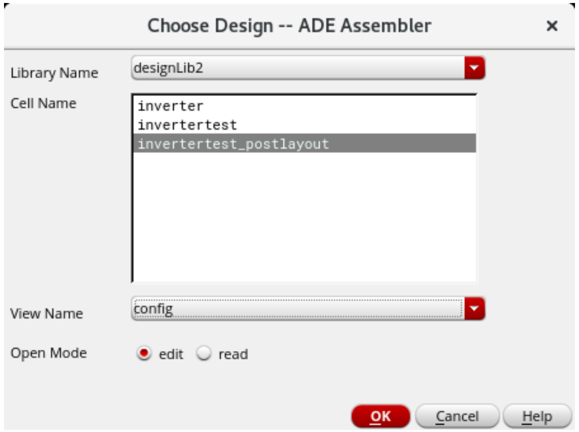
Using Configuration View for simulation

"Standard" simulation uses the schematic for the design under test.
Using a configuration gives the choice to select different design cellviews.
Of course the view must be simulatable (like schematic, Verilog-A, av_extracted ..)

The av_extracted view as the result of parasitic extraction is a simulatable view.

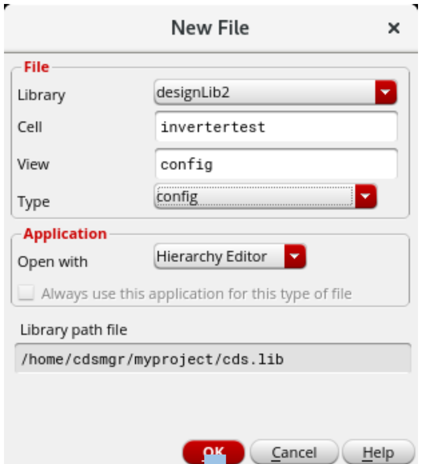
When a configuration is available it can be used to create a test in ADE-assembler.
(There may config cellview even for the initial simulation of the schematic.)

When Creating a new test change the view name to "config"

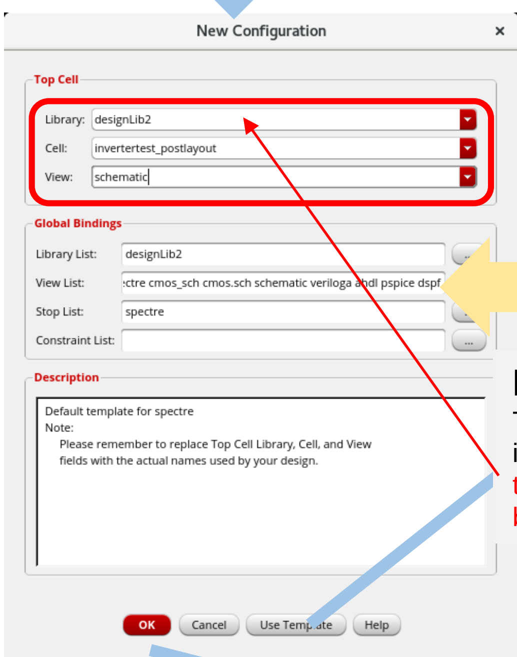


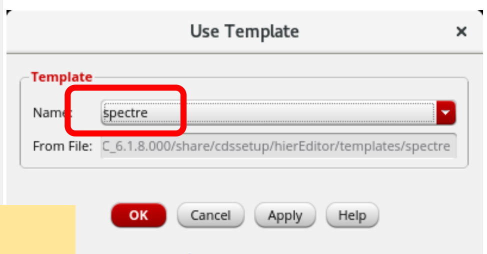
But before we can use the config cellview we have to create it.

Creating a Configuration (config) cellview for Post-Layout Simulation

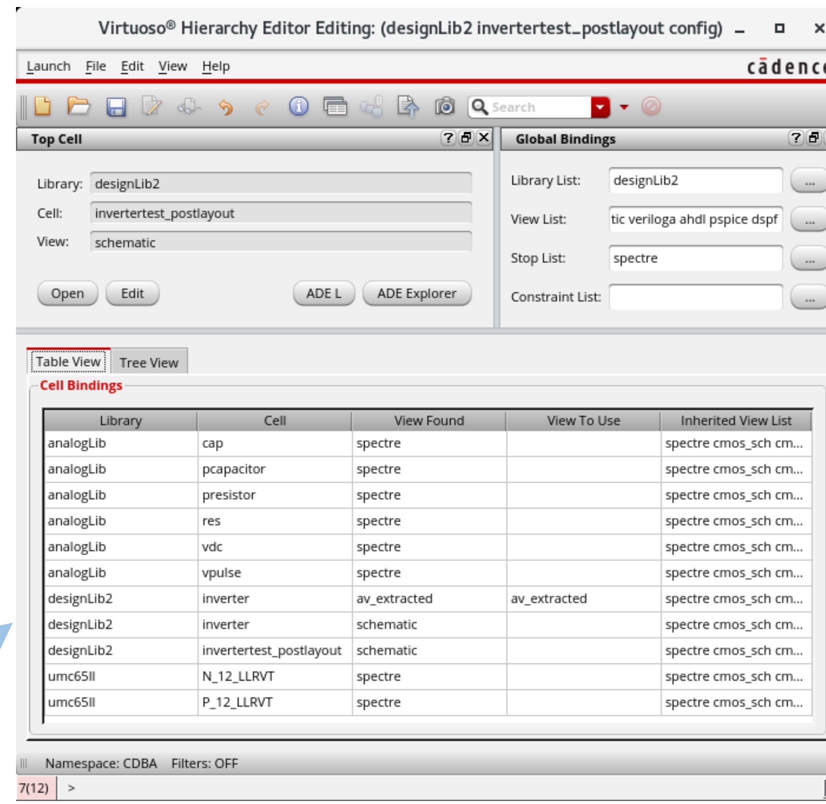


Configuration for the invertertest_postlayout schematic

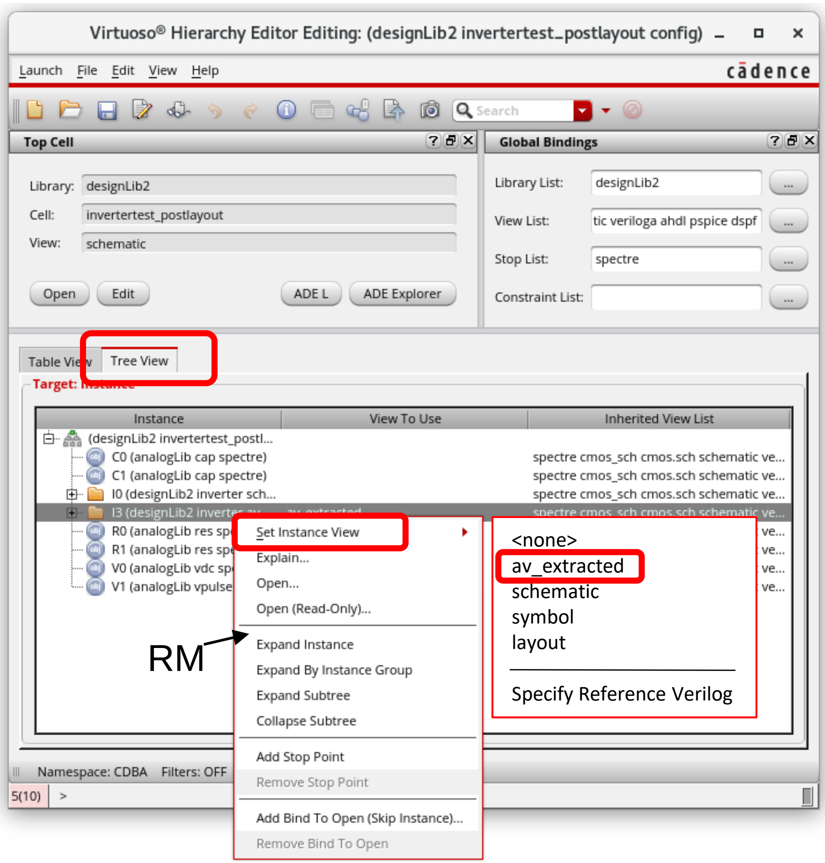




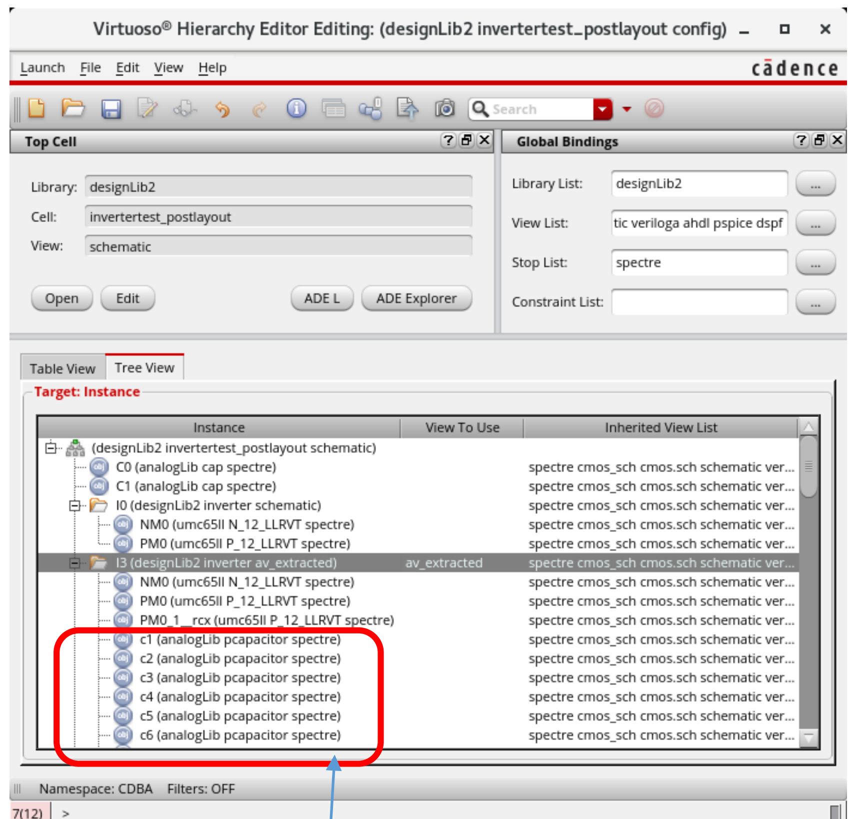
Begin here!
Template will fill the form in background, but changes the Library to mylib -> replace by your designlib name



Set one UUT instance as av_extracted (including parasitics)

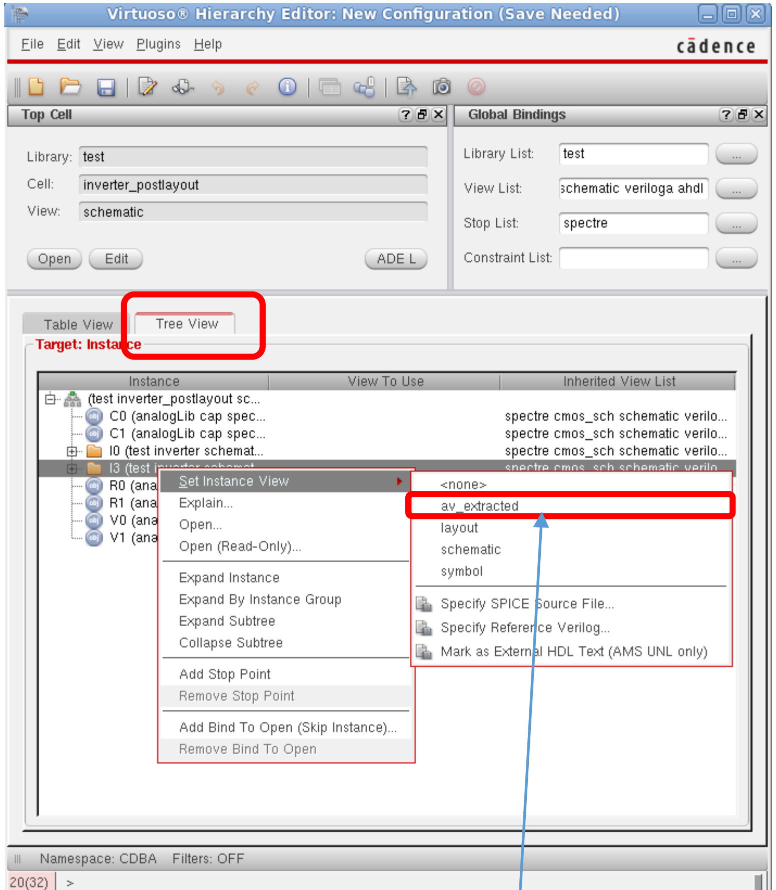


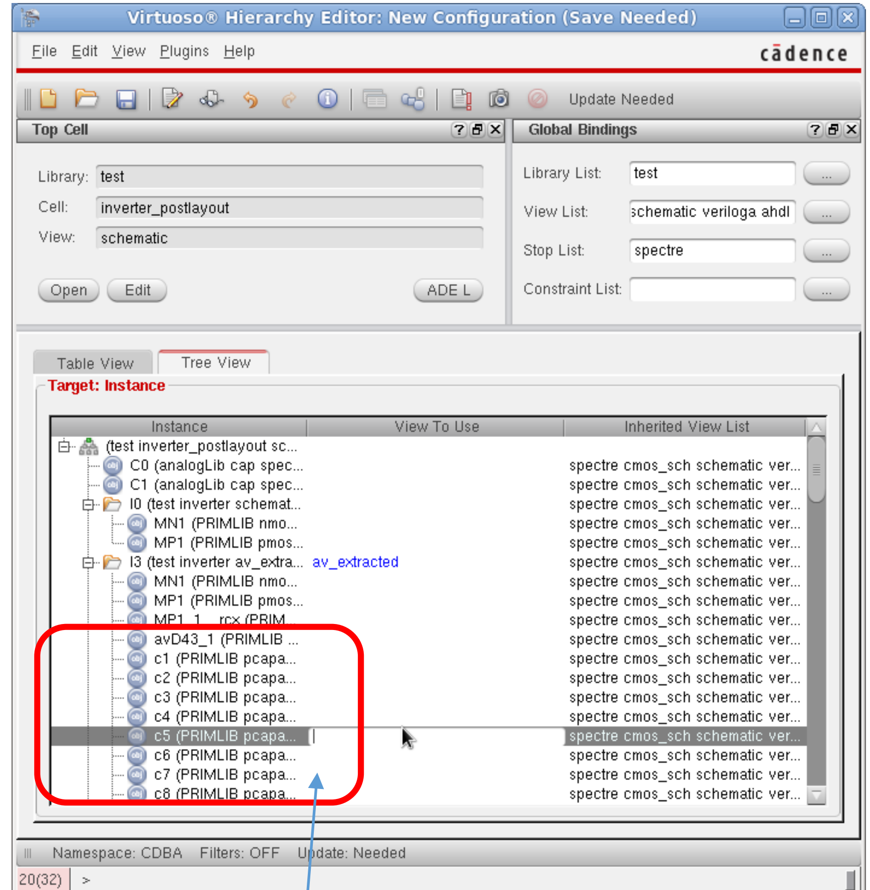
For one of the inverter instances we select "av_extracted" to use the model with parasitics



Opening the instance tree view we can see the transistors and all the parasitic elements

Cadence – Post-Layout Simulation (ADE-L/XL)





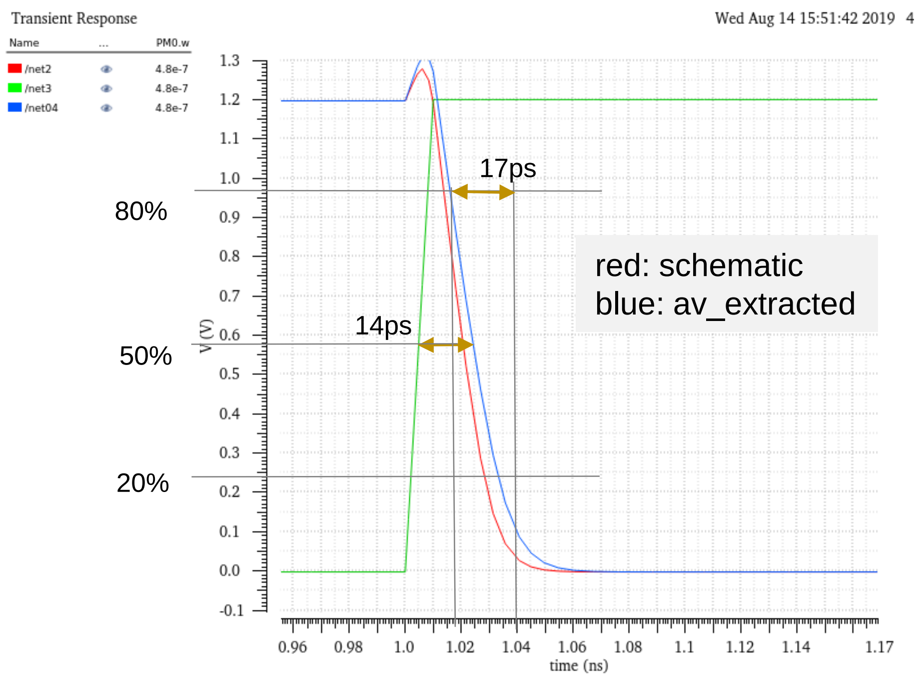
Parasitics

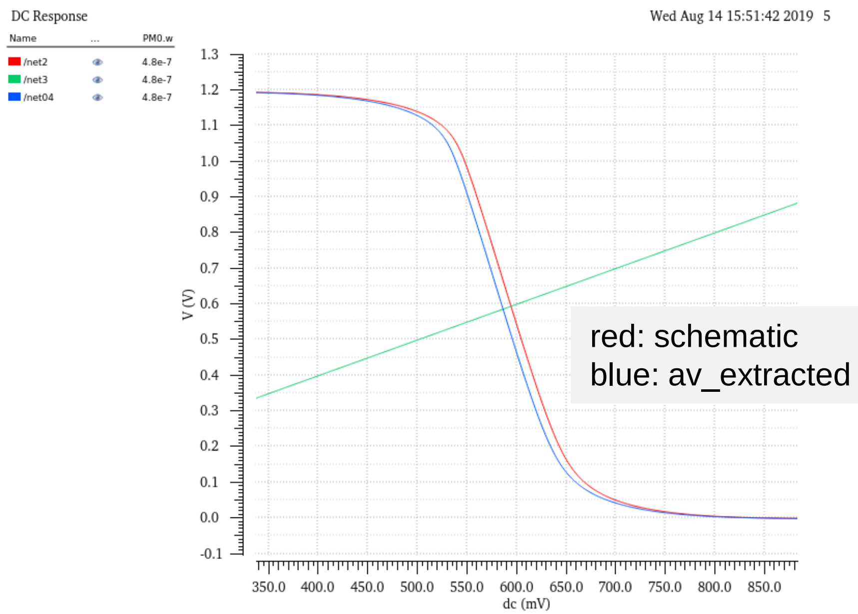
Replace (use RightMouse -> Set Instance View) schematic by av_extracted

Now we have two different models for the two Inverters.

Cadence – Post-Layout Simulation - Result

To perform simulation see slides above
from "Simulation ADE Assembler Start"





OK, delay increased by some ps

OK, marginal change

Shown for C_{Load}= 1 fF. Less difference for higher load.

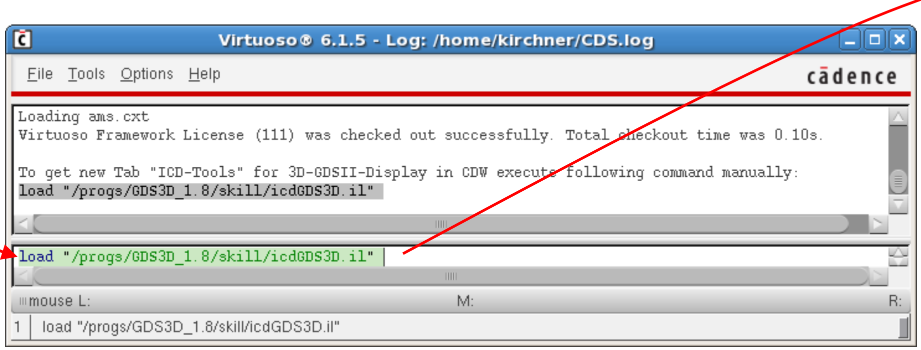
(if C_{Load} is higher parasitic capacitors have less influence)

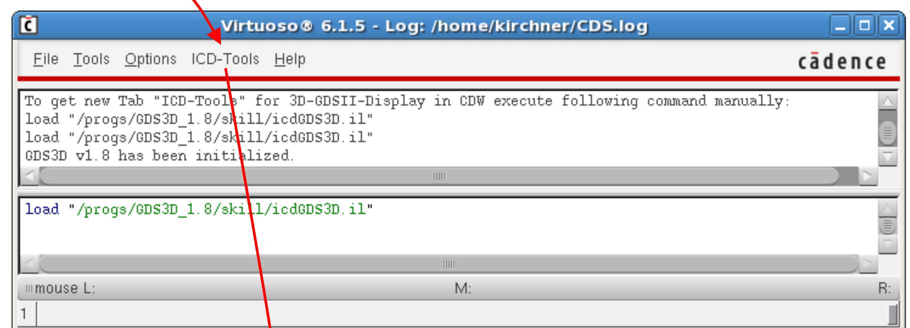
Layout 3D View

GDS3D from University of Twente, NL

Needs a technology file (layer thickness, pitches). Generated for umc65nm from process parameters.
Needs initialization for Cadence integration, but may be used standalone.
The technology library should be **"attached**" for the design library (not "referenced"). If we use "referenced" GDS3D needs a technology file named for the design library

Automatic initialization does not work yet. For manual initialization observe the last output after CIW start.

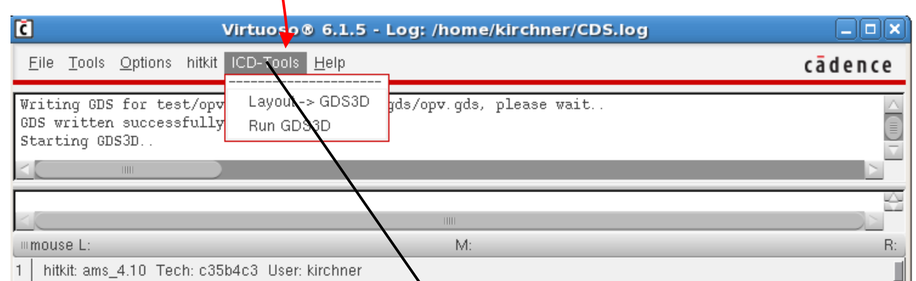




load "/progs/GDS3D_1.8/skill/icdGDS3D.il"

With a layout opened we can make a GDSII-Stream and run the GDS3D tool. ICD - Tools

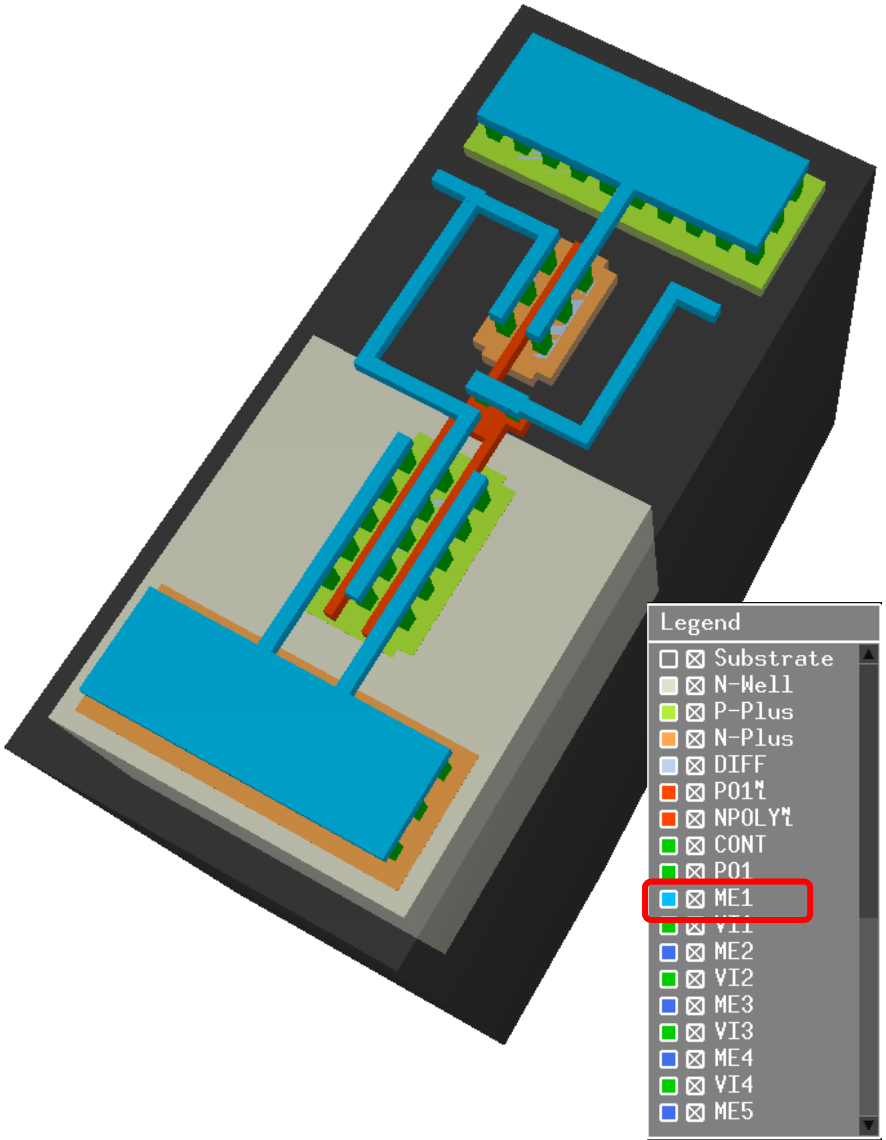
In the GDS3D-tool use F1 to see the (game like) usage.

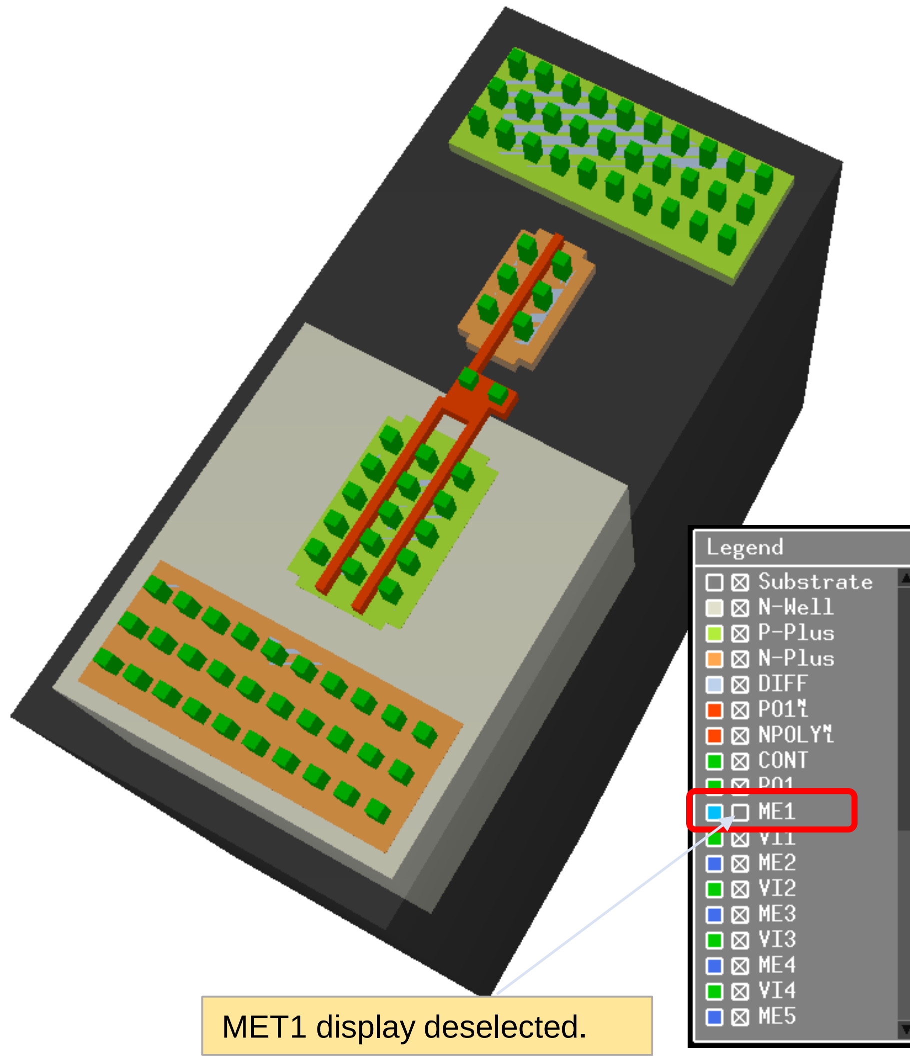


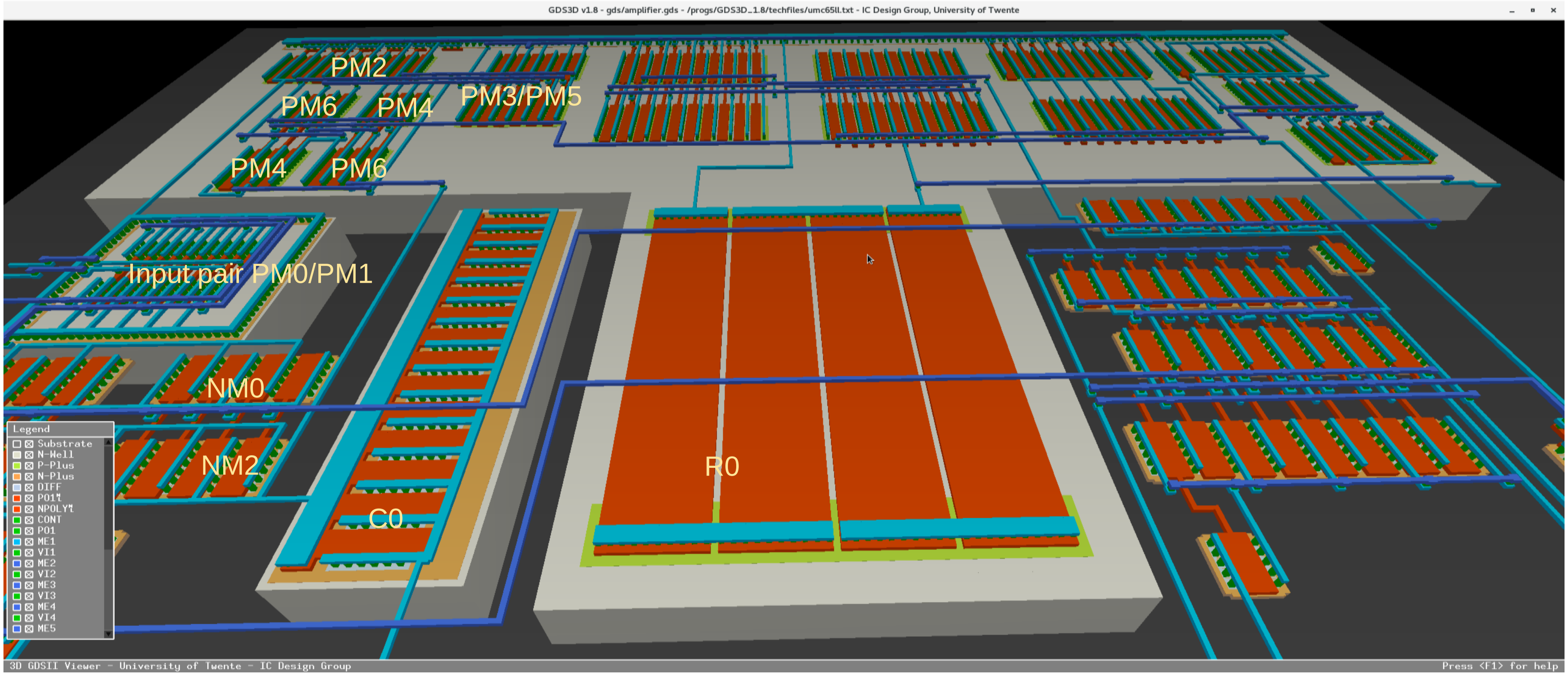
Do not use the normal streamout from CDW (File -> Export -> Stream) because GDS3D uses its own directory for the gds-File (~/GDS3D).

ICD is **I**ntegrated **C**ircuit **D**esign group at Twente University)

Layout 3D View







K.-P. Kirchner, Uni Rostock, Institut GS

Advanced Simulation

Monte Carlo simulation using matching parameter

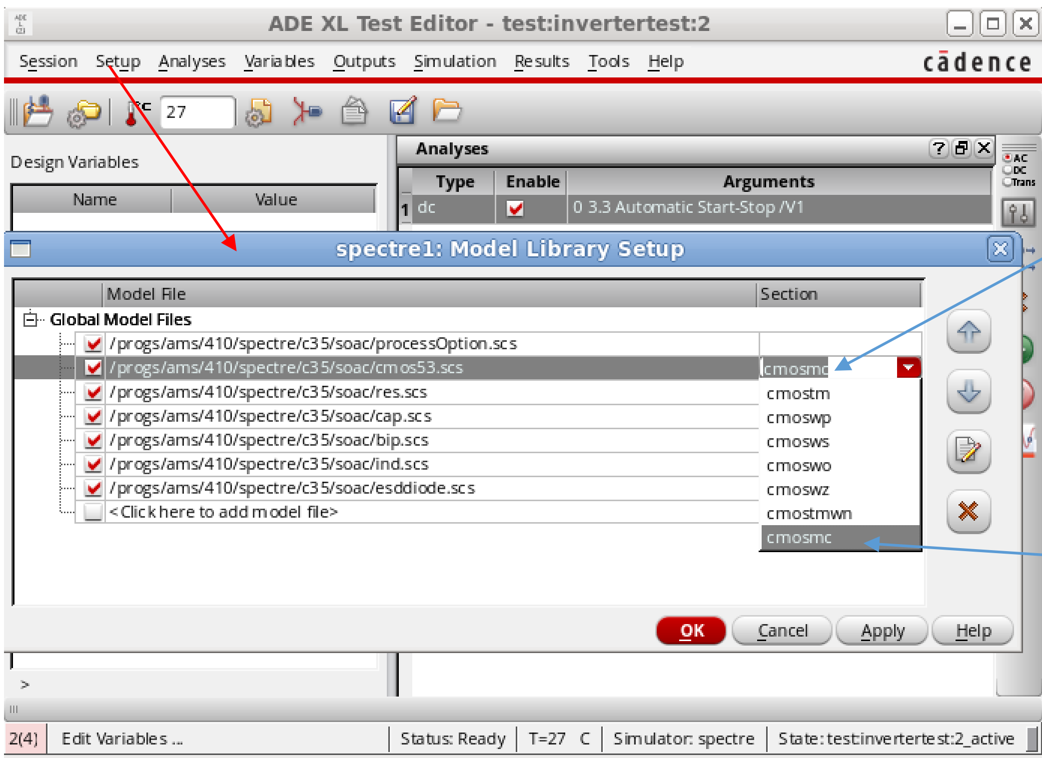
Open testbench schematic -> Launch ADE-XL -> existing view ADE-XL (assuming we have such a view already)

Use one of the tests from the normal schematic simulation

Here we use the dc-analyses to investigate the transfer characteristic.

RM over name of the test (e.g. test:name:19 → "Open Test Editor"

With Setup-> Model Libraries: Select cmosmc for the transistors (we have only transistors here we could select …mc for other pa



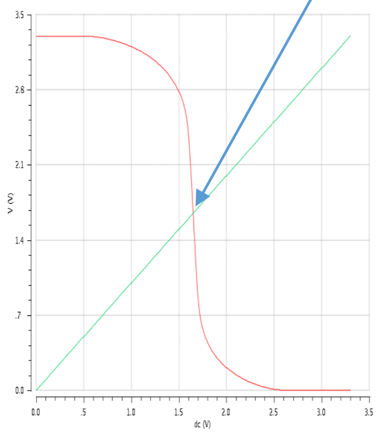
Dubbleclick here to open choices

…mc – Monte Carlo

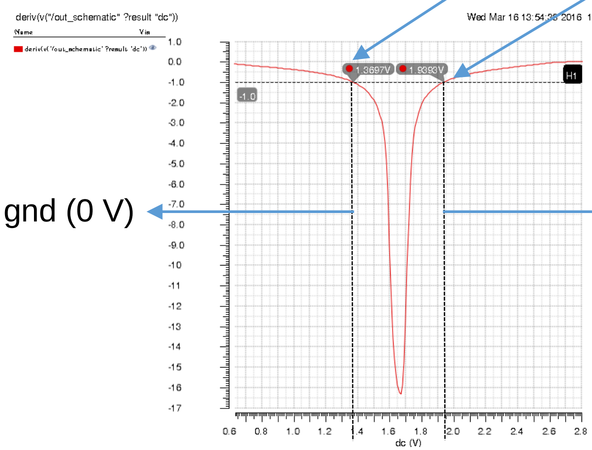
Advanced Simulation

Monte Carlo simulation using matching parameter

We want to see the switching point ($V_{out} = V_{in}$) and the noise margin for "low" and "high" output.



Difference to gnd (0 V)



Difference to vdd (3.3 V)

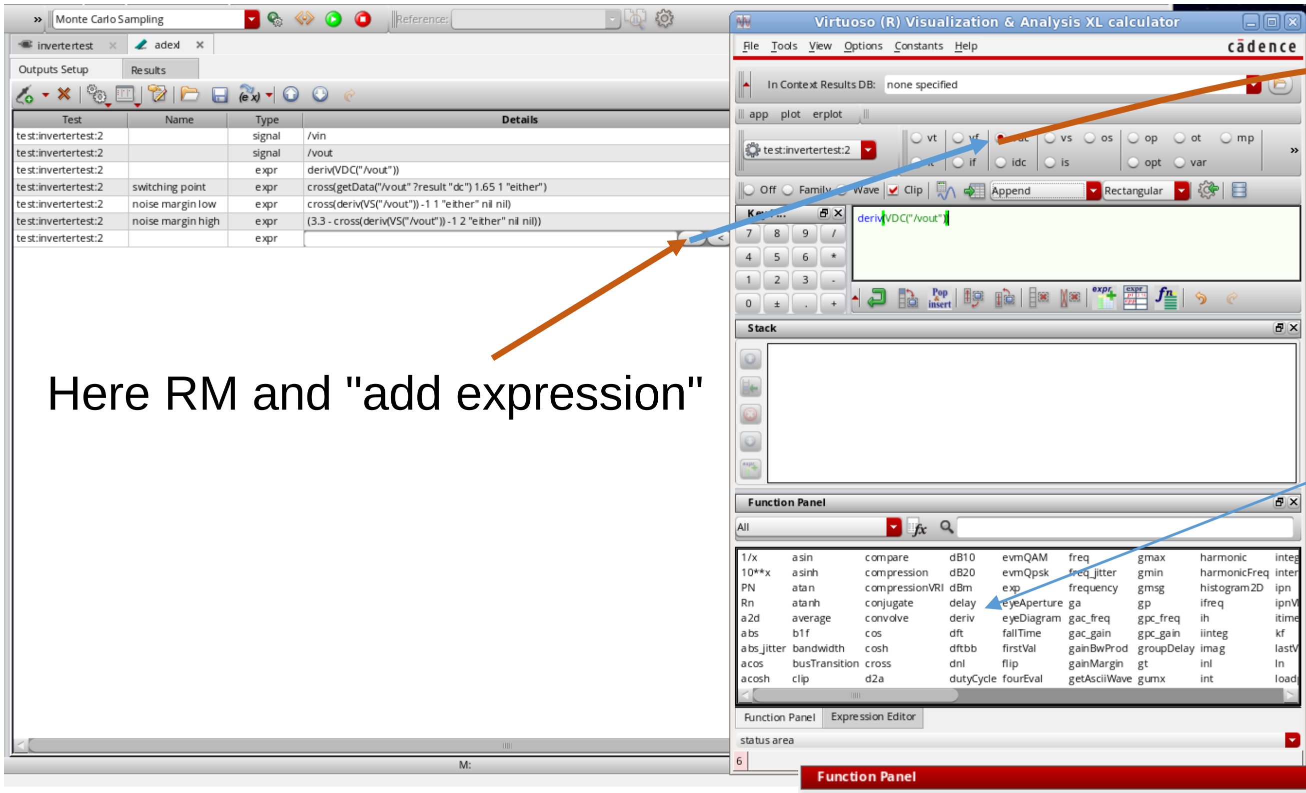
We investigate the following expressions:

Switching point: cross(VDC("/vout") 1.65 1 "either" nil)
Low logic value noise margin: cross(deriv(VDC("/vout")) -1 1 "either" nil)
High logic level noise margin: (3.3 - cross(deriv(VDC("/vout")) -1 2 "either" nil))

To set the output expressions: "add expression" - > open calculator

Advanced Simulation

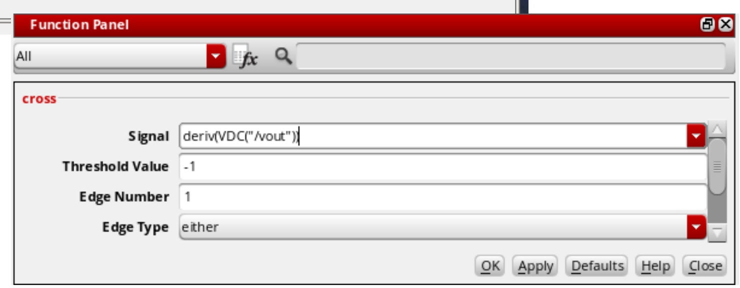
Monte Carlo simulation using matching parameter



Here RM and "add expression"

Select DC-signal vout from schematic

Use the **deriv** and **cross** functions

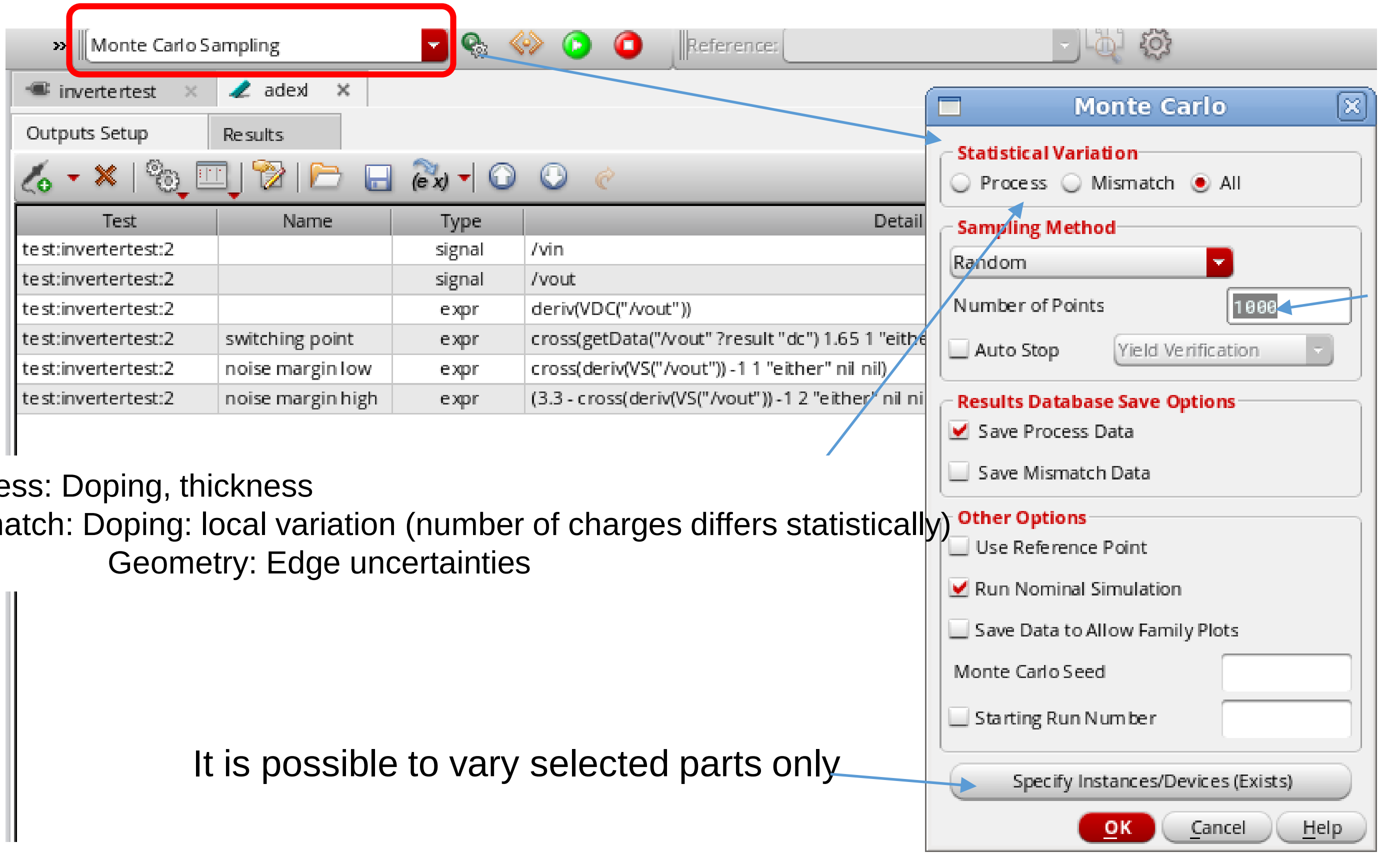


cross gives the vin-value if deriv(vout)= -1 first time

For the LOW noise margin we use the "cross"-value, for the HIGH noise margin we use 3.3-"cross"-value

Advanced Simulation

Monte Carlo simulation using matching parameter

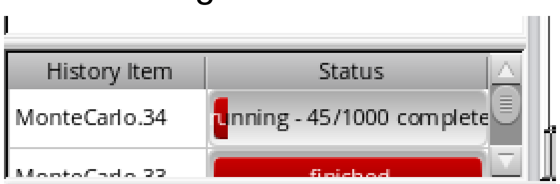


n simulation runs with random values

Process: Doping, thickness
Mismatch: Doping: local variation (number of charges differs statistically)
Geometry: Edge uncertainties

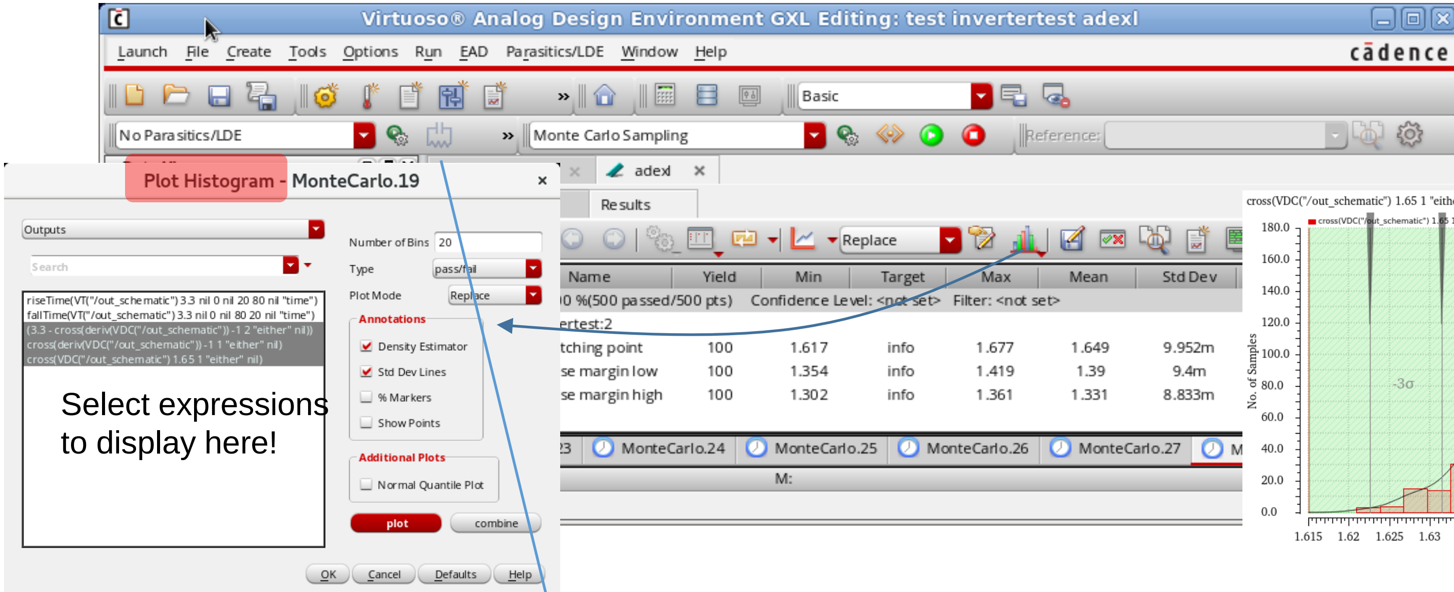
It is possible to vary selected parts only

See it running in the buttom left corner

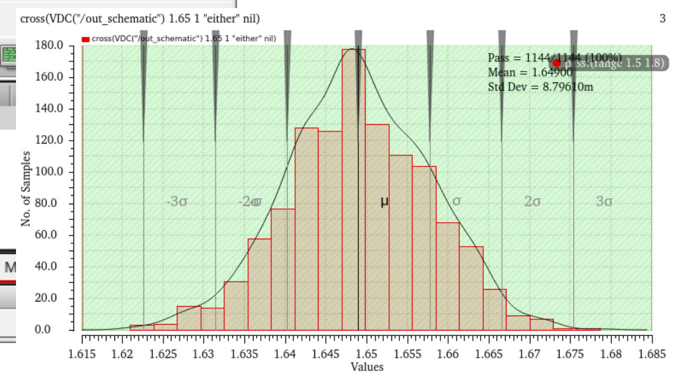


Advanced Simulation

Monte Carlo simulation using matching parameter



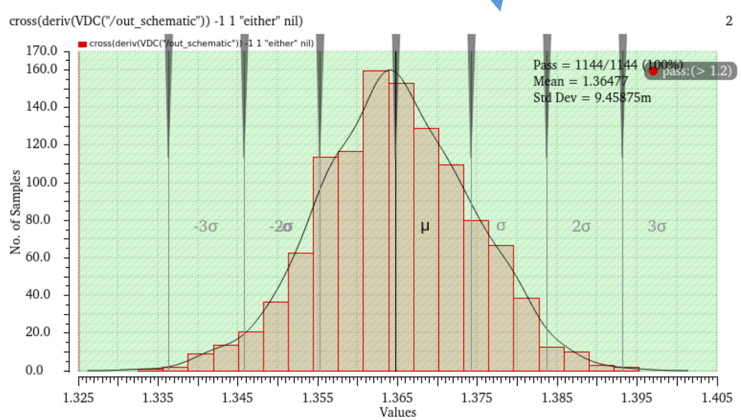
switching point



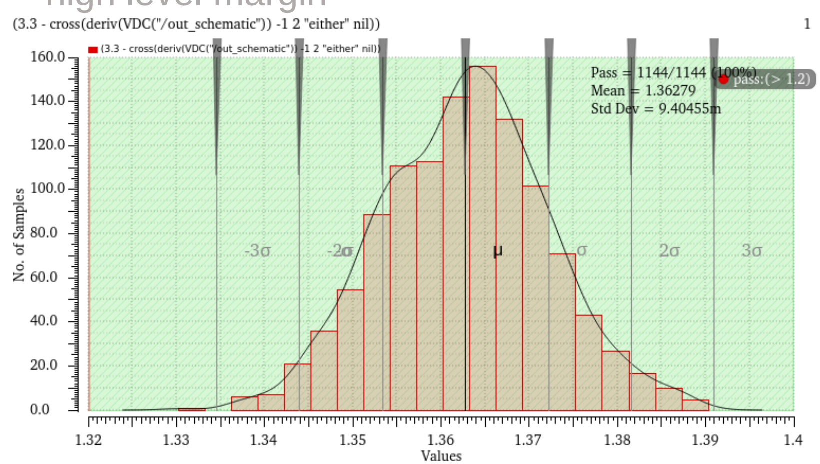
Number of intervals,
plots show 20 intervals

Result: Histogram

low level margin



high level margin

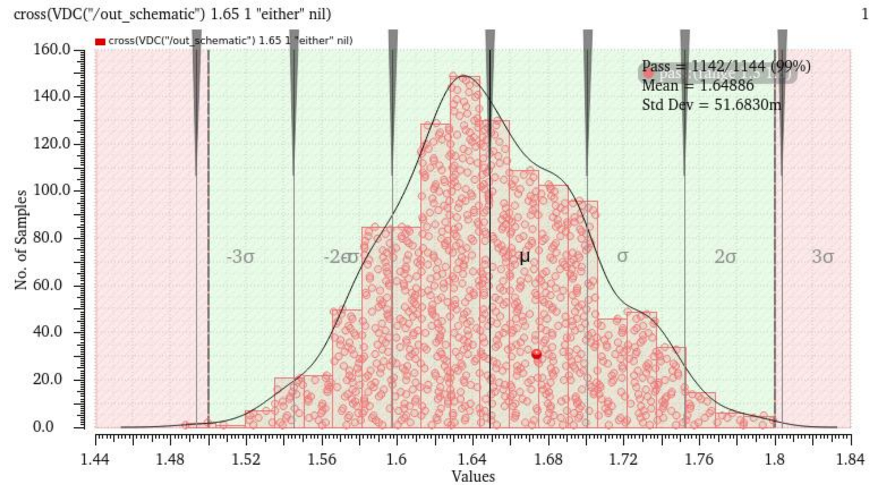


Distribution of noise margins (mismatch only)

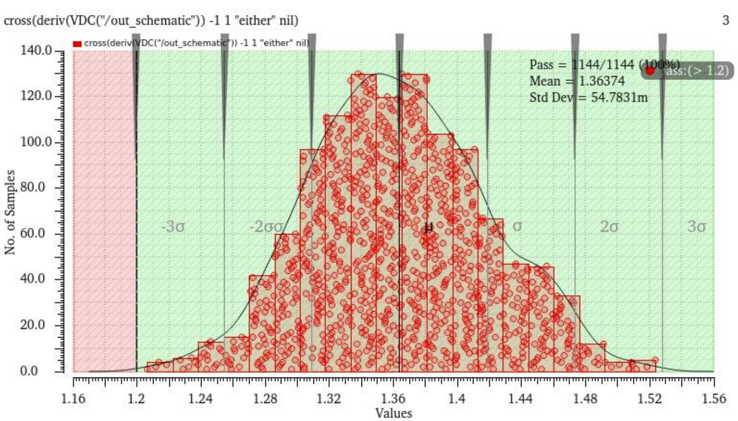
Advanced Simulation

Monte Carlo simulation using matching parameter

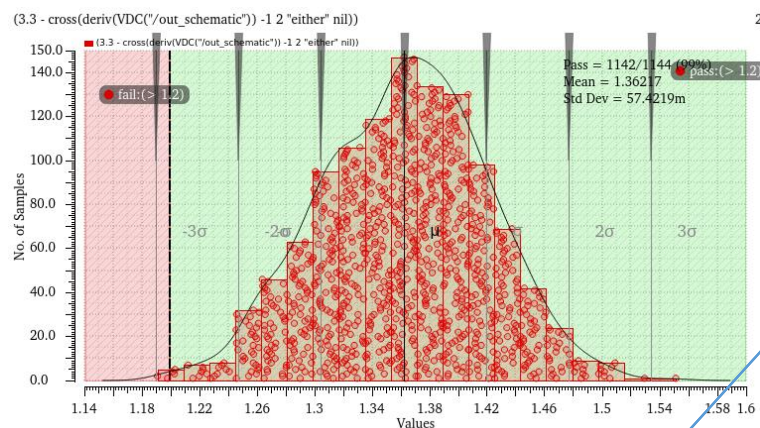
Switching Point

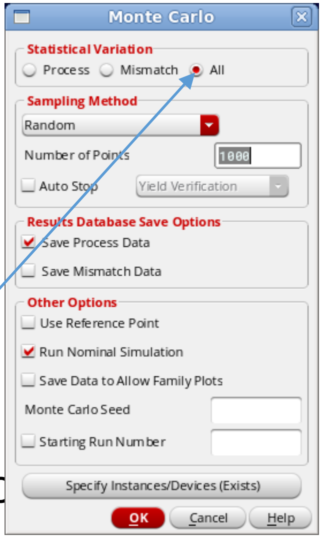


Low level margin



High level margin



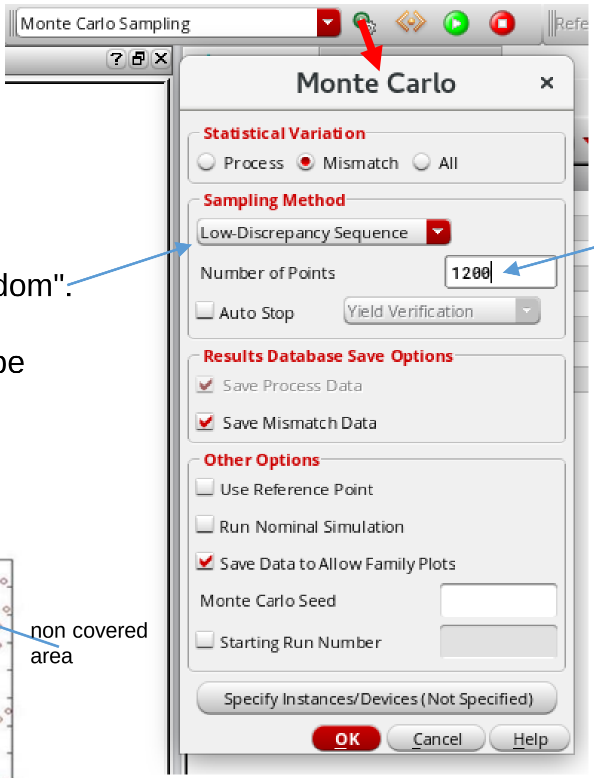


Distribution of noise margins (combined mismatch and process variat

Monte Carlo simulation and yield prediction

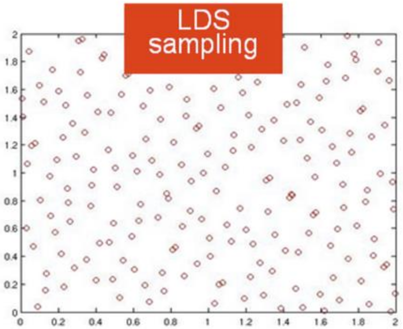
Define (realistic) specs for characteristics: Inverter: $V_{inHL}, V_{mHigh}, V_{mLow}$ ...

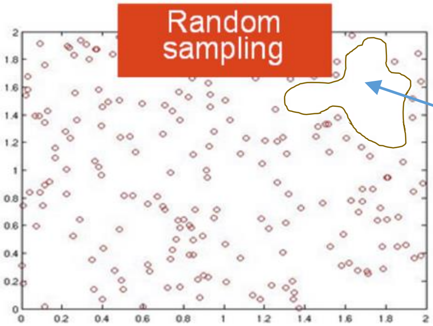
Set Options for Monte Carlo Simulation:



Recommended to
have enough san

Better than simple "Random".
Better covering of the
n-dimensional input scope





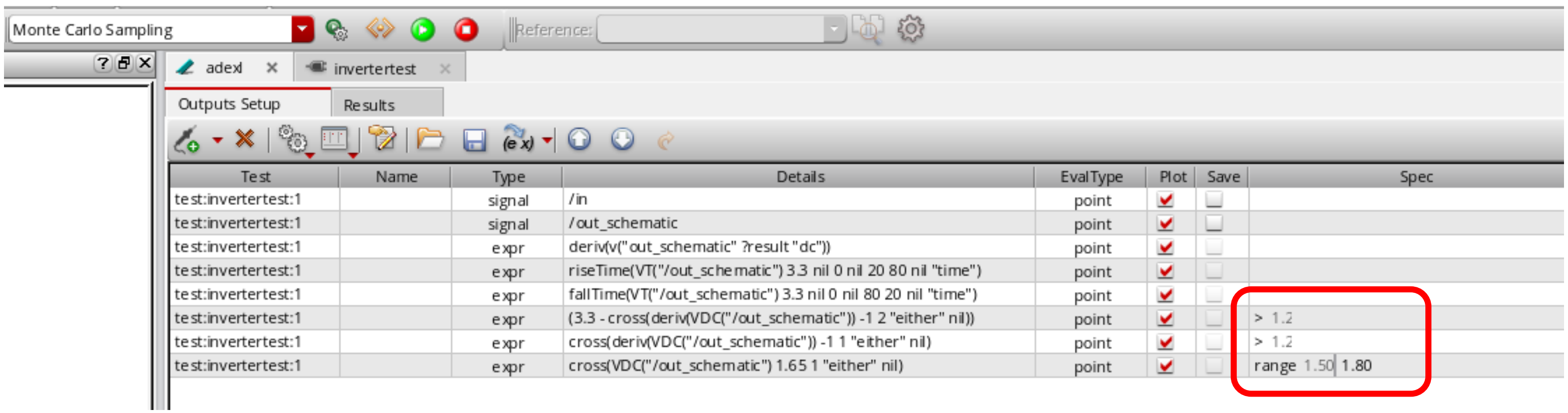
non covered
area

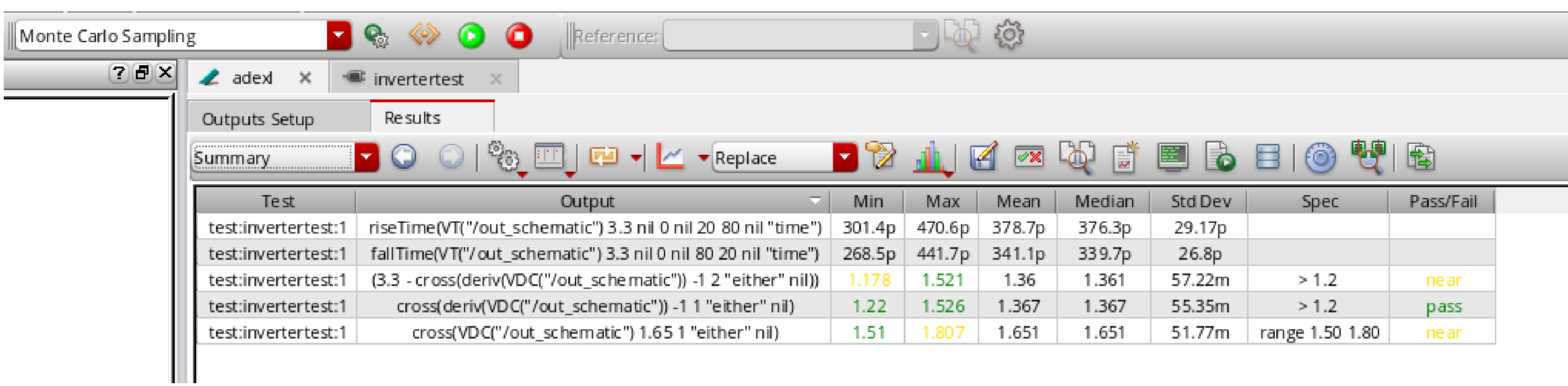
From: https://community.cadence.com/cadence_blogs_8/b/cic/posts/the-art-of-analog-design-part-2-monte-carlo-sampling

Advanced Simulation

Monte Carlo simulation and yield prediction

Set specs in adexl outputs window: low level margin > 1.2 V
high level margin > 1.2 V
1.50 V < switching voltage < 1.80 V

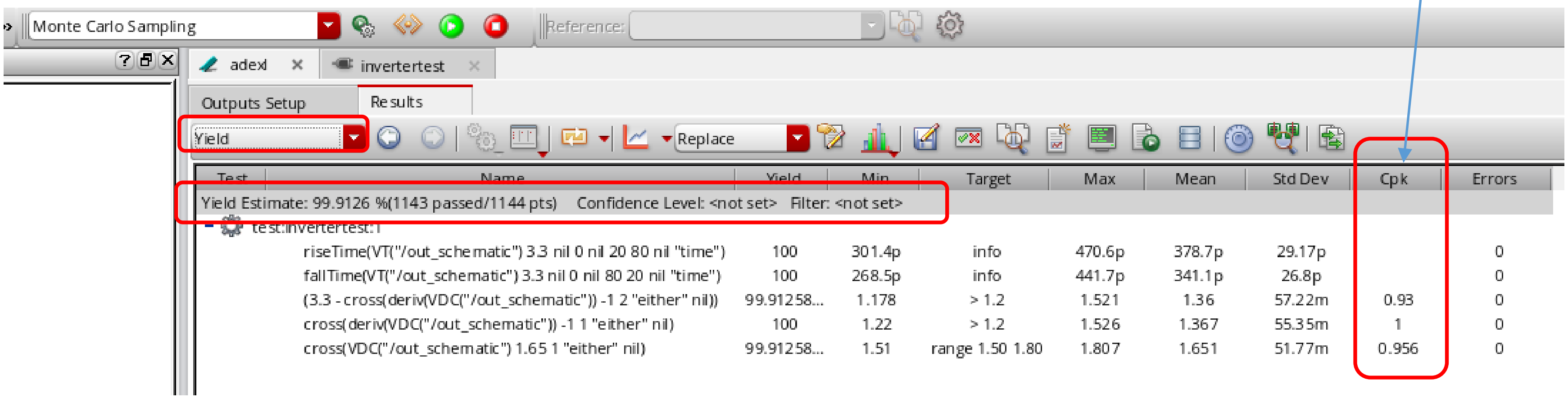




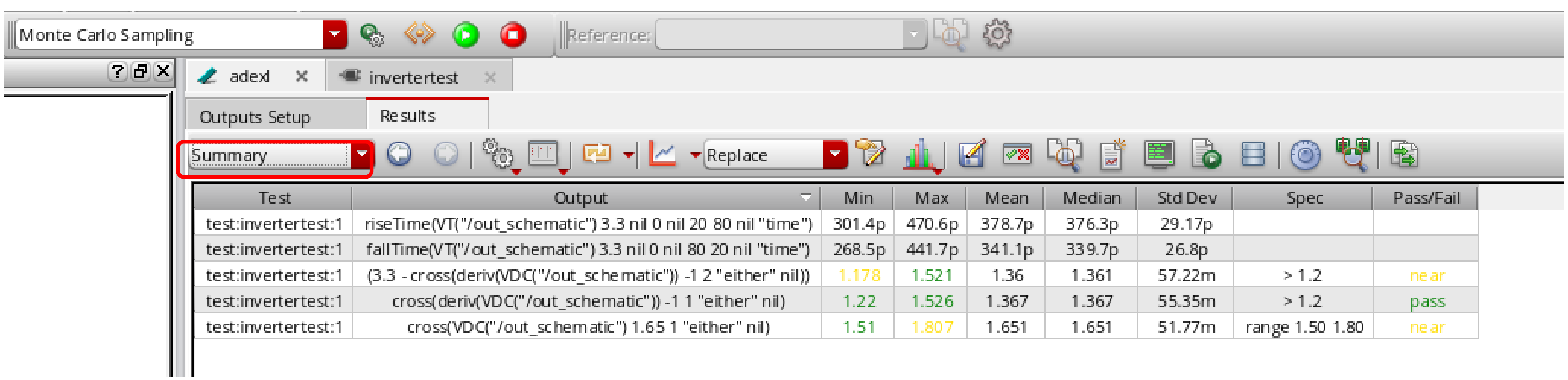
Advanced Simulation

Monte Carlo simulation and yield prediction

Next slide!



Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Yield Estimate: 99.9126 %(1143 passed/1144 pts) Confidence Level: <not set> Filter: <not set>									
test:invertertest:1									
	riseTime(VT("/out_schematic") 3.3 nil 0 nil 20 80 nil "time")	100	301.4p	info	470.6p	378.7p	29.17p		0
	fallTime(VT("/out_schematic") 3.3 nil 0 nil 80 20 nil "time")	100	268.5p	info	441.7p	341.1p	26.8p		0
	(3.3 - cross(deriv(VDC("/out_schematic")) -1 2 "either" nil))	99.91258...	1.178	> 1.2	1.521	1.36	57.22m	0.93	0
	cross(deriv(VDC("/out_schematic")) -1 1 "either" nil)	100	1.22	> 1.2	1.526	1.367	55.35m	1	0
	cross(VDC("/out_schematic") 1.65 1 "either" nil)	99.91258...	1.51	range 1.50 1.80	1.807	1.651	51.77m	0.956	0



Test	Output	Min	Max	Mean	Median	Std Dev	Spec	Pass/Fail
test:invertertest:1	riseTime(VT("/out_schematic") 3.3 nil 0 nil 20 80 nil "time")	301.4p	470.6p	378.7p	376.3p	29.17p		
test:invertertest:1	fallTime(VT("/out_schematic") 3.3 nil 0 nil 80 20 nil "time")	268.5p	441.7p	341.1p	339.7p	26.8p		
test:invertertest:1	(3.3 - cross(deriv(VDC("/out_schematic")) -1 2 "either" nil))	1.178	1.521	1.36	1.361	57.22m	> 1.2	near
test:invertertest:1	cross(deriv(VDC("/out_schematic")) -1 1 "either" nil)	1.22	1.526	1.367	1.367	55.35m	> 1.2	pass
test:invertertest:1	cross(VDC("/out_schematic") 1.65 1 "either" nil)	1.51	1.807	1.651	1.651	51.77m	range 1.50 1.80	near

Be aware that we did not change opereating environment (temperature, supply voltage

Monte Carlo simulation and yield prediction

C_p und C_{pk} Process Capability Statistics

C_p shows the relation beetwen Engeneering Tolerance (specs) to the Natural Tolerance (process) for a process with a distribution centered beetween specifications. For the Natural Tolerance the 6σ-interval $\pm 3\sigma$ is commonly used

Upper Spec Limit

Lower Spec Limit

$$C_p = \frac{USL - LSL}{6\sigma}$$

what the designer wants (specs)

what the process can deliver with high certainty

upper or lower limit only

$$C_p = \frac{USL - \mu}{3\sigma}$$

$$C_p = \frac{\mu - LSL}{3\sigma}$$

Mostly the distribution is not centered and the corrected value is used (maybe used for centered distribution as well):

Mittelwert

$$C_{pk} = \frac{minimum(USL - \mu), (\mu - LSL)}{3\sigma}$$

If $C_{pk} = 1$ then 99.73% of the samples will meet specs, for $C_{pk} = 2$: 99.9999998%

"Recommended": Values beetween 1.25 to 2.0 dependend on safety demands
> 2.5 area reduction possible? cost!

Advanced Simulation

Corner Simulation

FEOL (front end of line) Corners: Transistors n-channel , p-channel

all processes used for individual parts (transistors, resistors)

"**Fast**" means "best":

-thin oxide
-smaller length
-higher width

TT : typical n-channel- and p-channel-FETs
SS : slow n-channel-FET, slow p-channel-FET
SF : slow n-channel-FET, fast p-channel-FET
FS : fast n-channel-FET, slow p-channel-FET
FF : fast n-channel-FET, fast p-channel-FET

transistors affected in sim

Corners affect not only speed but also transfer characteristics (dc-dc)

BEOL (back end of line) Corners: Wires

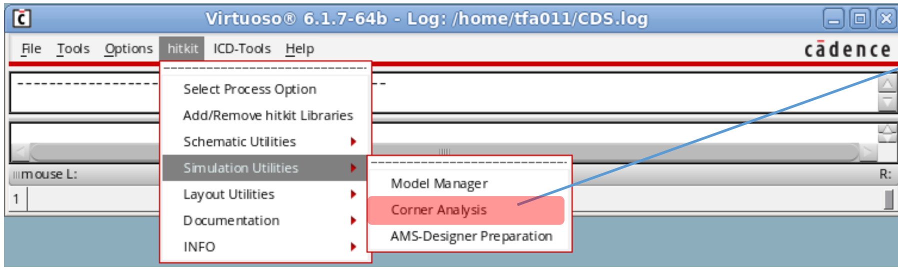
all processes after making parts -> wires, vias

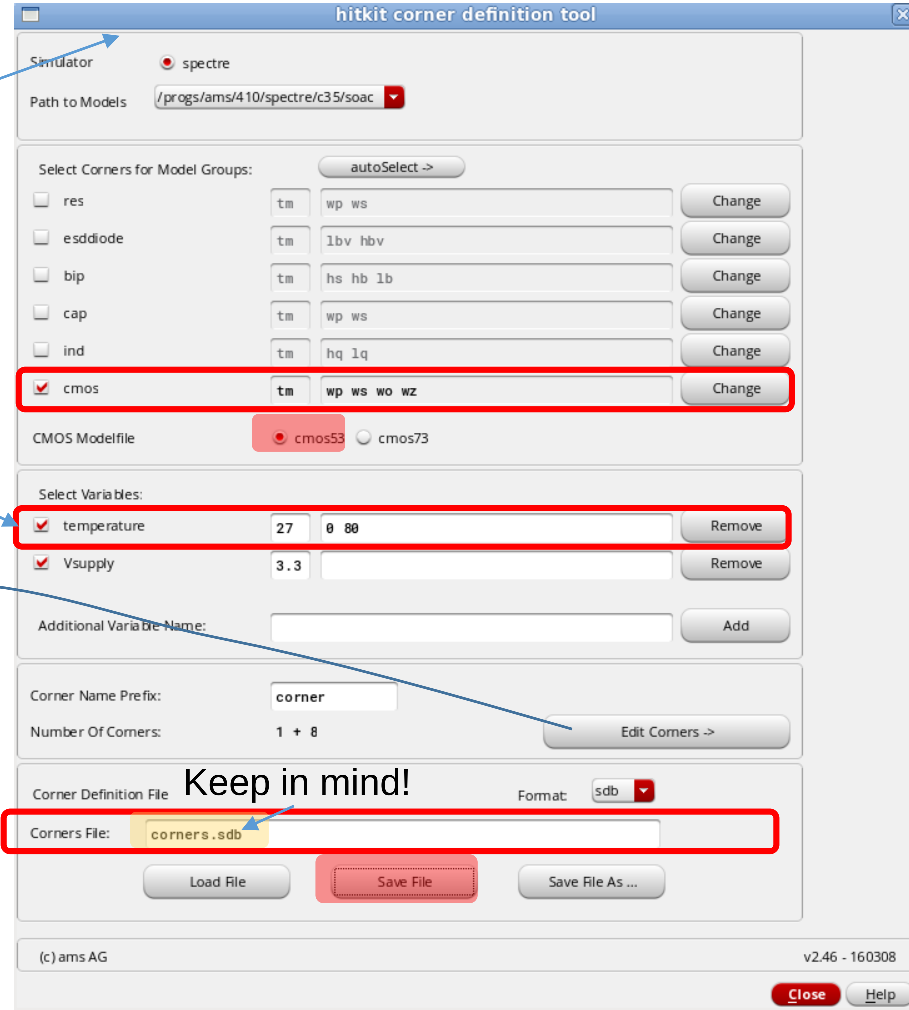
Temperature, Voltage

ams (and others):

TM: typical values - corresponds to TT
WS : worst slow – slow n and slow p corresponds to SS
WP: worst power – fast n and fast p corresponds to FF
WO : worst One – slow n, fast p corresponds to SF
WZ: worst Zero – fast n, slow p corresponds to FS

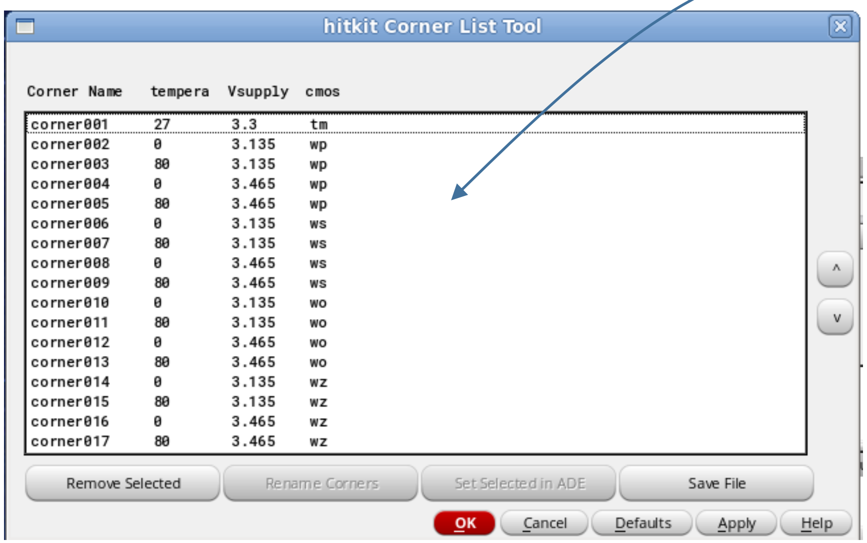
Corner Simulation – Corner Preparation





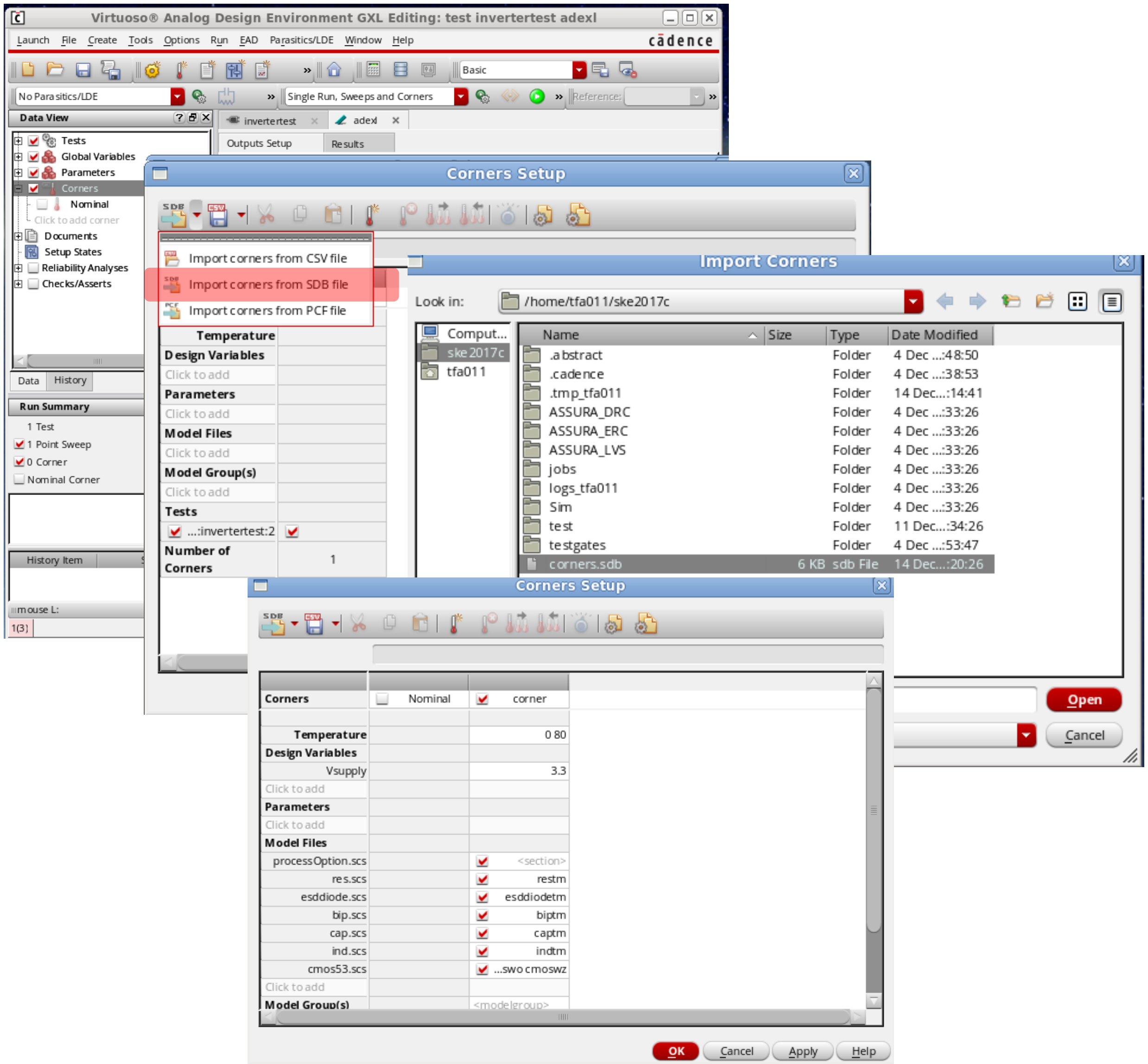
Modification of temperature

Keep in mind!

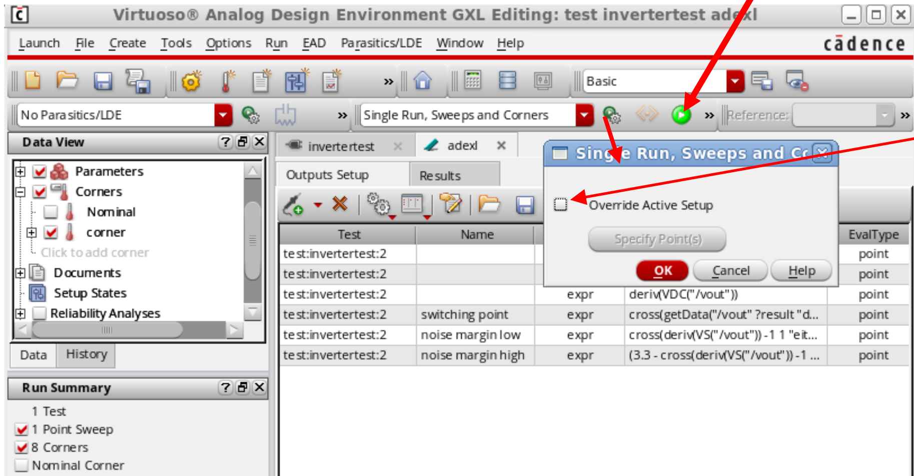


Here we can see all the selected corners

Corner Simulation – Import Corners



Corner Simulation – Run and Results

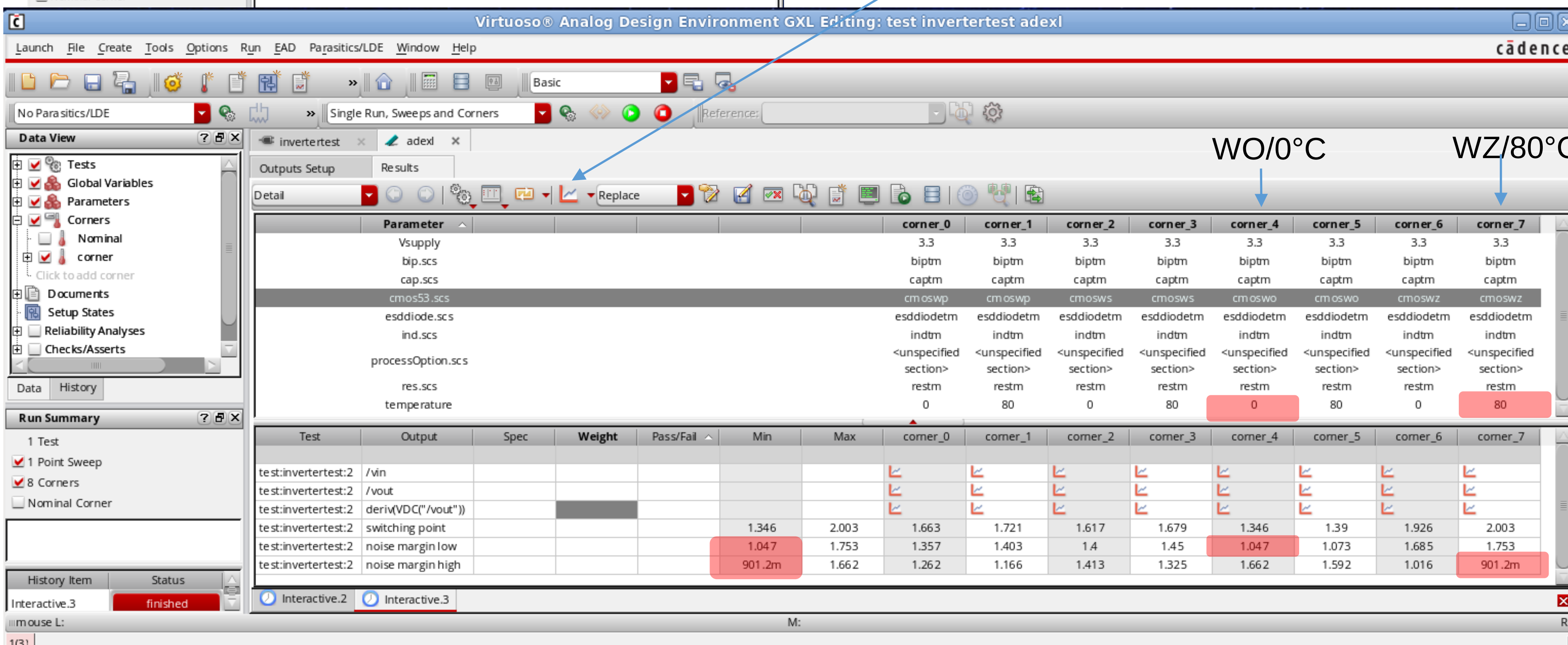


Do NOT select!

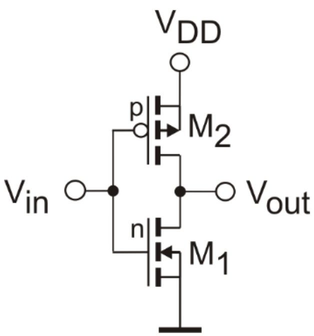
Press to see waveforms!
Not shown here.

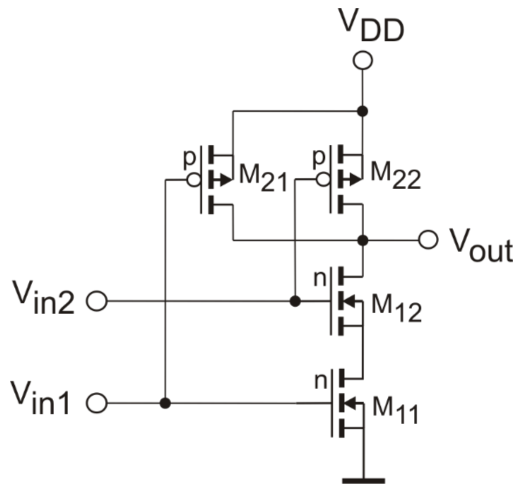
WO/0°C

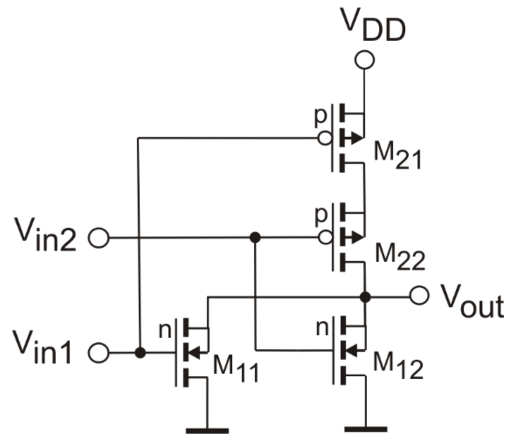
WZ/80°C



Lab: Circuit Design using Cadence and AMS CMOS 0.35µ Technology   Inverter, 2-Input-NAND, 2-Input-NOR







$V_{DD} = 3.3\ \mathrm{V}$
$V_{thn} = 0.48\ \mathrm{V}$
$V_{thp} = 0.67\ \mathrm{V}$
$KP_n = 170\ \dfrac{\mu\mathrm{A}}{\mathrm{V}^2}$
$KP_p = 58\ \dfrac{\mu\mathrm{A}}{\mathrm{V}^2}$

$(t_{ox} = 7.6\ \mathrm{nm})$

Calculate the geometry in ($W/_L$-ratio for p- und n-transistors) so that you get a good compromise for the switching point and noise margin for the cases:

- Both inputs change at the same moment (connected)
- One input is preassigned to a fixed value so that the second one causes output change. (May be different for both inputs!)

Switching point should be near $\frac{V_{DD}}{2}$ for all cases.

Design (schematic, layout) a 2-Input-NAND or a 2-Input-NOR using short channel length transistors (0.35µ) or long channel length transistors (3.5µm]. Selection of the specific design is done with the tutor.

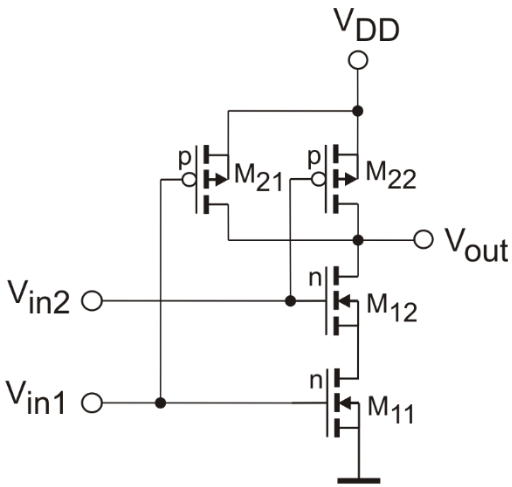
Cell hight shall be 20 µm.

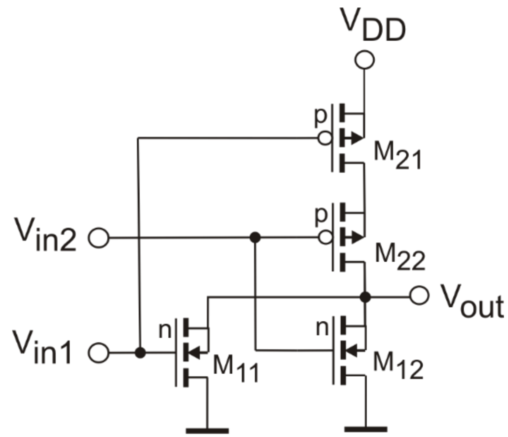
From Simulation (including postlayout) find out the following values:
Switching points and noise margin for the different cases at the two inputs.
Delay times (50%), falltime, risetime (20%-80%) dependend on load capacitance (4.. 100 fF) and temperature (-40 to 80°C

Lab: Circuit Design using Cadence and UMC 65 nmTechnology
Inverter, 2-Input-NAND, 2-Input-NOR

$V_{DD} = 1.2$ V
$V_{thn} = 0.36$ V
$V_{thp} = 0.36$ V
$KP_n = 440\ \frac{\mu A}{V^2}$
$KP_p = 150\ \frac{\mu A}{V^2}$

$(t_{ox} = 2.6$ nm$)$





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$(t_{ox} = 2.6$ nm$)$

Calculate the geometry in ($^{W}/_{L}$-ratio for p- und n-transistors) so that you get a good compromise for the switching point and noise margin for the cases:
- Both inputs change at the same moment (connected)
- One input is preassigned to a fixed value so that the second one causes output change. (May be different for both inputs!)

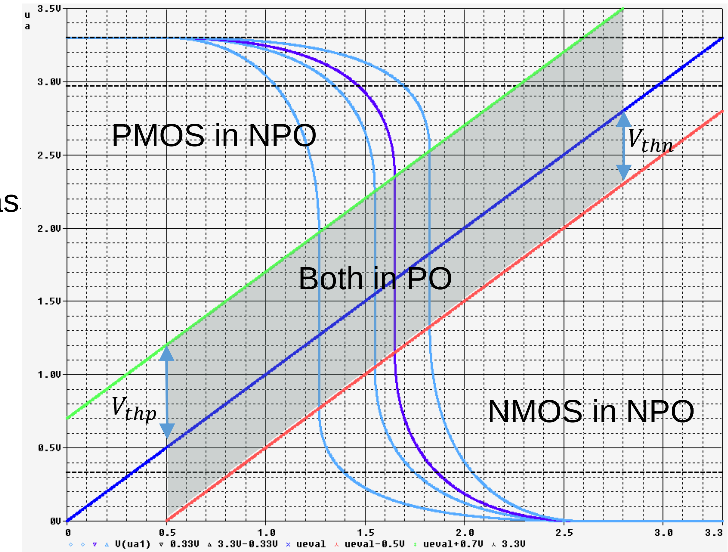
Switching point should be near $\frac{V_{DD}}{2}$ for all cases.

Design (schematic, layout) a 2-Input-NAND or a 2-Input-NOR using short channel length transistors (0.35µ) or long channel length transistors (3.5µm]. Selection of the specific design is done with the tutor.

Cell hight shall be 2.4 µm.

From Simulation (including postlayout) find out the following values:
Switching points and noise margin for the different cases at the two inputs.
Delay times (50%), falltime, risetime (20%-80%) dependend on load capacitance (1.. 50 fF) and temperature (-40 to 80°C)

Lab: Circuit Design using Cadence and UMC 65nm CMOS 0.35µ Technology
Inverter, 2-Input-NAND, 2-Input-NOR



V_{inSP}

At the switching point both transistor-pairs are in the pinch-off state and pas by the same current :

$$\frac{\beta_n}{2}(V_{inSP} - U_{thn})^2 = \frac{\beta_p}{2}(V_{DD} - V_{inSP} - V_{thp})^2$$

But: There is a difference in the number of involved switching transistors for the different input cases!

Common Equation for **CMOS-Inverter**:

$$\frac{\beta_n}{2}(V_{inSP} - V_{thn})^2 = \frac{\beta_p}{2}(V_{DD} - V_{inS} - V_{thp})^2$$

Dissolve for V_{inSP}:

$$\sqrt{\beta_n}(V_{inSP} - V_{thn}) = \sqrt{\beta_p}(V_{DD} - V_{inSP} - V_{thp})$$

$$V_{inSP} = \frac{\sqrt{\frac{\beta_n}{\beta_p}} \cdot V_{thn} + V_{DD} - V_{thp}}{1 + \sqrt{\frac{\beta_n}{\beta_p}}}$$

Für $V_{inSP} = \frac{V_{DD}}{2}$

$$\frac{\beta_n}{\beta_p} = \left(\frac{\frac{V_{DD}}{2} - V_{thp}}{\frac{V_{DD}}{2} - V_{thn}}\right)^2$$

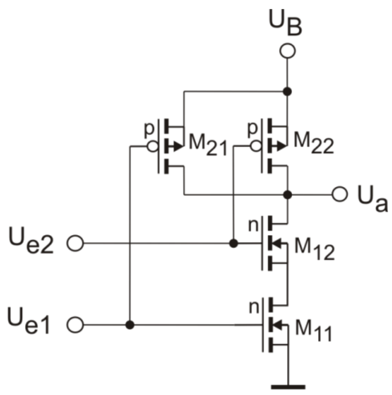
$$\beta = KP \cdot \frac{W}{L}$$

$$mit: \; KPN = 3 \cdot \mathrm{KPP}$$

$$\frac{\left[\frac{W}{L}\right]_n}{\left[\frac{W}{L}\right]_p} = \frac{1}{3} \cdot \left(\frac{\frac{V_{DD}}{2} - V_{thp}}{\frac{V_{DD}}{2} - V_{thn}}\right)^2$$

$$\frac{\left[\frac{W}{L}\right]_p}{\left[\frac{W}{L}\right]_n} \approx 3$$

NAND-Gate



For $V_{inSP} = \frac{V_{DD}}{2}$ when both inputs change:

$$\frac{\beta_{nges}}{\beta_{pges}} = \left(\frac{\frac{V_{DD}}{2} - V_{thp}}{\frac{V_{DD}}{2} - V_{thn}}\right)^2$$

$\beta_{nges} = KPN \cdot \frac{1}{2}\left[\frac{W}{L}\right]_n$ One transistor of double length

$\beta_{pges} = KPP \cdot 2\left[\frac{W}{L}\right]_p$ One transistor of double width

$KPN = 3 \cdot \mathrm{KPP}$

$$\frac{\left[\frac{W}{L}\right]_n}{\left[\frac{W}{L}\right]_p} = \frac{4}{3} \cdot \left(\frac{\frac{V_{DD}}{2} - V_{thp}}{\frac{U_B}{2} - V_{thn}}\right)^2$$

$$\frac{\left[\frac{W}{L}\right]_n}{\left[\frac{W}{L}\right]_p} = \frac{4}{3} \cdot \left(\frac{\frac{V_{DD}}{2} - V_{thp}}{\frac{V_{DD}}{2} - V_{thn}}\right)^2 = \frac{4}{3}\left(\frac{0.6 - 0.36}{0.6 - 0.36}\right)^2 = 0.908 = 1.33$$

$$\frac{\left[\frac{W}{L}\right]_p}{\left[\frac{W}{L}\right]_n} = 0.75$$

Lab: Circuit Design using Cadence and AMS CMOS 0.35µ Technology

2-Input-NAND, 2-Input-NOR

$$\frac{\beta_n}{2}(V_{inSP} - V_{thn})^2 = \frac{\beta_p}{2}\left(V_{DD} - V_{inSP} - V_{thp}\right)^2$$

$$\sqrt{\beta_n}(V_{inSP} - V_{thn}) = \sqrt{\beta_p}\left(V_{DD} - V_{inSP} - V_{thp}\right)$$

One p-transistor is off and remains off.
One n-Transistor is on and remains on.
Only an inverter is switching.

Lower transistor in NAND during switching phase:

$$V_{DS11} = \frac{I_D}{\beta_N(V_B - V_{GSon})} = \frac{\frac{\beta_N}{2}(V_{eS} - V_{GSon})^2}{\beta_N(V_B - V_{GSon})} = \frac{(V_{eS} - V_{GSon})^2}{2(V_B - V_{GSon})} = \frac{(0.6 - 0.36)^2}{2 \cdot (1.2 - 0.36)} = 0{,}034\ \mathrm{V}$$

$$V_{eS} = \frac{\sqrt{\frac{\beta_n}{\beta_p}} \cdot (V_{thn} + V_{DS11}) + V_{thn_B} - V_{thp}}{1 + \sqrt{\frac{\beta_n}{\beta_p}}}$$

See if the values are acceptable with regard to noise margin.
Try to change W/L ratios to find a compromise.

For $V_{inSP} = \frac{V_{DD}}{2}$

$$\beta_{nges} = KPN \cdot \frac{W}{L}$$

$$\beta_{pges} = KPN \cdot \frac{W}{L}$$

$$KPN = 3 \cdot \mathrm{KPP}$$

$$\frac{\beta_n}{\beta_p} = \left(\frac{\frac{V_{DD}}{2} - V_{thp}}{\frac{V_{DD}}{2} - V_{thn} - V_{DS11}}\right)^2$$

$$\frac{\left[\frac{W}{L}\right]_n}{\left[\frac{W}{L}\right]_p} = \frac{1}{3} \cdot \left(\frac{\frac{V_{DD}}{2} - V_{thp}}{\frac{V_{DD}}{2} - V_{thn} - V_{DS11}}\right)^2$$

$$\frac{\left[\frac{W}{L}\right]_n}{\left[\frac{W}{L}\right]_p} = \frac{1}{3} \cdot \left(\frac{0.6 - 0.36}{0.6 - 0.36 - 0.034}\right)^2 = 0.45 \qquad \frac{\left[\frac{W}{L}\right]_p}{\left[\frac{W}{L}\right]_n} = 2.2$$

$$\frac{\left[\frac{W}{L}\right]_n}{\left[\frac{W}{L}\right]_p} = \frac{1}{3} \cdot \left(\frac{0.6 - 0.36}{0.6 - 0.36}\right)^2 = 0.33 \qquad \frac{\left[\frac{W}{L}\right]_p}{\left[\frac{W}{L}\right]_n} = 3$$

Interesting, but we can implement one geometry only and must configure a compromise.

Lab: Circuit Design using Cadence and UMC 65nm Technology
2-Input-NAND, 2-Input-NOR

$$V_{inSP} = \frac{\sqrt{\frac{\beta_n}{\beta_p}} \cdot (V_{Thn} + V_{DS11}) + V_{DD} - V_{thp}}{1 + \sqrt{\frac{\beta_n}{\beta_p}}}$$

Using the Relation for switching both transistors ${}^{W}/_{L} - relation = 1.1$:

KPN/KPP

$$V_{inSPu} = \frac{\sqrt{\frac{3}{1.1}} \cdot 0.5\ \text{V} + 3.3\ \text{V} - 0.7\ \text{V}}{1 + \sqrt{\frac{3}{1,1}}} = 1.29\ \text{V}$$

$$\left[\frac{W}{L}\right]_p \Big/ \left[\frac{W}{L}\right]_n$$

$$V_{inSPo} = \frac{\sqrt{\frac{3}{1.1}} \cdot 0.5\ \text{V} + 0.236\ \text{V} + 3.3\ \text{V} - 0.7\ \text{V}}{1 + \sqrt{\frac{3}{1,1}}} = 1.38\ \text{V}$$

Not considered:
Conducting NMOS has a resistance ($U_{DS} > 0$)
Body-/Backgate-Effect (U_{Th} dependent of subtrate voltage)

${}^{W}/_{L} - Relation$PMOS/NMOS	V_{inSP} when switching both inputs [Volts] and difference to V_{DD}	V_{inSP} only "upper NMOS" switching [Volts] and difference to V_{DD}	V_{inSP} only "lower NMOS" switching [Volts] and difference to V_{DD}
1.1	1.65 / 1.65	1.36 / 1. 94	1.30 / 2.00
2.2	1.83 / 1.47	1.55 / 1.75	1.47 / 1.83
3.3	1.92 / 1.38	1.66 / 1.64	1.58 / 1.72
4.4	1.98 / 1.32	1.74 / 1.56	1.66 / 1.64

Switching input voltage $\boldsymbol{V_{inSP}}$ for different ${}^{W}/_{L}$ - relations of the transistors from **Simulation**

Lab: Circuit Design using Cadence and AMS CMOS 0.35µ Technology

2-Input-NAND, 2-Input-NOR

Practical hint to safe time:

Starting from the inverter schematic:

You may open the inverter schematic as the origin for the new design.

Then create a new schematic cellview (from CDW or Library Manager). **File -> New -> Cellview**

When you have the new empty schematic and the inverter schematic side by side you can copy the inverter circuit or parts of it to the new schematic.

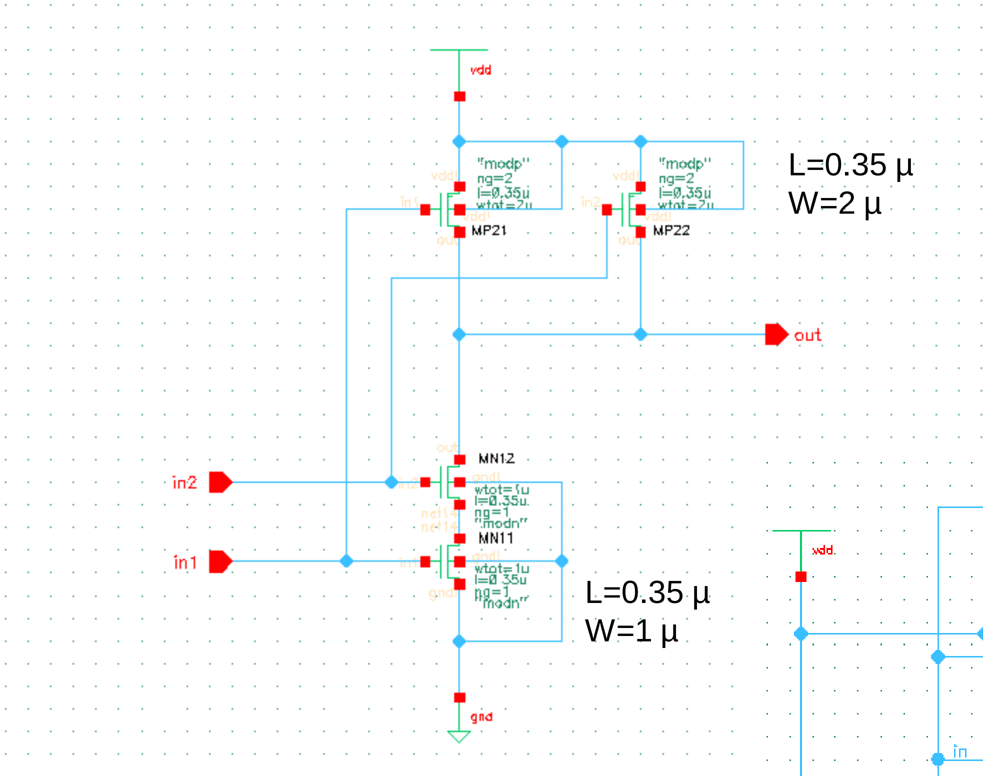
Use copy: "**c**" -> select the desired part using the left mouse button
-> click on the selected part
-> drag to the new place (may be the same schematic window or the new one)

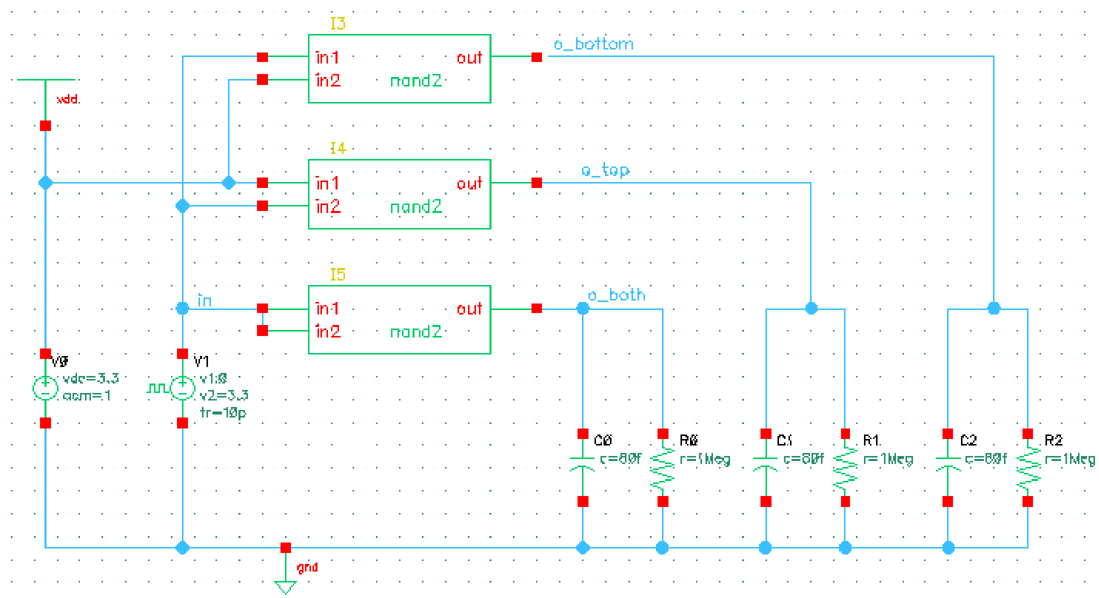
Copy object from one schematic window to another: In target no command should be active! (Use ESC sometimes!)

You may proceed in the same manner for the testbench.

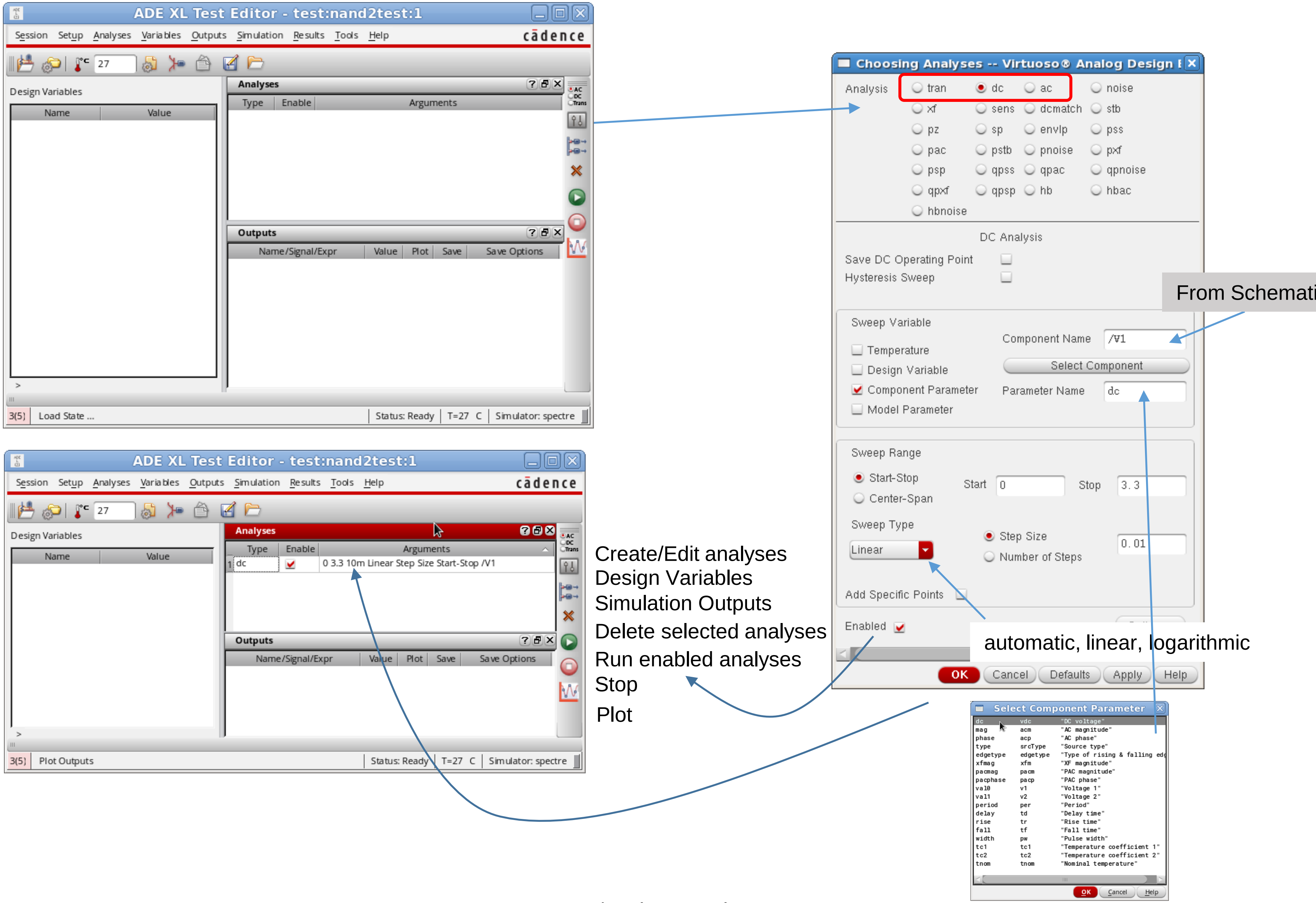
Edit the new schematic to build the NAND or NOR design.

2-input-NAND Schematic and associated Testbench

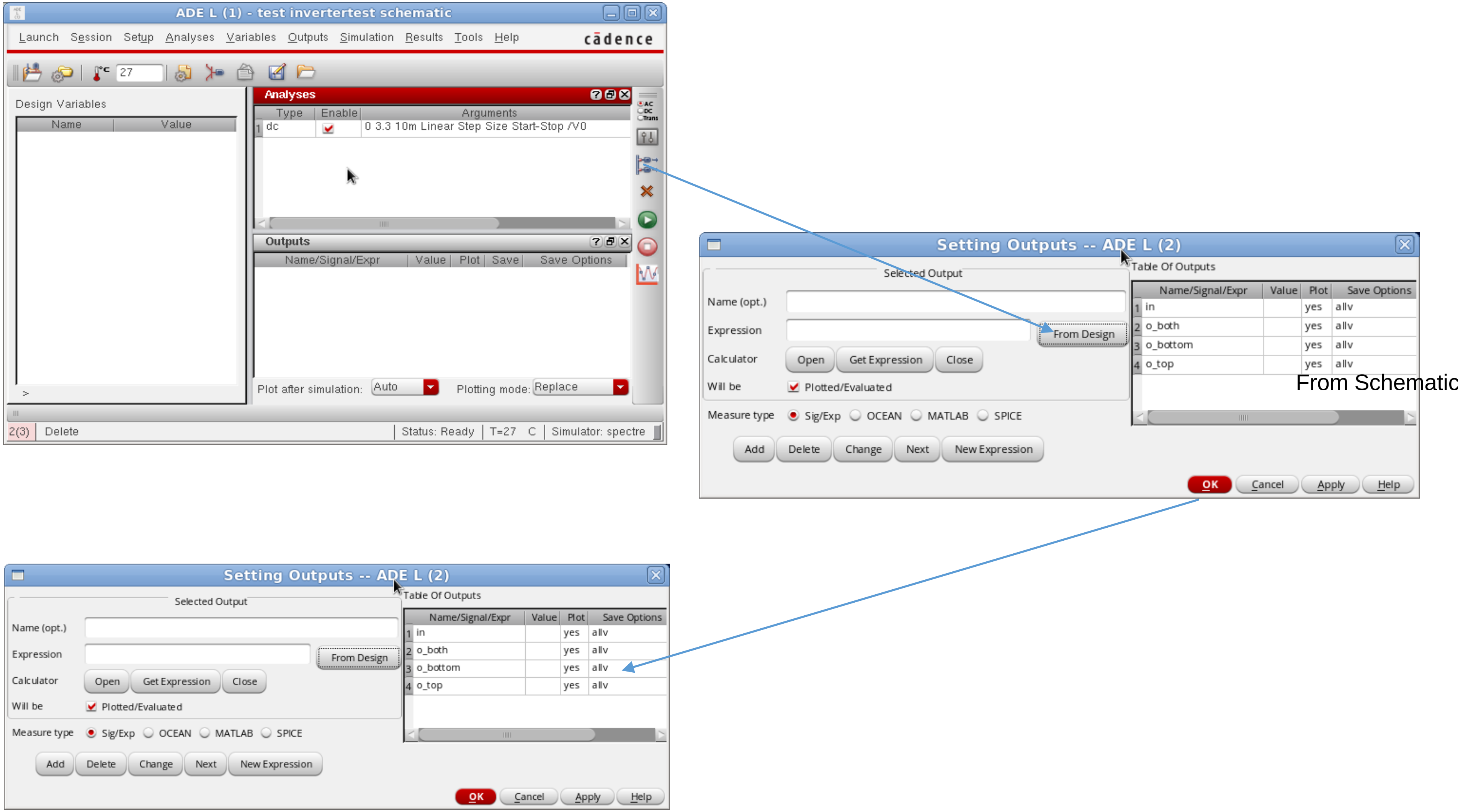




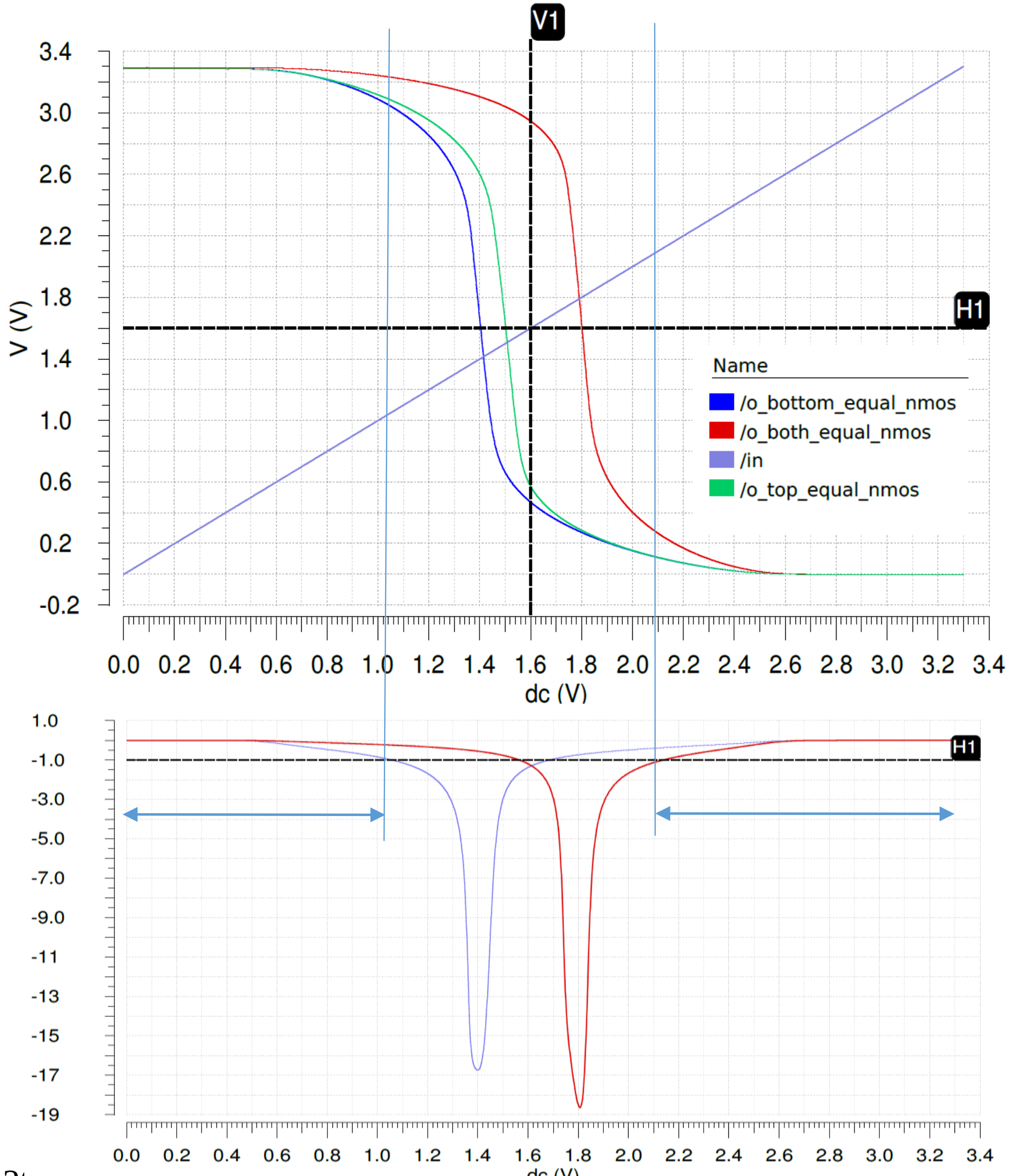
2-Input-NAND, 2-Input-NOR – Use steps as before in inverter simulation



2-Input-NAND, 2-Input-NOR – Use steps as before in inverter simulation

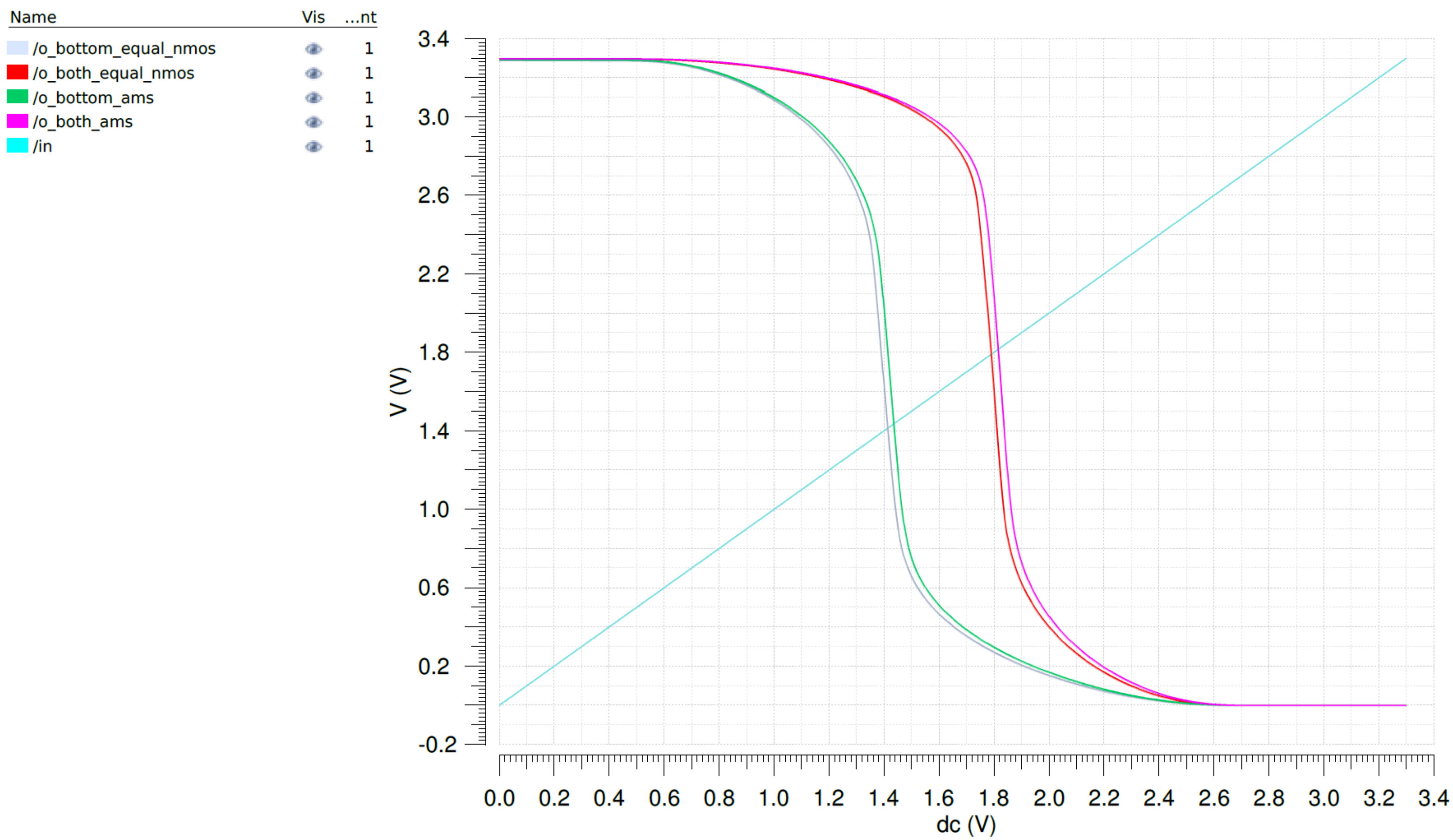


Transfer characteristics and noise margin for 2-Input-CMOS-NAND-Gate



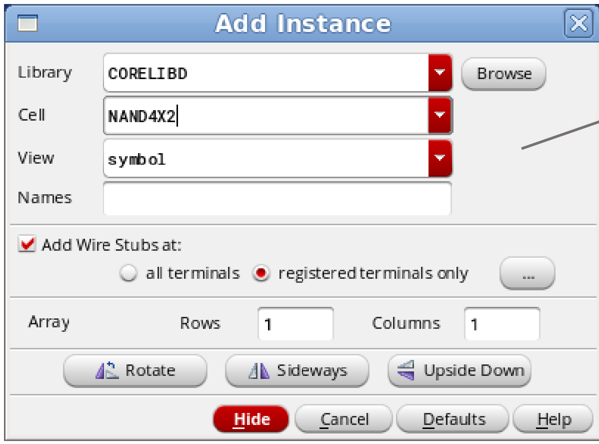
$$\frac{dV_{out}}{dV_{in}}$$

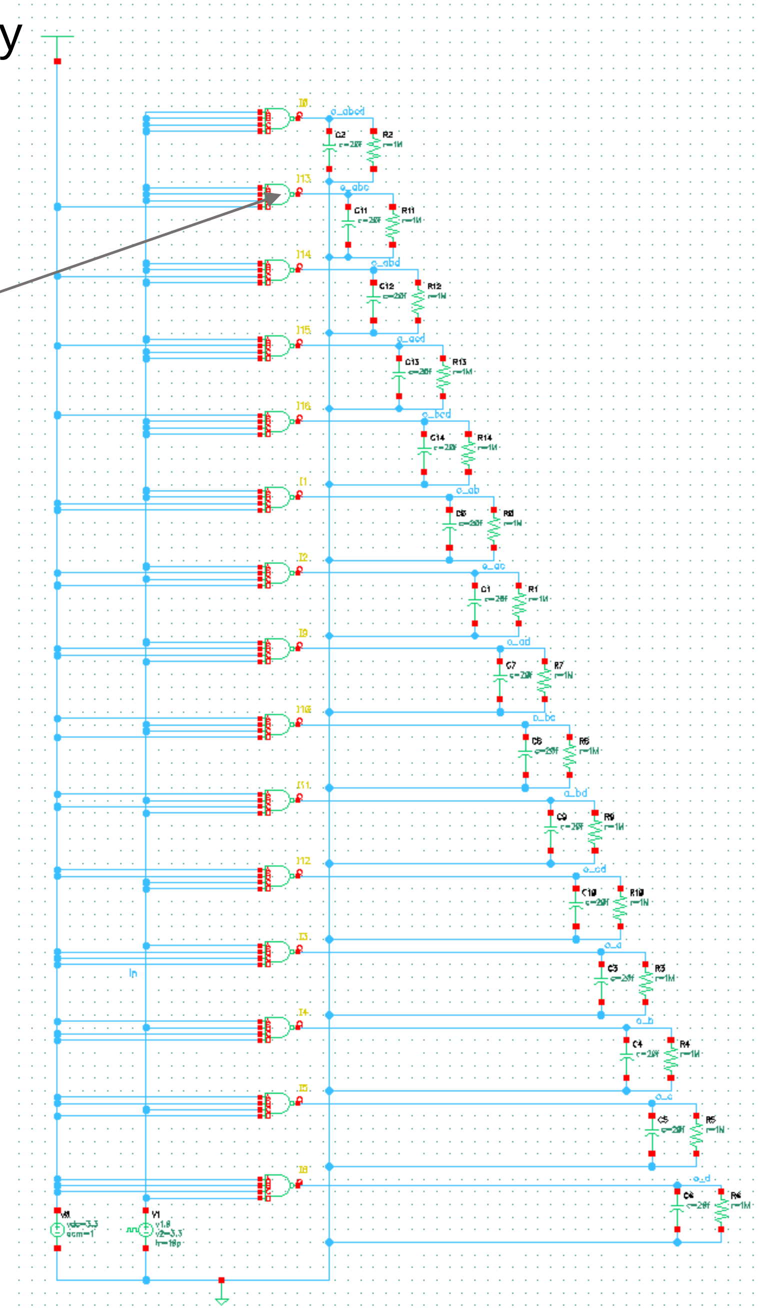
Transfer characteristic for 2-Input-CMOS-NAND-Gates



Transfer characteristics of 2-input-NAND-Gates
from AMS-0.35µ-library with $L_{NMOS,PMOS} = 0.35\mu$, $W_{NMOS} = 1.325\mu, W_{PMOS} = 2.1\mu$
and
self-made gate with $L_{NMOS,PMOS} = 0.35\mu$, $W_{NMOS} = 1.0\mu, W_{PMOS} = 1.6\mu$
(Characteristics for switching top N-transistor only not shown, always between bottom and both cases)

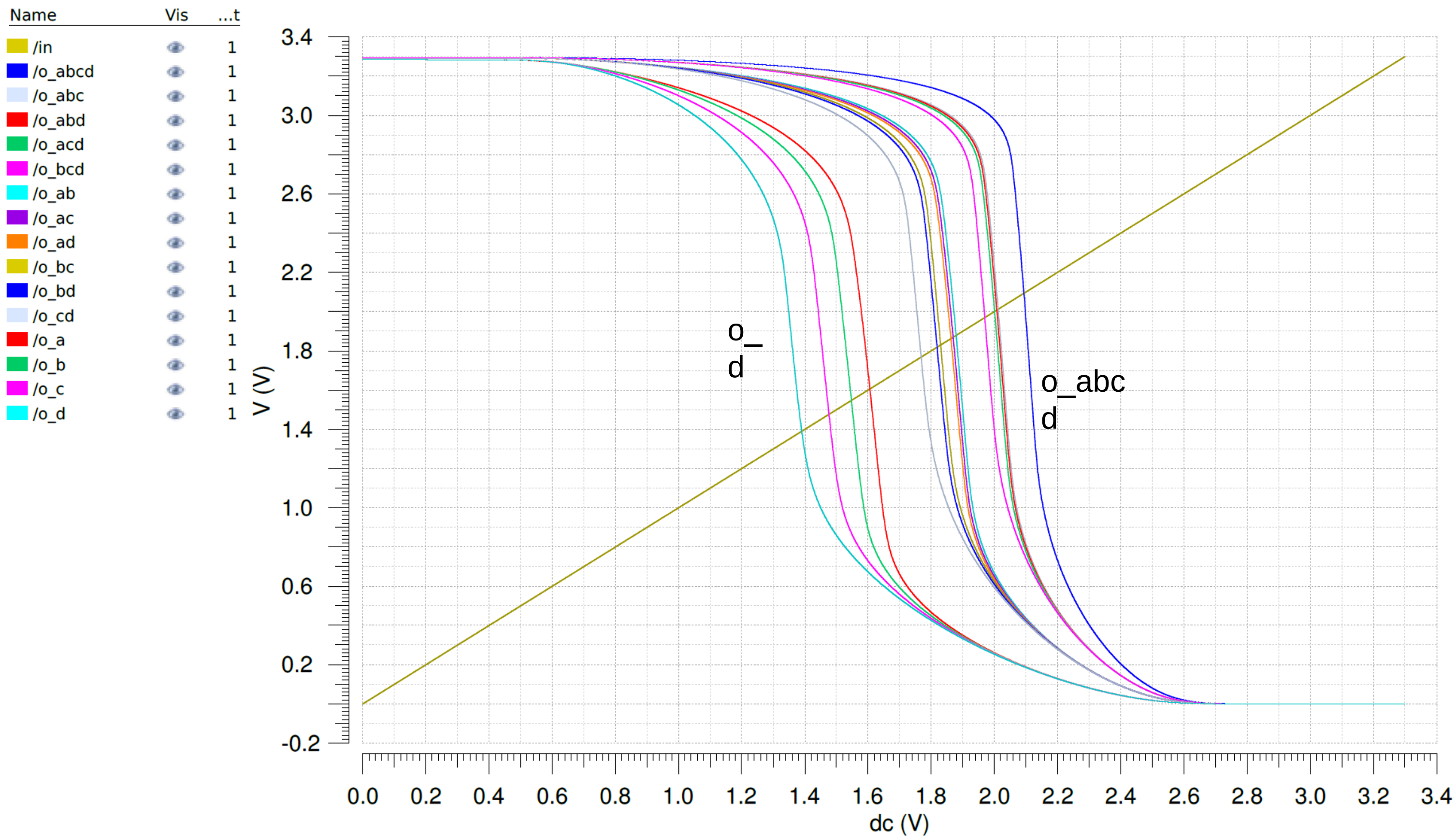
Simulation of 4-input-NAND from 0.35µm library





D-input uses NMOS-transistor nearest to GND, A-input is top NMOS-transistor.

4-Input-NAND-Gate from AMS-0.35µ-library

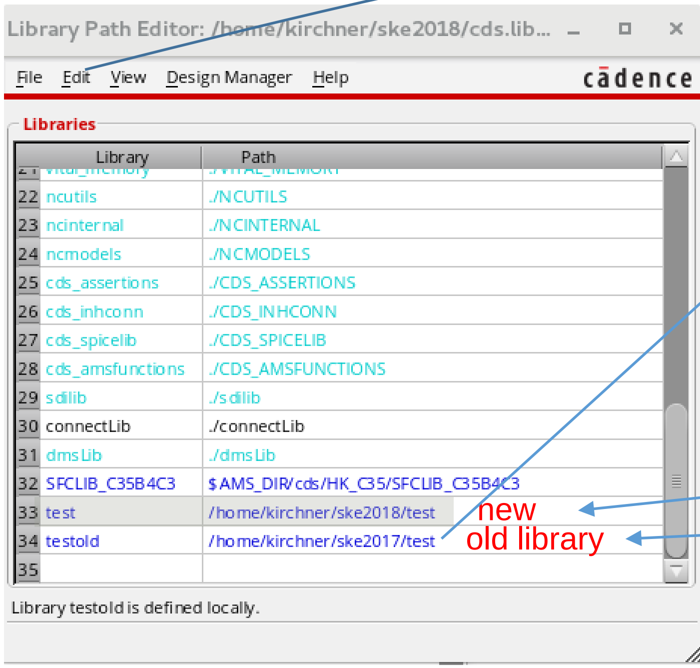


Transfer characteristic of 4-input-NAND-Gate from AMS-0.35µ-library
with $L_{NMOS,PMOS} = 0.35\mu$ $W_{NMOS} = 1.05\mu, W_{PMOS} = 1.4\mu$
Leftmost line for switching lowest (nearest to ground)nmos-transistor only, rightmost line for all 4 inputs switched simultaneously

Cadence Transfer Library Content to a new Library

Use Cadence tools always! DO NOT copy in linux.

Edit -> Copy Wizard

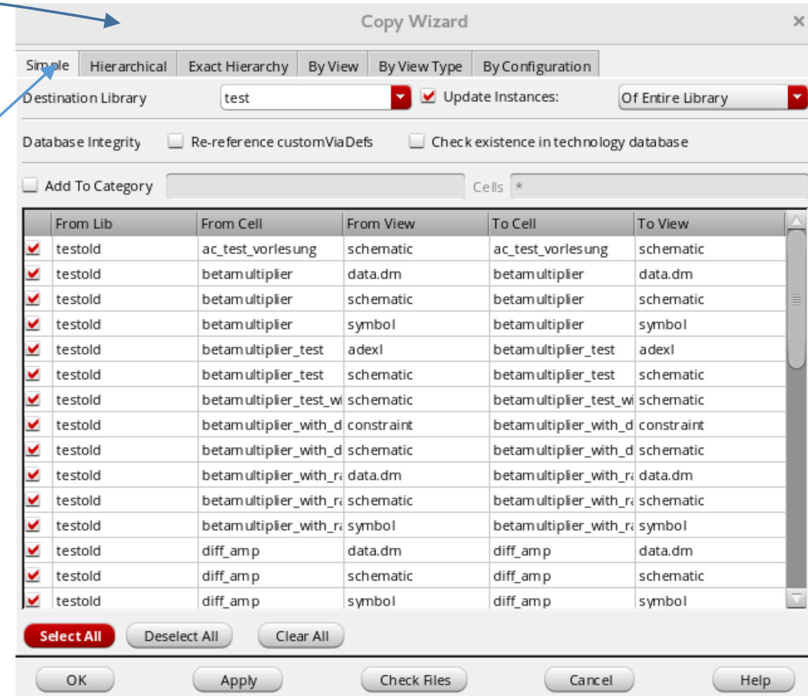


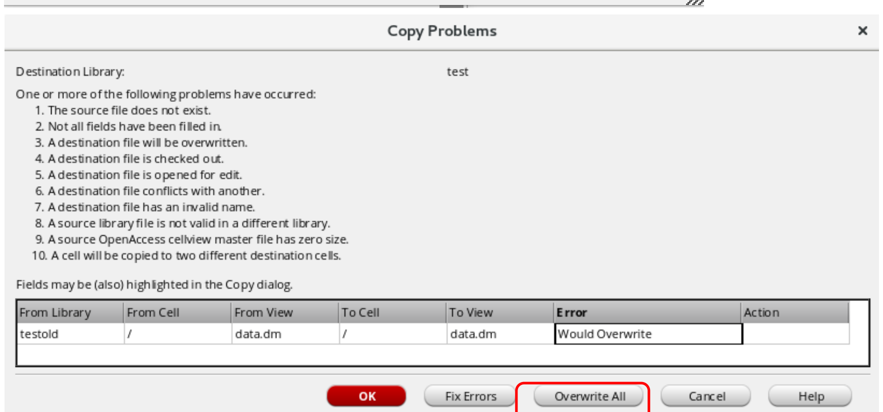
new

old library

Create before!

Enter path!

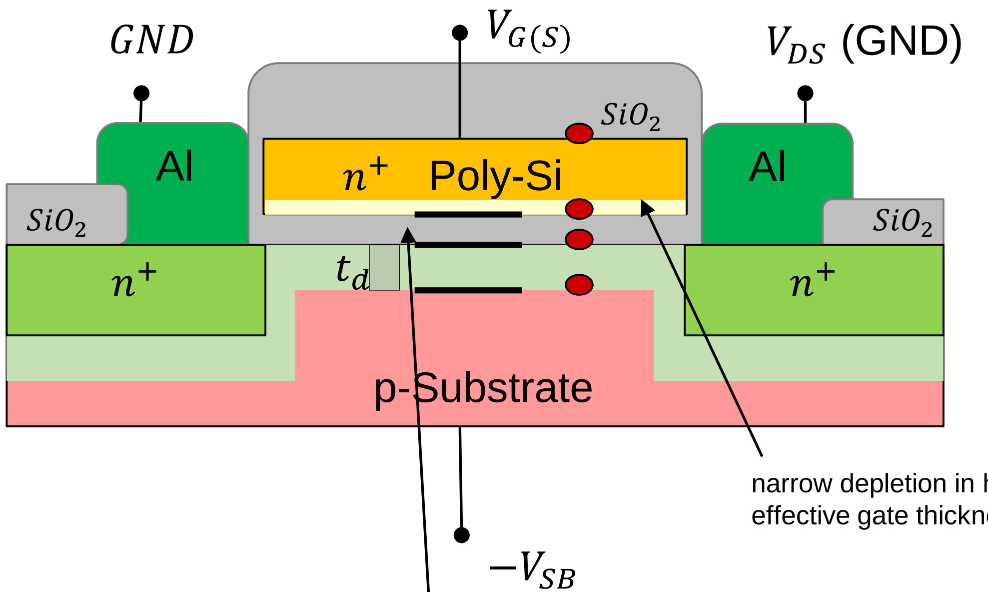




We get a warning because we are coying the library completely.
Acknowledge with "Overwrite All" !

When we will have finished copying we can delete the old libraries er

MOSFET: Threshold Voltage



$$\Phi_{F\ gate} = \frac{kT}{e} \ln \frac{N_{Dpoly}}{n_i}$$

Remember that the gate is used as the mask for the doping process making source and drain (self alignment)

$$\Phi_{F\ substr} = V_{fp} = -\frac{kT}{e} \ln \frac{N_{Asubstr}}{n_i}$$

narrow depletion in heavely doped gate
effective gate thickness increases

flatband voltage

$$V_{fb} = (V_G - V_{OX}) + (V_{OX} - V_{fp}) = V_G - V_{fp} = \frac{kT}{e} \ln \frac{N_{Dpoly}}{n_i} + \frac{kT}{e} \ln \frac{N_{ASubstr}}{n_i} = \frac{kT}{e} \ln \frac{N_{Dpoly} \cdot N_{ASubstr}}{n_i}$$

inversion voltage

Voltage necessary to achieve channel inversion
2x the potential in the depletion area
(from p-conduction to zero to n-conduction)

contact voltage(s))

$$V_{TH} = \frac{Q_b' - Q_{SS}'}{C_{OX}'} - 2V_{fp} - V_{contact}$$

Voltage over gate oxide (observe that the charge quantity on gate is equal to bulk $Q_g = -Q_b$)

MOSFET: Threshold Voltage

inversion voltage

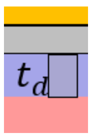
contact voltage(s)

Compare with pn-junction, but here one side weakly doped: $u_i < \; 2 \cdot 300mV$

$$V_{TH} = \frac{Q'_b - Q'_{SS}}{C'_{OX}} - \boxed{2V_{fb}} - V_c$$

voltage over gate oxide

charge on gate = - charge on substrate



Charge per area in channel volume element

$$Q'_{b0} = eN_A t_d = eN_A \sqrt{\frac{2\varepsilon_{Si} \cdot |V_G - v_{fb}|}{e \cdot N_A}} = \sqrt{2eN_A \varepsilon_{Si} \cdot |-2\, v_{fb}|}$$

without bulk voltage

$$V_G = -v_{fb}$$

Thickness of depletion area -> pn junction

$$t_d = \sqrt{\frac{2\varepsilon_{Si}|V_G - v_{fb}|}{e \cdot N_A}}$$

$$Q'_b = eN_A D_d = eN_A \sqrt{\frac{2\varepsilon_{Si} \cdot |V_S - v_{fb}|}{e \cdot N_A}} = \sqrt{2eN_A \varepsilon_{Si} \cdot |-2v_i + V_{SB}|}$$

with bulk voltage

$$\frac{Q'_b - Q'_{SS}}{C'_{OX}} = \frac{Q'_{b0} - Q'_{SS}}{C'_{OX}} - \frac{Q'_{b0} - Q'_b}{C'_{OX}}$$

made to define a fixed part and a V_{SB}-dependent part

$$V_{TH} = -2V_{fb} - V_c + \frac{Q'_{b0} - Q'_{SS}}{C'_{OX}} - \frac{Q'_{b0} - Q'_b}{C'_{OX}} = -2V_{fb} - V_c + \frac{Q'_{b0} - Q'_{SS}}{C'_{OX}} + \frac{\sqrt{2eN_A\varepsilon_{Si}}}{C'_{OX}} \left(\sqrt{|-2v_{fb} + V_{SB}|} - \sqrt{|-2v_{fb}|} \right)$$

MOSFET: Threshold Voltage

$-2v_{fp}$

surface potential

does not depend on V_{SB}

contact voltage

$$V_{TH} = -2v_{fb} - V_c + \frac{Q'_{b0}-Q'_{SS}}{C'_{OX}} + \frac{\sqrt{2eN_A\varepsilon_{Si}}}{C'_{OX}}\left(\sqrt{|-2v_{fp} + V_{SB}|} - \sqrt{|-2v_{fp}|}\right)$$

Q'_{b0} – charge per (top) area under gate oxide in length ΔL

Q'_{SS} – charge stacked at surface

γ – body effect parameterr

$$V_{TH} = -2V_{fp} - V_c - \frac{Q'_{SS}}{C'_{OX}} + \frac{Q'_{b0}}{C'_{OX}} + \gamma\left(\sqrt{|-2v_{fp} + V_{SB}|} - \sqrt{|-2v_{fp}|}\right)$$

$$V_{TH} = \boxed{- V_c - \frac{Q'_{SS}}{C'_{OX}}} - 2v_{fb} + \frac{Q'_{b0}}{C'_{OX}} + \gamma\left(\sqrt{|-2v_{fp} + V_{SB}|} - \sqrt{|-2v_{fp}|}\right)$$

V_{FB}- Flat Band Voltage

$\sqrt{2eN_A\varepsilon_{Si} \cdot |-2\, v_{fp}|}$

$$V_{TH} = \boxed{\boxed{V_{FB}} - 2v_{fp} + \frac{Q'_{b0}}{C'_{OX}}} + \gamma\left(\sqrt{|-2v_{fp} + V_{SB}|} - \sqrt{|-2v_{fp}|}\right)$$

$$V_{TH} = V_{GS0} = \boxed{V_{TH0}} + \gamma\left(\sqrt{|-2v_{fp} + \boxed{V_{SB}}|} - \sqrt{|-2v_{fp}|}\right)$$

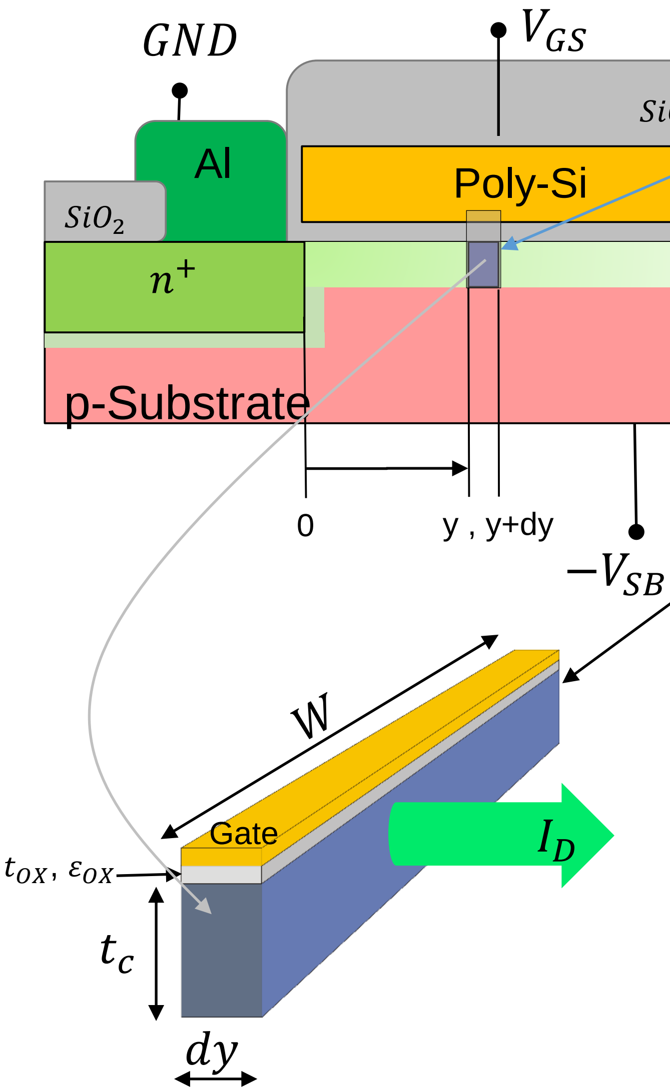
V_{TH} depends on:

substrate doping Q'_{b0} (used for adjustment)
steep retrograde doping (light at surface, strong in body)
thin channel, less dependent on V_{SB})

oxide thickness (some nm, 2..20)
substrate voltage (consider transistor location in circuit relative to GND/VCC)
gate material (metal, n+-Si, p+-Si)
surface charge density Q'_{SS}(Si, Na – ions)

MOSFET: Control Equation

We do not know the profile of the channel in advance
but we can evaluate the conditions in a small volume



$$Q'(y) = C'_{OX}(V_{GS} - V(y) \; - V_{th})$$

Charge of gate capacitor per area $\Delta y \cdot W$

$$\rho = \frac{1}{e \cdot n \cdot \mu} = \frac{1}{\frac{Q'(y)}{t_c} \cdot \mu}$$

Charge in volume $\Delta y \cdot W \cdot t_c$

$$r_\square = \frac{\rho}{t_c} = \frac{1}{\mu \cdot Q'(y)}$$

$$dR = \underbrace{\boxed{\frac{1}{\mu_n Q'(y)}}}_{r_\square} \cdot \frac{dy}{W}$$

$$dV(y) = I_D \cdot dR = \frac{I_D}{W \cdot \mu_n Q'(y)} \cdot dy$$

$$I_D \cdot dy = W \cdot \mu_n \cdot C'_{OX}(V_{GS} - V(y) \; - V_{th})dV(y)$$

$$\mu_n \cdot C'_{OX} = \mu_n \cdot \frac{\varepsilon_{OX}}{t_{OX}} = \boxed{KP_n}$$

Effective mobility in inverted channel less than mobility in table for n-Si : 450 vs. 1650 $\frac{cm^2}{Vs}$

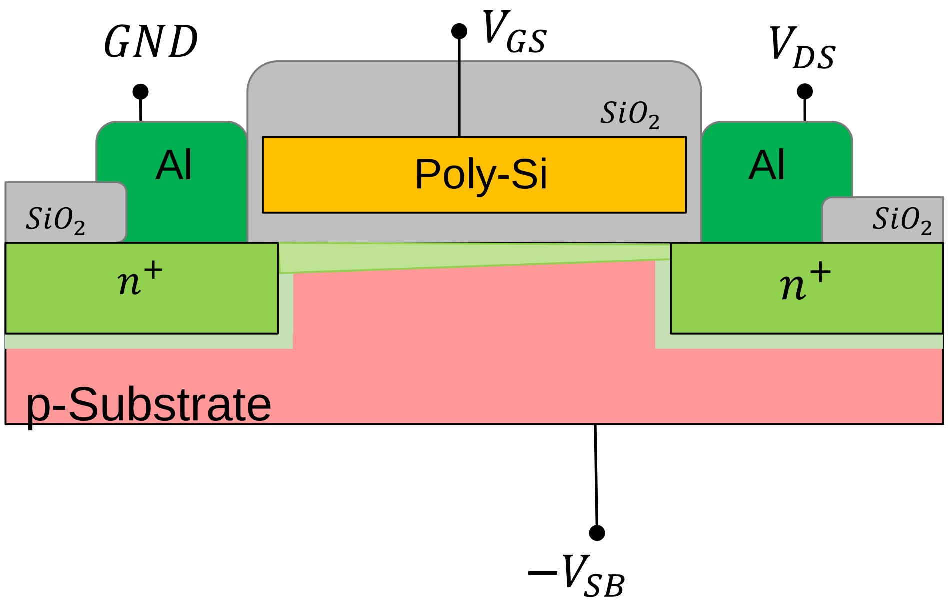
$$I_D \cdot dy = W \cdot KP_n \cdot (V_{GS} - V(y) \; - V_{th})dV(y)$$

MOSFET: Control Equation, non pinched off

$$I_D \cdot dy = W \cdot KP_n \cdot (V_{GS} - V(y) - V_{th}) \cdot dV(y)$$

$$I_D \int_0^L dy = W \cdot KP_n \int_0^{V_{DS}} (\underline{V_{GS} - V_{th}} - \underline{V(y)}) \cdot dV(y)$$

$$I_D = \boxed{\frac{W}{L} \cdot KP_n} \cdot \left[(V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2} \right] = \boxed{\beta} \cdot \left[(V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2} \right]$$

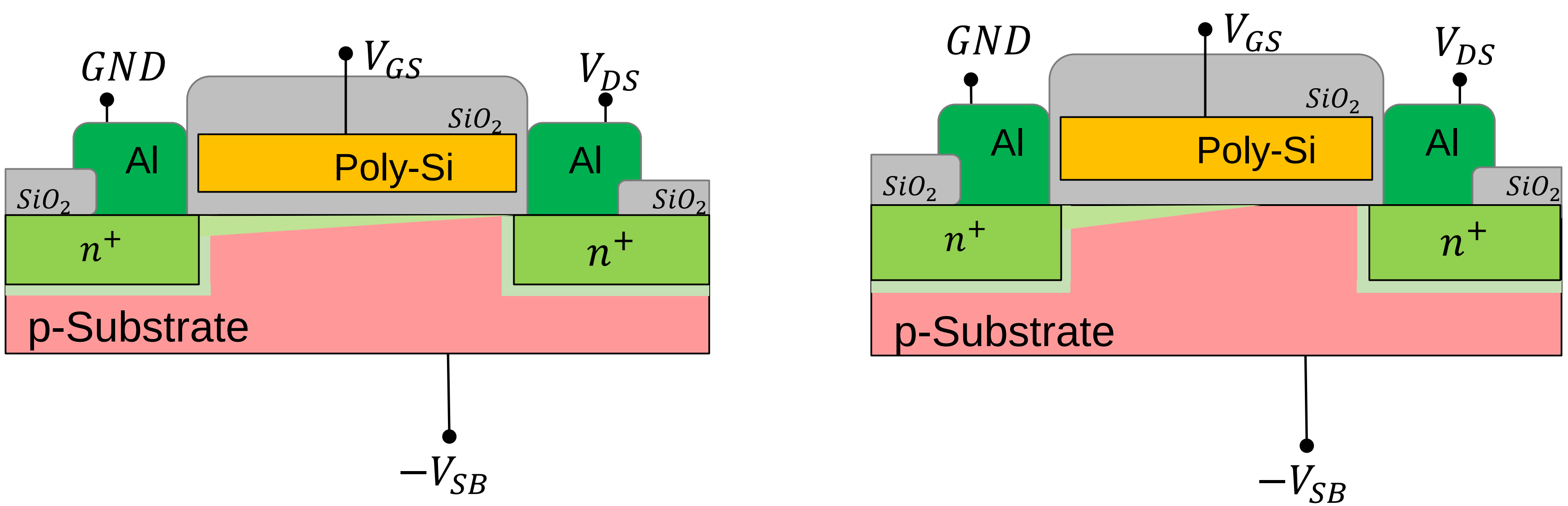


MOSFET: Control Equation, pinched off

$$I_D = \frac{W}{L} \cdot KP_n \cdot \left[(V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2} \right]$$

Pinch off when: $V_{GS} - V_{th} \leq V_{DS}$

$$I_D = \frac{W}{L} \cdot KP_n \cdot \frac{(V_{GS} - V_{th})^2}{2} = \frac{\beta}{2} \cdot (V_{GS} - V_{th})^2$$

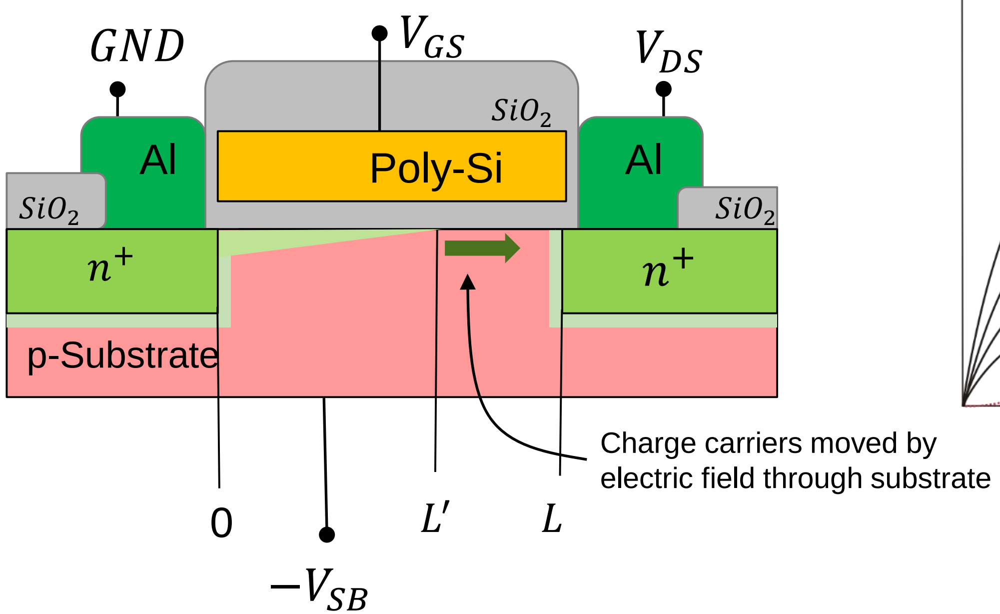


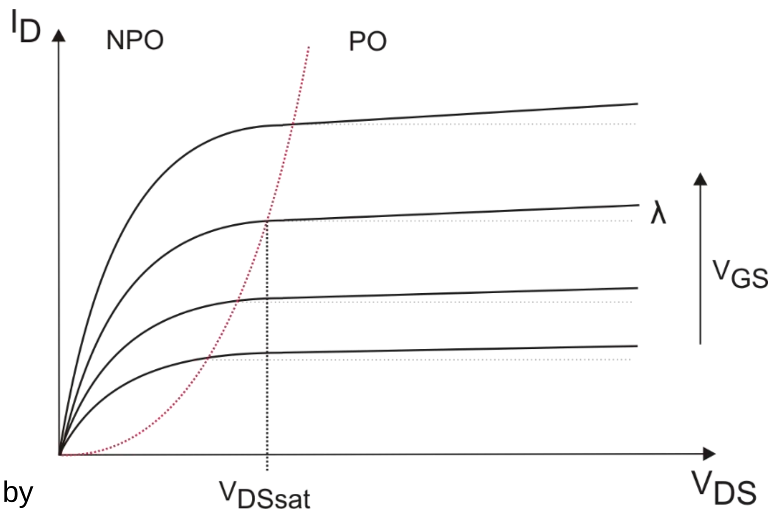
MOSFET: Control Equation, pinched off

$$I_D = \frac{W}{L} \cdot KP_n \cdot \frac{(V_{GS} - V_{th})^2}{2} = \frac{\beta}{2} \cdot (V_{GS} - V_{th})^2$$

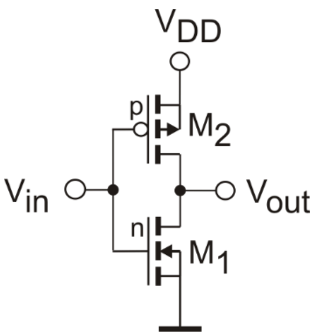
Channel-length modulation

$$I_D = \frac{\beta}{2} \cdot (V_{GS} - V_{th})^2 \quad \rightarrow \frac{\beta}{2} \cdot (V_{GS} - V_{th})^2 \cdot [1 + \lambda \cdot (V_{DS} - V_{DSsat})]$$





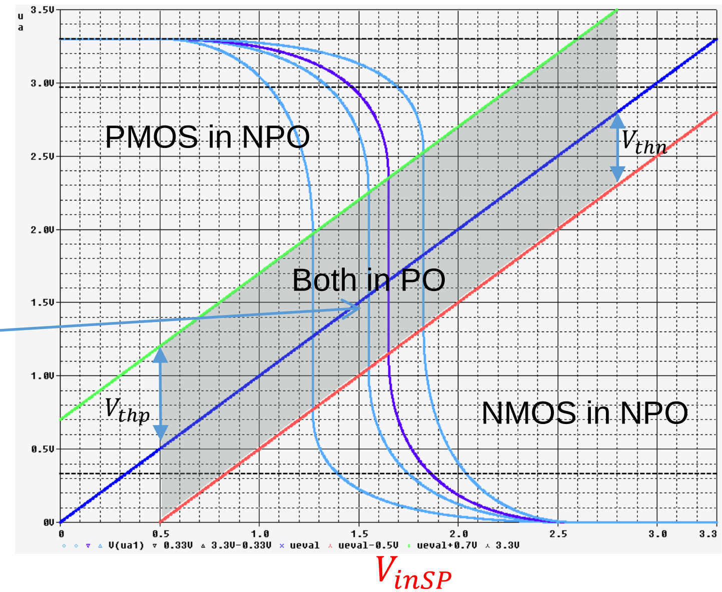
CMOS Inverter Calculation



There is the same current flowing through both transistors.Beide. For ideal switching point is

$V_{in} = V_{GSn} = V_{SGp} = V_{out} = V_{DSn} = V_{SDp} = \frac{V_{DD}}{2}$.

Because of $|V_{DS}| > |V_{GS} - V_{th}|$ both transistors are in the pinch-off state and we can assume:



$$\frac{\beta_n}{2}(V_{inSP} - V_{thn})^2 = \frac{\beta_p}{2}\left(V_{DD} - V_{inSP} - V_{thp}\right)^2$$

$V_{thn} = 0.5\ \mathrm{V}$,	$\mathrm{KPN} = 170\ \frac{\mu\mathrm{A}}{\mathrm{V}^2}$
$V_{thp} = 0.65\ \mathrm{V}$,	$\mathrm{KPP} = 58\ \frac{\mu\mathrm{A}}{\mathrm{V}^2}$

Dissolving for V_{inSP}:

$$\sqrt{\beta_n}(V_{inSP} - V_{thn}) = \sqrt{\beta_p}(V_{DD} - V_{inSP} - V_{thp})$$

$$V_{inSP} = \frac{\sqrt{\frac{\beta_n}{\beta_p}} \cdot V_{thn} + V_{DD} - V_{thp}}{1 + \sqrt{\frac{\beta_n}{\beta_p}}}$$

For desired $V_{inSP} = \frac{V_{DD}}{2}$ we calculate the relation $\frac{\beta_n}{\beta_p}$

$$\frac{\beta_n}{\beta_p} = \left(\frac{\frac{V_{DD}}{2} - V_{thp}}{\frac{V_{DD}}{2} - U_{thn}}\right)^2$$

$$\beta = KP \cdot \frac{W}{L}$$

using: $KPN = 3 \cdot KPP$

$$\frac{\left[\frac{W}{L}\right]_n}{\left[\frac{W}{L}\right]_p} = \frac{1}{3} \cdot \left(\frac{\frac{V_{DD}}{2} - V_{thp}}{\frac{V_{DD}}{2} - V_{thn}}\right)^2 \approx \boxed{\frac{1}{3}}$$

for $|V_{DD}| \gg |V_{thp} - V_{thn}|$